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16 October 1963

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16 November 1959

MILITARY STANDARD

DIMENSIONING AND TOLERANCING

JAMES E. DRAPER

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MILITARY STANDARD
DIMENSIONING AND TOLERANCING

DEPARTMENT OF DEFENSE
DEFENSE SUPPLY AGENCY

Dimensioning and Tolerancing
MIL–STD–8C 16 October 1963

1. This standard has been approved by the Department of Defense and is mandatory for use by the Department of the Army, the Navy, and the Air Force, effective 16 October 1963.

2. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Defense Supply Agency, Washington 25, D. C.

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CONTENTS

Paragraph		*Page*
1.	SCOPE	1
1.1	Scope	1
1.2	Purpose	1
1.3	Figures and notes	1
2.	REFERENCED DOCUMENTS	1
3.	DEFINITIONS	1
3.1	Allowance	1
3.2	Clearance	1
3.3	Dimension	1
3.4	Dimensioning	1
3.5	Eccentricity	1
3.6	Feature	1
3.7	Fit	1
3.8	Interchangeability	1
3.9	Interference	2
3.10	Limits	2
3.11	Typical	2
4.	GENERAL REQUIREMENTS	2
4.1	Reference to this standard	2
4.2	Fundamental rules	2
4.3	Use of standards	2
5.	DETAIL REQUIREMENTS	2
5.1	Drafting methods	2
5.2	Units of measurement	7
6.	DIMENSIONS	9
6.1	General remarks about dimensioning	9
6.2	Expressing size and position	10
6.3	Dimensions from datum	10
6.4	Reference dimensions	14
6.5	Basic dimension	14
6.6	Rules for dimensioning special features	14
7.	TOLERANCES AND LIMITS	28
7.1	General	28
7.2	Expressing positional and form tolerances	32
7.3	Significance of MMC and RFS	35
7.4	Virtual size	36
7.5	True position tolerancing	37
7.6	Tolerances of form	45
8.	NOTES	64
	APPENDIX	65
10.	True position	65
20.	The meaning of MMC as related to positional tolerances	69
30.	Differences in American, British, and Canadian standards	73
40.	Interrelated geometric characteristics and the Two Datum System	73
	INDEX	80

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FIGURES

Figure		*Page*
1.	Two ways to indicate dimensions	2
2.	Dimension lines	3
3.	Arranging dimension lines, etc., for orderly appearance	3
4.	Numerals staggered for clarity	3
5.	Use of dimension lines as extension lines	4
6.	Use of extension lines	4
7.	Extension lines	4
8.	Breaks in extension lines	4
9.	Extension lines pass through points they locate if the points are not on outlines	5
10.	Use of leaders	5
11.	Leaders to circles should be in radial directions	5
12.	Leaders should not make small angles with outlines	5
13.	Dimensioning in limited space	6
14.	Dimensions from a datum in a single line	6
15.	Unidirectional dimensional expressions	6
16.	Aligned dimensional expressions	6
17.	Do not place aligned dimension numerals in the crosshatched area	7
18.	Dimensioning within the outline of a view	7
19.	Expressing angular dimensions	8
20.	Optional ways to show dimensions in common fractions	8
21.	Decimal dimensioning	9
22.	Locating round holes by linear distances	10
23.	Locating round holes by rectangular coordinates	11
24.	Locating holes on circle by angular dimensioning	11
25.	Locating holes on circles by radius or diameter and "EQUALLY SPACED"	11
26.	"EQUALLY SPACED" holes in a line	11
27.	Dimensions from datums	12
28.	Ordinate dimensioning	12
29.	Tabular dimensioning	13
30.	A reference dimension	14
31.	Dimensioning diameters of concentric elements	14
32.	An end view of cylindrical surfaces	15
33.	How to dimension a circular arc	15
34.	True radius dimensioning	15
35.	Indicating a spherical surface	15
36.	Dimensioning along a curved surface	16
37.	Dimensioning parts with rounded ends	16
38.	Dimensioning curves made up of circular arcs	16
39.	Dimensioning a noncircular curve	16
40.	Dimensioning round holes	17

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Check the source to verify that this is the current version before use.

FIGURES (continued)

Figure		Page
41.	A counterbored hole	17
42.	A countersunk hole	17
43.	A spot-faced hole	18
44.	A shaft center	18
45.	Dimensioning chamfers	18
46.	Method of dimensioning Woodruff keyslots	19
47.	Keyways for stock keys	19
48.	A knurl for decoration or gripping	19
49.	A knurl for a press-fit	19
50.	Dimensioning taper, taper per inch method	20
51.	Dimensioning taper, angular method	20
52.	Dimensioning taper, BASIC diameter method	21
53.	Dimensioning taper where diameter and taper are important	22
54.	Dimensioning taper, BASIC length method	22
55.	Taper on a flat piece	23
56.	Cumulative angular tolerancing	23
57.	Basic angle tolerancing	24
58.	Dovetails: Basic angle tolerancing to control width at contacting surface	24
59.	Dovetails: Basic angle tolerancing to show tolerance of pin measurements	25
60.	Dovetails: Basic angle tolerancing	25
61.	Dimensioning bent or formed sheet metal parts	26
62.	Indicating dimensioning intersections	26
63.	Dimensioning stiffening rib sections	27
64.	Indicating sheet metal cross sections	27
65.	Dimensioning eyelets	27
66.	Dimensioning overlaps for welding	28
67.	Locating round holes	29
68.	Giving a tolerance by a plus and minus figure	29
69.	Using a combined plus and minus sign	29
70.	Limit dimensioning	30
71.	Expressing a single tolerance or limit	30
72.	Indicating dimensions of surfaces that are to fit closely	31
73.	Dimensioning noninterchangeable parts that are to fit closely	31
74.	"Basic hole" fits	32
75.	"Basic shaft" fits	32
76.	Geometric characteristic symbols and modifiers	33
77.	Feature control symbol	33
78.	Datum identifying symbol	34
79.	Feature control symbol incorporating datum reference	34
80.	Combined feature control symbol and datum identifying symbol	35

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

FIGURES (continued)

Figure		Page
81.	Interrelated geometric characteristics	35
82.	Square tolerance zone resulting from rectangular coordinate dimensioning	37
83.	Wedge-shaped tolerance zone resulting from angular dimensioning	37
84.	Circular tolerance zone resulting from true position dimensioning	37
85.	Projected tolerance zone interpretation	39
86.	Projected tolerance zone for tapped holes	39
87.	Projected tolerance zone for fixed pins or studs	40
88.	Methods of specifying true position	42
89.	Application of positional tolerances	43
90.	Hole patterns at true position	44
91.	Flatness	46
92.	Straightness of axis	47
93.	Straightness of elements RFS	47
94.	Straightness of elements MMC	48
95.	Angularity	49
96.	Angularity tolerance for solids of revolution	49
97.	Perpendicularity	50
98.	Perpendicularity of cylinder to datum plane	51
99.	Perpendicularity of cylinder to datum cylinder	52
100.	Parallelism	53
101.	Concentricity	54
102.	Concentricity (Example 1)	54
103.	Concentricity (Example 2)	55
104.	Symmetry	56
105.	Roundness of rigid parts	57
106.	Roundness of non-rigid parts (free-state variation)	57
107.	Free-state variation	59
108.	Perfect form at MMC (Example 1)	60
109.	Perfect form at MMC (Example 2)	60
110.	Specifying that a hole must be parallel with a flat surface	61
111.	Specifying that a hole must be located from and is parallel to another hole	62
112.	Methods of zone tolerancing	63
113.	Tolerance zone or boundary within which form may vary when no tolerance of form is given	63
10–1 thru 10–10.	True position	65
20–1 thru 20–8.	The meaning of MMC as related to positional tolerances	70
30–1	Differences in American-British-Canadian Standards	74
40–1	True position and form tolerances using double datums	76
40–2	Controlling geometrical relationships of multiple surfaces	78

TABLE

Table I.	Standard Conversion Table	79

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MILITARY STANDARD

DIMENSIONING AND TOLERANCING

1. SCOPE

1.1 *Scope.* This standard establishes the rules, principles, and methods of dimensioning and tolerancing applicable to defining geometric characteristics of objects delineated on drawings.

1.2 *Purpose.* The purpose of this standard is to establish uniform practices of stating and interpreting requirements shown on drawings.

1.3 *Figures and notes.* Excepting the figure establishing geometric characteristic symbols, the figures given in this standard are to illustrate meanings only. Notes printed in capital letters are intended to appear on drawings. Notes in lower case letters are explanatory only and do not appear on drawings.

2. REFERENCED DOCUMENTS

NOT APPLICABLE

3. DEFINITIONS

3.1 *Allowance.* An allowance is a prescribed difference between the maximum material condition of mating parts. It is the minimum clearance (positive allowance) or maximum interference (negative allowance) between such parts (see figs 74 and 75) (see definition of fit).

3.2 *Clearance.* Clearance is the total space between mating parts. (see figs 74 and 75).

3.3 *Dimension.* A dimension is a numerical value expressed in appropriate units of measure. It is indicated on drawings in conjunction with lines, symbols, and notes to define the geometrical characteristics of an object.

3.4 *Dimensioning.*

3.4.1 *Angular dimensioning.* The angular dimensioning system is a method for indicating the position of a point, line; or surface by means of linear dimensions and angles, other than 90° angles (see 7 and 19).

3.4.2 *Rectangular dimensioning.* The rectangular dimensioning system is a method for indicating distances, locations, and sizes by means of linear dimensions measured parallel to reference lines or planes which are perpendicular to each other (see fig. 1).

3.5 *Eccentricity.* Eccentricity is a condition where the axis of a particular feature is parallel to, but offset from, the axis of another feature; or where the axis of a rotating part mounted in an assembly does not coincide with the axis of the part about which it turns.

3.6 *Feature.* Features are specific characteristics or component portions of a part, which may include one or more surfaces such as holes, screw threads, profiles, or rabbets.

3.7 *Fit.* Fit is the general term used to signify the range of looseness or tightness which may result from the application of a specific combination of allowances or tolerances in the design of mating parts.

3.7.1 *Actual fit.* The actual fit between two or more mating parts is the relation existing between them with respect to the amount of clearance or interference which is present when they are assembled.

3.7.2 *Clearance fit.* A clearance fit is one having limits of size so prescribed that clearance always results when mating parts are assembled.

3.7.3 *Interference fit.* An interference fit is one having limits of size so prescribed that an interference always results when mating parts are assembled.

3.7.4 *Transition fit.* A transition fit is one having limits of size so prescribed that either a clearance or an interference may result when parts are assembled.

3.8 *Interchangeability.* Interchangeability is a condition of design wherein any and all mating parts will assemble and function properly without the need for any part modification at assembly.

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3.9 *Interference.* Interference is the total amount of deformation required to force an internal member into a smaller external member.

3.10 *Limits.* Limits are the maximum and minimum values prescribed for a specific dimension. Limits may be of size or of location depending upon the application.

3.11 *Typical.* The term "typical," when associated with a dimension or feature, means that this dimension or feature applies to the locations that appear to be identical in size and configuration. The tolerance stated for a dimension labeled "typical" also applies to each identical feature.

Note. Additional definitions will appear where applicable.

4. GENERAL REQUIREMENTS

4.1 *Reference to this standard.* When drawings are based on this standard, this fact shall be noted on the drawing or in a specification referenced on the drawing.

4.2 *Fundamental rules.*

4.2.1 Show enough dimensions so that the intended sizes, shapes, and locations can be determined without assuming any distances.

4.2.2 State each dimension clearly, so that it can be interpreted in only one way.

4.2.3 Show the dimensions between points, lines, or surfaces which have a necessary and specific relation to each other or which control the location of other components or mating parts.

4.2.4 Select and arrange dimensions to avoid accumulations of tolerances that may permit various interpretations and cause unsatisfactory mating of parts and failure in use.

4.2.5 Show each dimension only once except when necessary to clarify a section or view. Repeated dimensions shall be noted as REF. (see 6.4—Reference dimensions)

4.2.6 When possible, dimension each feature in the view where it appears in profile or where its true shape appears.

4.2.7 Each dimension, except those labeled "REF", "MAX", "MIN" or "BASIC", shall have a specified tolerance, either applied directly to the dimension or indicated by a general note or referenced document.

4.3 *Use of standards.* Wherever possible, specify dimensions to make use of readily available materials, parts, tools, and gages. Savings are often possible when drawings specify:

(a) Commonly used materials in stock sizes;

(b) Sizes that can be produced with standard tools and inspected with standard gages;

(c) Tolerances from acceptable published standards.

5. DETAIL REQUIREMENTS

5.1 *Drafting methods.*

5.1.1 *How dimensions are indicated.* Dimensions that indicate sizes and positions are given as linear distances or as angles. They may be given with dimension lines and extension lines that show their direction and extent, or in the form of notes with leaders that direct them to intended places on the drawing (see fig. 1).

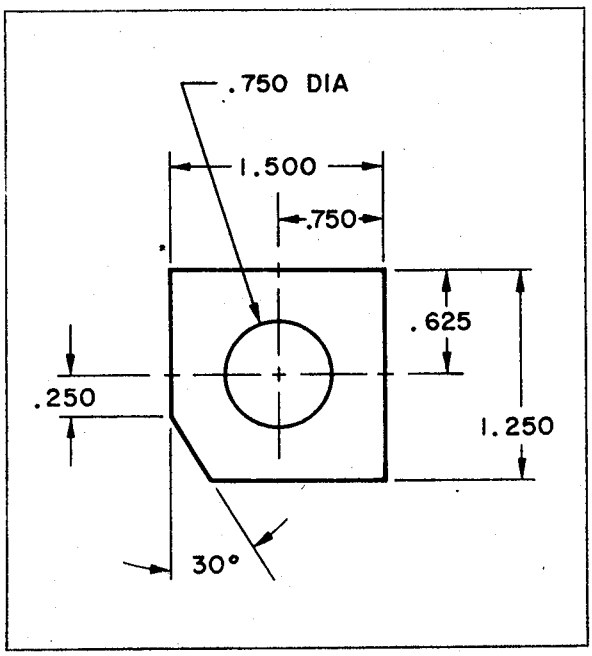


FIGURE 1. *Two ways to indicate dimensions.*

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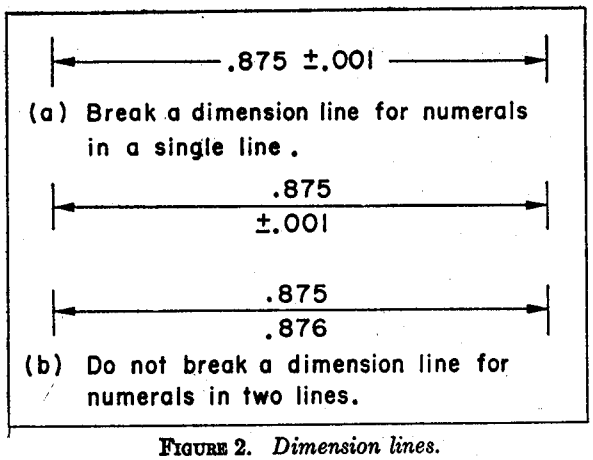


FIGURE 2. *Dimension lines.*

5.1.1.1 *Dimension lines.* A dimension line, with its arrowheads, shows the direction and extent of a dimension. Numerals indicate the number of units of a measurement. Dimension lines shall not pass through these numerals. If the numerals are in a single line, the dimension line shall be broken as in figure 2a. If the numerals are in two lines, one may be placed above and the other below the line, as in figure 2b. In crowded spaces, special methods are used (see 5.1.1.4). Dimension lines should be aligned if possible, and should be grouped for uniform appearance (see fig. 3). Parallel dimension lines should be placed not less than ¼ inch apart; no dimension line should be closer to the outline of the object than ⅜ inch. Where there are several parallel dimension lines, the numerals should be staggered to make them easier to read (see fig. 4). Do not use centerlines, extension lines, or lines that are part of the outline of the object, or a continuation thereof, as dimension lines. Do not use dimension lines as extension lines. Figure 5 illustrates a permissible exception to this rule which may be applied to the dimensioning of irregular curves only. Avoid crossing dimension lines if possible. The dimension line of an angle is the arc drawn with its center at the apex of the angle and terminating at the two sides (see fig. 6).

5.1.1.2 *Extension lines.* It is usually undesirable to terminate dimension lines directly at lines that represent surfaces of the actual object portrayed by the drawing. To avoid this, extension lines are used to show where numerical or

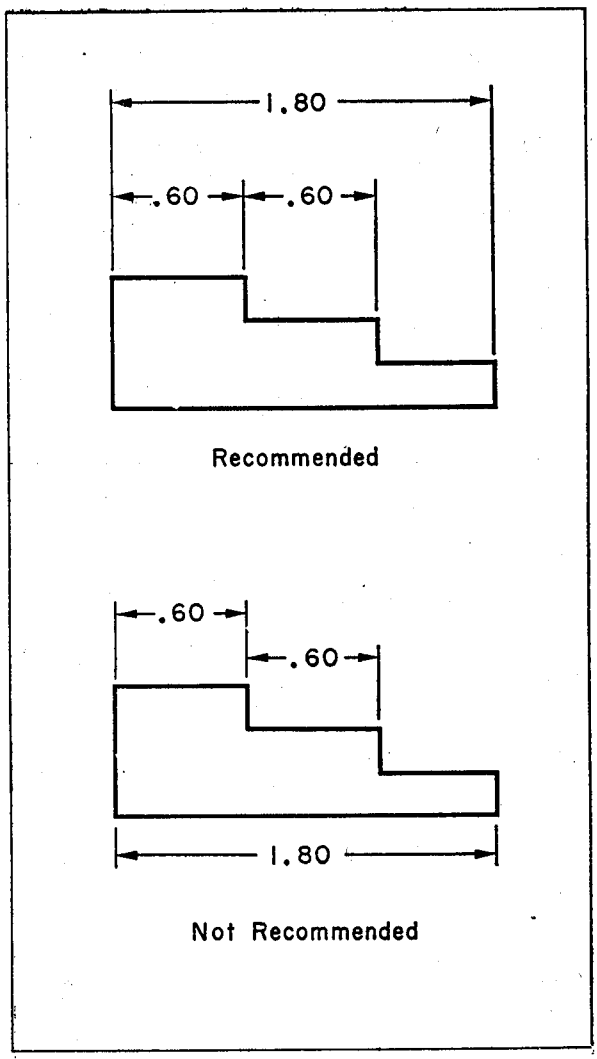


FIGURE 3. *Arranging dimension lines, etc., for orderly appearance.*

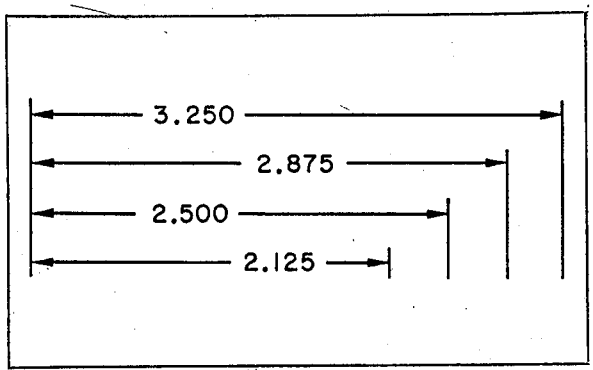


FIGURE 4. *Numerals staggered for clarity.*

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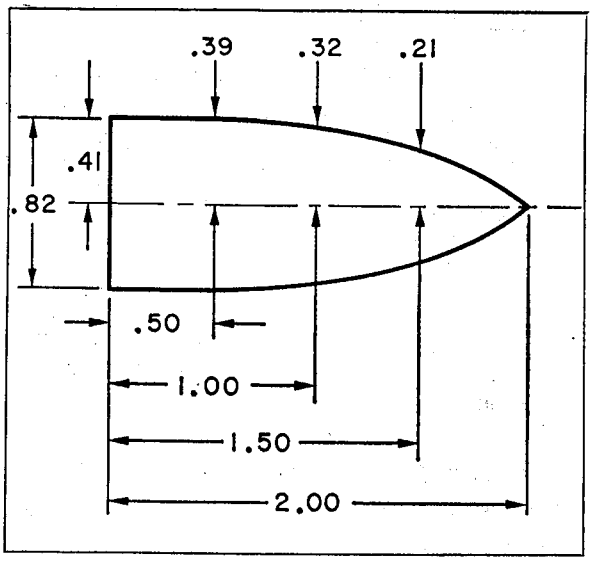


FIGURE 5. *Use of dimension lines as extension lines.*

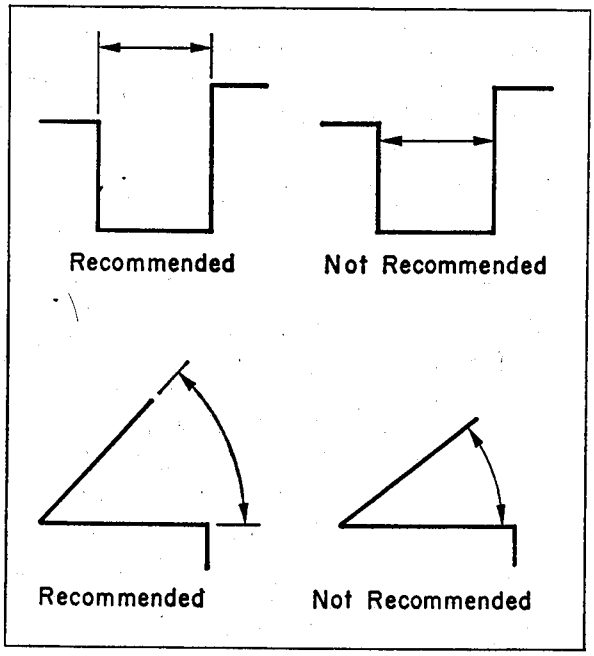


FIGURE 6. *Use of extension lines.*

other expressions are intended to apply (see fig. 6). An extension line drawn to an outline representing a surface shall start with a visible gap from the outline and extend about ⅛ inch beyond the outermost dimension line (see fig. 7). Extension lines are usually drawn perpendicular to dimension lines, but may be at other angles if their meaning is clear (see fig. 13).

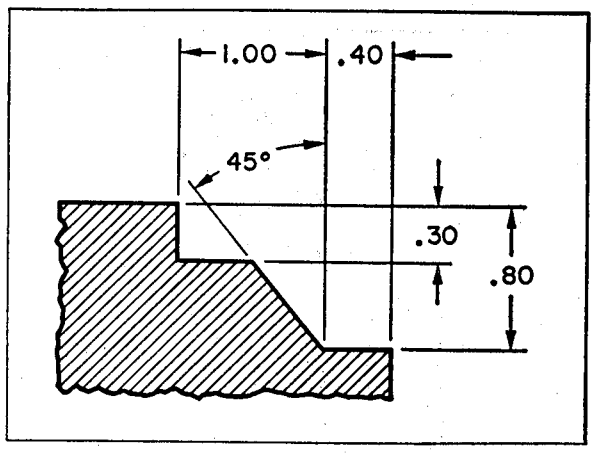


FIGURE 7. *Extension lines.*

5.1.1.2.1 *Crossing extension lines.* Wherever possible, extension lines should neither cross one another nor cross dimension lines. To reduce crossings to a minimum, draw the shortest dimension line nearest the outline of the object, and adjacent parallel dimension lines in order of their length, with the longest line the outermost (see fig. 27). Where extension lines cross other extension lines, dimension lines, or object lines they should not be broken. However, if an extension line crosses a dimension line close to an arrowhead, a break in the extension line is recommended (see fig. 8).

5.1.1.2.2 *Extension lines through point locations.* Where a point is located by extension lines alone, the extension lines shall pass through the point (see fig. 9).

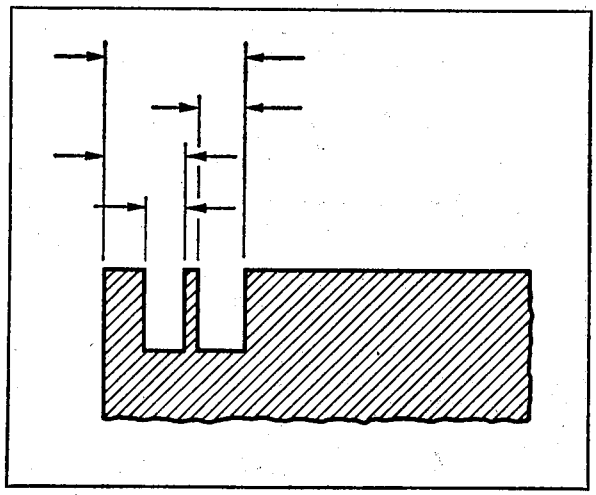


FIGURE 8. *Breaks in extension lines.*

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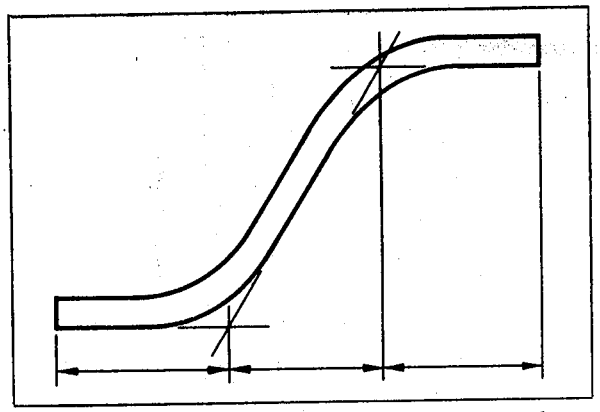


FIGURE 9. *Extension lines pass through points they locate if the points are not on outlines.*

5.1.1.3 *Leaders.* A leader is used to direct notes or symbols to the intended place on the drawing. It should terminate in an arrowhead or dot. Arrowheads should always terminate on a line; dots should be within the outline of the object. A leader should generally be an oblique straight line except for a short horizontal portion extending to midheight, preferably of the first or last letter or digit of the note. The horizontal portion is optional; if used, it should not underline the note. Two or more leaders to adjacent places should be drawn parallel if practicable (see fig. 10).

5.1.1.3.1 *Leaders to circles and arcs.* Where a leader is directed to a circle or circular arc, its direction shall be radial (see fig. 11).

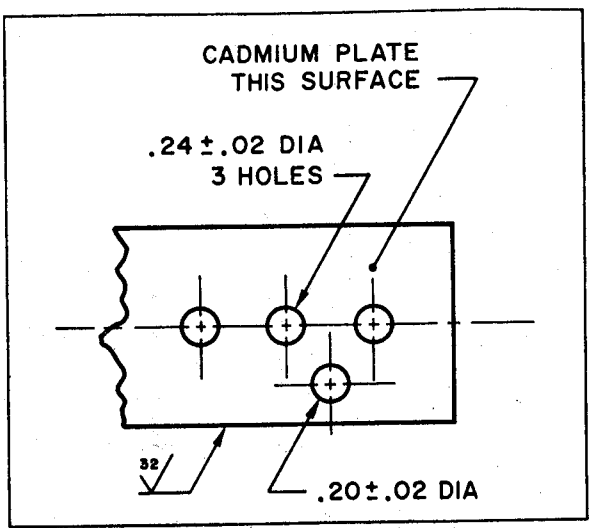


FIGURE 10. *Use of leaders.*

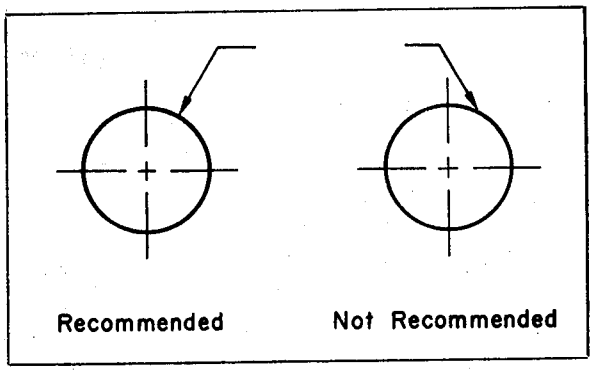


FIGURE 11. *Leaders to circles should be in radial directions.*

5.1.1.3.2 *Improper use of leaders.* Avoid the following if possible:

(a) Crossing leaders;
(b) Long leaders;
(c) Leaders in a horizontal or vertical direction;
(d) Leaders parallel to adjacent dimension lines, extension lines, or section lines;
(e) Small angles between leaders and lines where they terminate (see fig. 12).

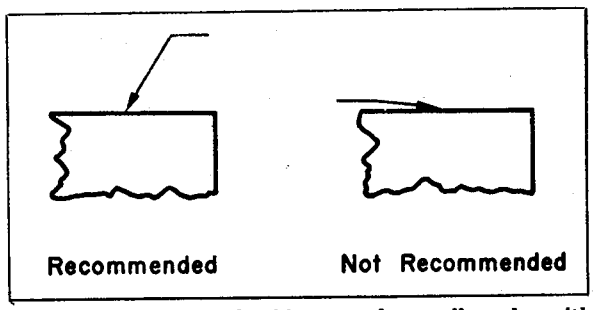


FIGURE 12. *Leaders should not make small angles with outlines.*

5.1.1.4 *Dimensioning in limited space.* For dimensioning in limited space, any of the methods shown in figure 13 may be used.

5.1.1.4.1 *Numerals in line from a datum.* When space is limited, a group of numerals giving dimensions from a common datum line may be placed in a straight line with single arrowheads, each indicating a distance from the datum line. The numerals should be kept close to the arrowheads and a dimension line used only between each arrowhead and its numeral. If practicable, the longest dimension line should be separate and complete (see fig. 14).

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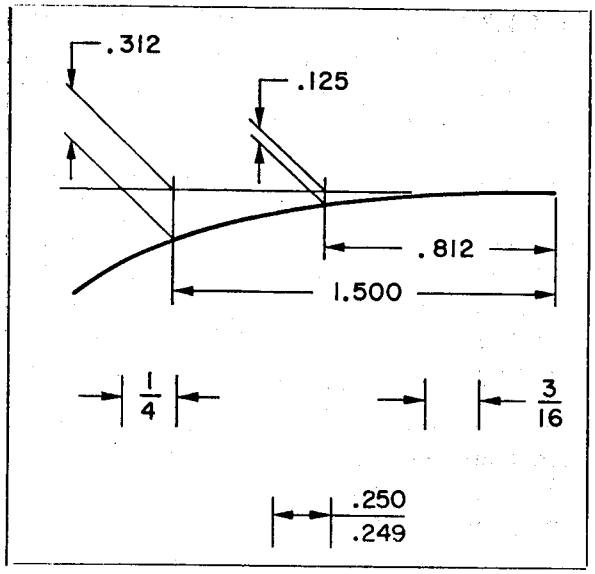


FIGURE 13. *Dimensioning in limited space.*

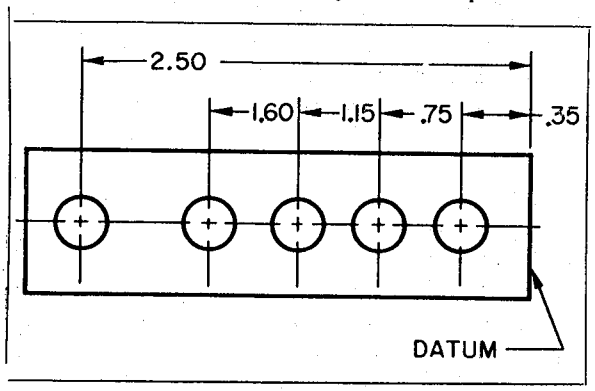


FIGURE 14. *Dimensions from a datum in a single line.*

5.1.1.5 *Reading direction.* Dimensions on drawings are shown by either of two methods, unidirectional or aligned.

5.1.1.5.1 Unidirectional dimensions are placed to be read from the bottom of the drawing (see fig. 15). Common fractions are written with the bar parallel to the bottom of the drawing.

5.1.1.5.2 Aligned expressions are placed along their dimension lines. Numerals should be readable from the bottom or right side of a drawing (see fig. 16). For this reason, dimension lines in the 45° zone indicated by shading in figure 17 should be avoided.

5.1.1.5.3 In both methods, dimensions and notes shown with leaders should be aligned with the bottom of the drawing.

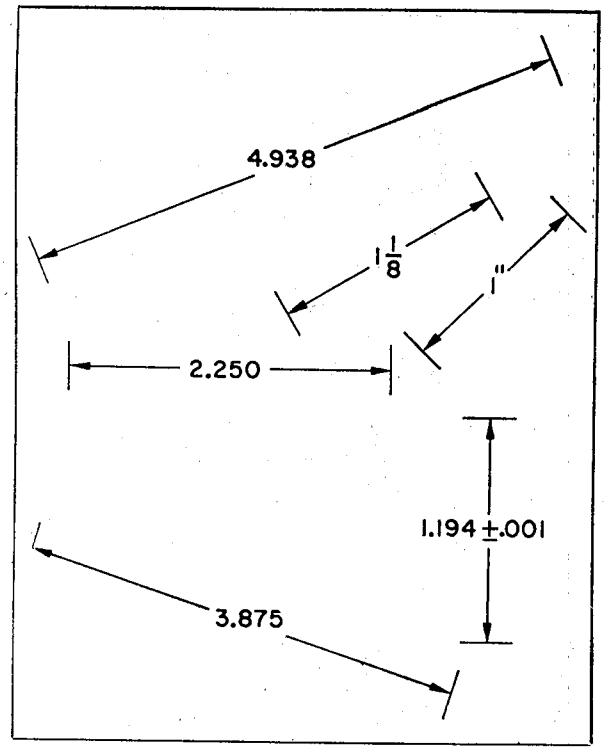


FIGURE 15. *Unidirectional dimensional expressions.*

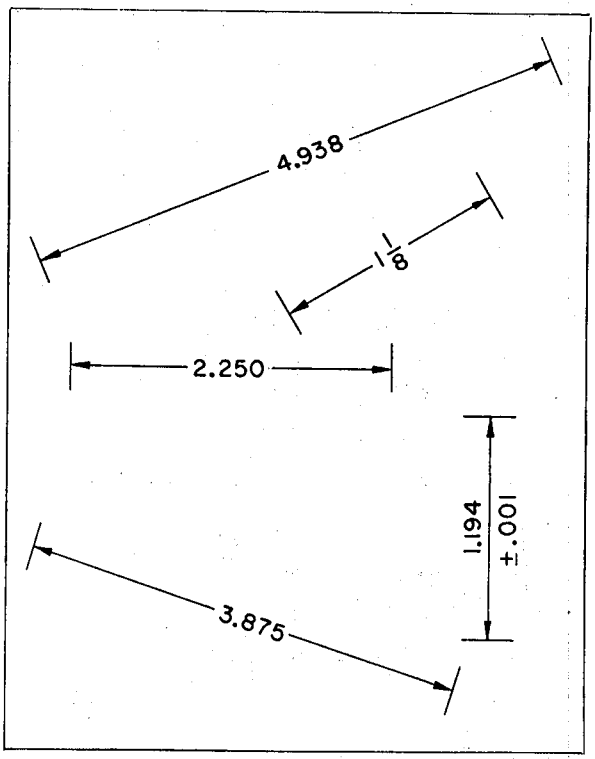


FIGURE 16. *Aligned dimensional expressions.*

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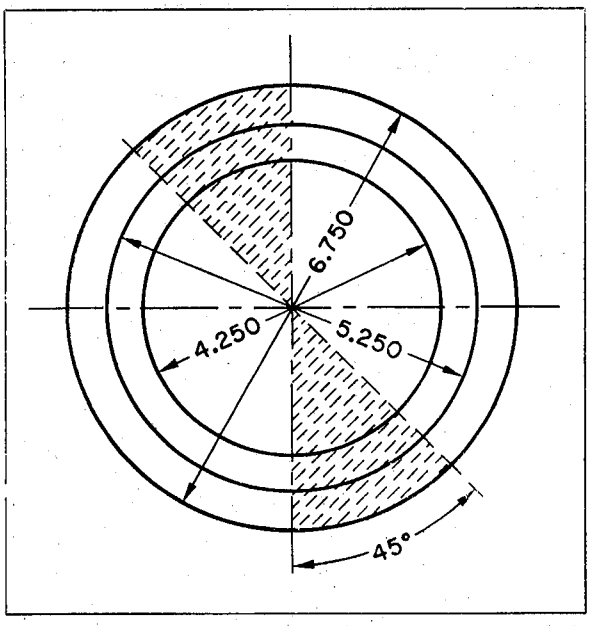


FIGURE 17. *Do not place aligned dimension numerals in the crosshatched area.*

5.1.1.6 *Dimensioning within the outline of a view.* Dimensions should preferably be placed outside the views. When directness of application makes it desirable, dimensions are sometimes placed within the outline of a view. When dimensions are indicated within the sectioned portion of a sectional view, an area around each numeral should be kept free from section lines (see fig. 18).

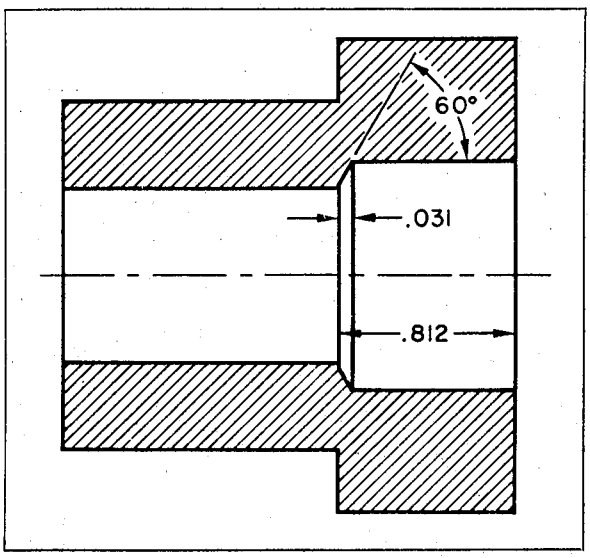


FIGURE 18. *Dimensioning within the outline of a view.*

5.1.1.7 *Dimensioning to hidden lines.* When practicable, avoid indicating dimensions to hidden lines; it is better to dimension another view. A sectonal view may be necessary.

5.1.1.8 *Abbreviations.* Certain abbreviations are used on drawings very commonly, and are universally understood. Examples are THD, DIA, MAX, etc. It is recommended that less common abbreviations be used as little as possible, because misinterpretations are less likely if words are fully spelled out.

5.1.1.9 *Decimal points.* Decimal points shall be definite, shall be given sufficient space for legibility, and shall be placed in line with the bottom edges of digits.

5.1.1.10 *Dashes.* Dashes are used between parts of certain expressions as ¼–20, or .500–.502. Dashes shall be definite, one letterspace in length, at midheight of the associated letters or digits, and parallel to the direction of the expression.

5.2 *Units of measurement.* Linear dimensions are most commonly expressed in inches, or in feet and inches. The symbol for the inch is ″, and for the foot, ′. Omit inch symbols if all the dimensions on a drawing are in inches, except when a dimension is likely to be misunderstood (see fig. 15). (Thus 1 Valve should be 1″ Valve, 1 Bore should be 1″ Bore, etc.) When used, dimensions in feet and inches should be expressed thus:

8′–7
8′–0¼
8′–0

5.2.1 *Metric system.* When all dimensions on a drawing are in the metric system a general note may be added to avoid using abbreviations throughout the drawing. When both inch and millimeter dimensions are expressed on a drawing the abbreviation MM shall be included with each millimeter dimension.

5.2.2 *Angular dimensions.* Angular dimensions are most commonly expressed in degrees, minutes, and seconds. These are indicated by their symbols; °, degree; ′, minute; ″, second. When de-

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grees are indicated alone, the numerical value may be followed by the symbol °, or by the abbreviation DEG. When minutes are indicated alone, the number of minutes should be preceded by 0° (see fig. 19). When desired, the angle may be given in degrees and decimal parts of a degree, and the tolerance in decimal parts of a degree. Angular dimensions and their tolerances shall contain the same units of measurement. For example, use 10° 30′ 0″ ± 0° 0′ 30″, do not use 10°-30′ ± 30″.

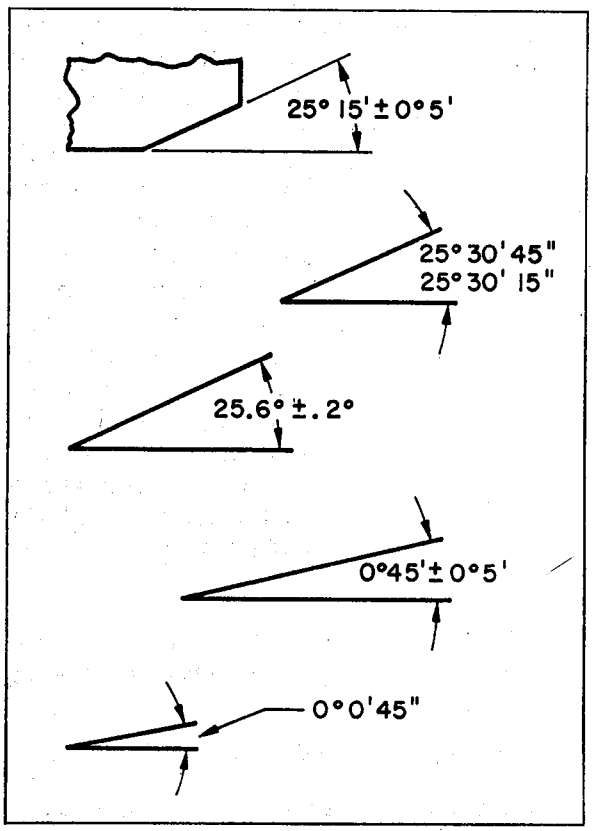


FIGURE 19. *Expressing angular dimensions.*

5.2.3 *Parts of units.* Parts of measurement units may be expressed by decimals or common fractions. The choice depends on the accuracy required, or on industrial practice.

5.2.3.1 *Decimals.* When the accuracy required is closer than ±1/64 inch, decimal dimensioning must be used. For the decimal equivalents of common fractions see table I. All dimensions expressed in terms of decimals shall be carried to the same number of places as the required tolerance. For example, use .050–.005; do not use .05–.005. Use .08–.01; do not use .080–.01. A cipher shall not be used alone before decimal points.

5.2.3.2 *Common fractions.* When common fractions are used on drawings, the following practices must be observed. The fraction bar must not be omitted and should not be slanted except when applied with a typewriting machine which does not have a horizontal fraction bar. Numerals should not touch the fraction bar. When the numerator is above a dimension line and the denominator below, the fraction bar should be indicated by making it heavier than the dimension line, and the dimension line broken to admit the fraction, as in figure 20. Common fractions may be used to indicate nominal sizes of materials or features such as holes, threads, keyways, etc. Example: ¼-20 UNC-2A; $\frac{5}{16}$ DIA; STOCK ⅝ x ⅞.

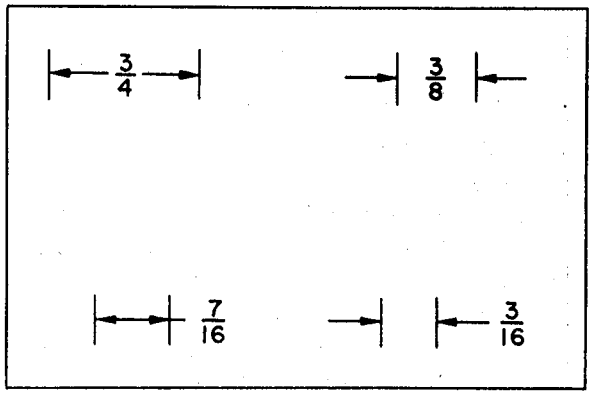


FIGURE 20. *Optional ways to show dimensions in common fractions.*

5.2.3.3 *Decimal dimensioning.* Preferably, decimals should be used for dimensioning except for materials or features covered in 5.2.3.2. When dimensions on drawings of existing parts are converted from fractions to decimals, three- or four-place decimals may be necessary to bring the maximum and minimum limits of the new dimensions into agreement with those of the old dimensions. The decimal system requires the use of the following conventions.

5.2.3.3.1 Two-place decimals are used for dimensions where tolerance limits of ±.01, or increments of .01, can be allowed (see fig. 21a).

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5.2.3.3.2 Decimals to three or more places shall be used for tolerance limits less than ±.010 (see fig. 21b).

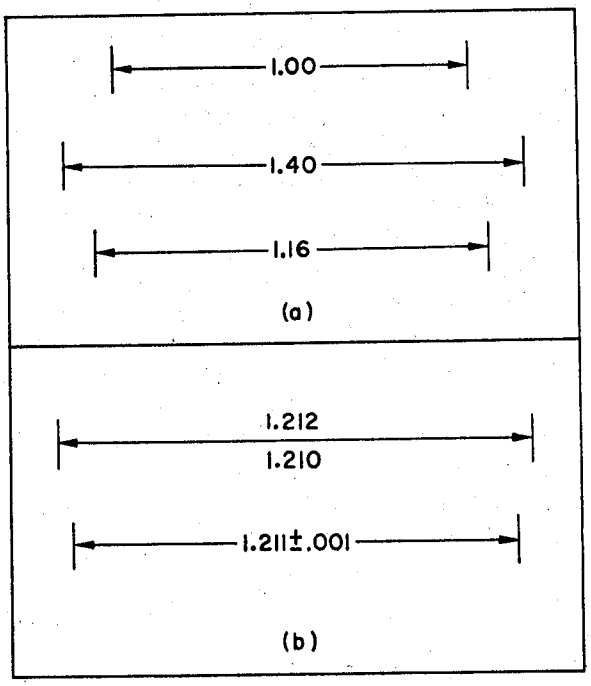


FIGURE 21. *Decimal dimensioning.*

5.2.3.3.3 The second decimal place in a two-place decimal should preferably be an even digit (.02, .04, .06 are preferred to .01, .03, .05, etc.), so that when divided by two, as in obtaining a radius from a diameter, the result will remain a two-place decimal (see fig. 21a). Odd two-place decimals are used where necessary for design reasons; as, to provide clearance, strength, smooth curves, etc. (see fig. 39).

5.2.3.4 Wires, cables, etc., which are manufactured to gage or code number shall be specified by the decimal indicating the diameter or thickness. The gage or code number may be placed in parentheses after the decimal if desired, e.g., .0508 (No. 16, AWG).

6. DIMENSIONS

In general, drawings furnished to the Government should show end product design only.

6.1 *General remarks about dimensioning.* Dimensions must clearly and explicitly indicate the size and position of all features of the part or assembly. In placing dimensions on the drawing, consideration should be given to the following:

(a) *Dimensions for end-product drawings.* On drawings of parts produced for interchangeable assembly, it is usual to give dimensions and notes for only the final forms and sizes, without reference to methods of manufacture. Thus, only the diameter of a hole is given, without indication as to whether it may be drilled, reamed, punched, or made by other operations.

(b) *Dimensions for process drawings.* When one or more preliminary operations are performed on a surface it may be desirable to give the dimension for each preliminary operation, with a proper allowance for stock removed and with suitable tolerances.

(c) In certain cases, it may be desirable to show some end-product requirements on one drawing and the remainder on a drawing of an intermediate step. On the latter, the process information should be distinguished from end-product requirements by notes (or symbols referring to a general note), such as "NON-MANDATORY—DESIGN MANUFACTURER'S PROCESS."

6.1.1 The choice between (a), (b), or (c) will depend on the purpose for which the drawing is intended.

6.1.2 *Effect of surface coatings.*

6.1.2.1 When a fit will be affected by plating, one of the following notes should be placed on the drawing:

(a) Dimensional limits apply before plating.

(b) Dimensional limits apply after plating.

(c) Dimensional limits and surface roughness designations apply before plating.

(d) Dimensional limits and surface roughness designations apply after plating.

(For other coatings, substitute the appropriate expression for "plating.")

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6.1.2.2 When a drawing does not specify whether dimensions apply before or after application of a surface coating, the following interpretation shall be used:

(a) Unless otherwise specified, dimensional limits and surface roughness designations apply after the application of inorganic finishes, such as platings, ceramic coatings, chromates, oxides, etc.

(b) Unless otherwise specified, where organic finishes, such as lacquers, varnishes, enamels, etc., are used, dimensional and surface roughness designations must be met prior to the application of the organic finish.

6.2 *Expressing size and position.*

6.2.1 *Definitions relating to size.* The following terms have been defined to provide a common interpretation in respect to their use in this standard.

6.2.1.1 *Size.* Size is a designation of magnitude. When a value is assigned to a dimension it is referred to hereinafter as the size of that dimension.

Note. It is recognized that the words "dimension" and "size" are both used at times to convey the meaning of magnitude.

6.2.1.2 *Nominal size.* The nominal size is the designation which is used for the purpose of general identification. Example, ½-inch pipe.

6.2.1.3 *Basic size.* The basic size is that size from which the limits of size are derived by the application of allowances and tolerances.

6.2.1.4 *Design size.* The design size is that size from which the limits of size are derived by the application of tolerances. When there is no allowance the design size is the same as the basic size.

6.2.1.5 *Actual size.* An actual size is a measured size.

6.2.1.6 *Limits of size.* The limits of size (commonly referred to simply as "limits") are the applicable maximum and minimum sizes.

6.2.2 *Position.* Dimensions that establish position generally require more analysis than dimensions that only state sizes. Either linear or angular expressions may locate features with respect to one another (point-to-point), or from a datum. Point-to-point distances may be adequate for describing simple parts. Dimensions from a datum may be necessary if a part with more than one critical dimension must mate with another part.

6.2.2.1 *Locating round holes.* Figures 22 through 27 illustrate the positioning of round holes by giving distances, or distances and directions to the hole centers. These methods can also be used to locate round pins and other features of symmetrical contour. Allowable variations for any of the positioning dimensions illustrated may be specified by giving a tolerance with each distance or angle by stating limits of dimensions or angles, or by the true position expressions of 7.5.1 and 7.5.5.

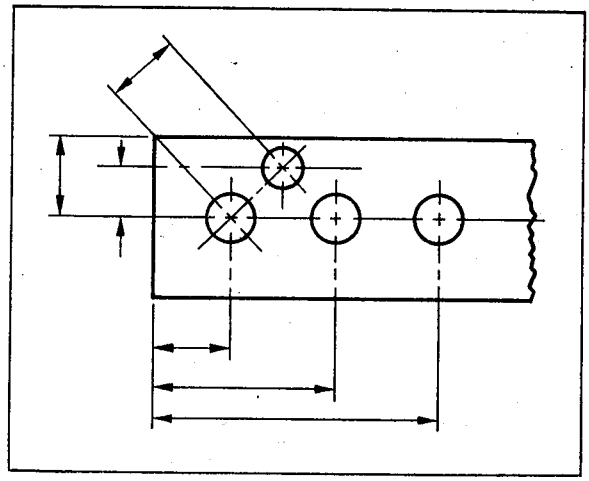


FIGURE 22. *Locating round holes by linear distances.*

6.3 *Dimensions from datum.* Datums are points, lines, planes, cylinders, etc., which are assumed to be exact for purposes of computation or reference, and from which the location of features of a part may be established. Where positions are specified by dimensions from a datum, different features of the part are located with respect to this datum, and not with respect to one another (see fig. 27). For a primary datum surface which is nominally flat, the datum shall be assumed to be a plane established by the con-

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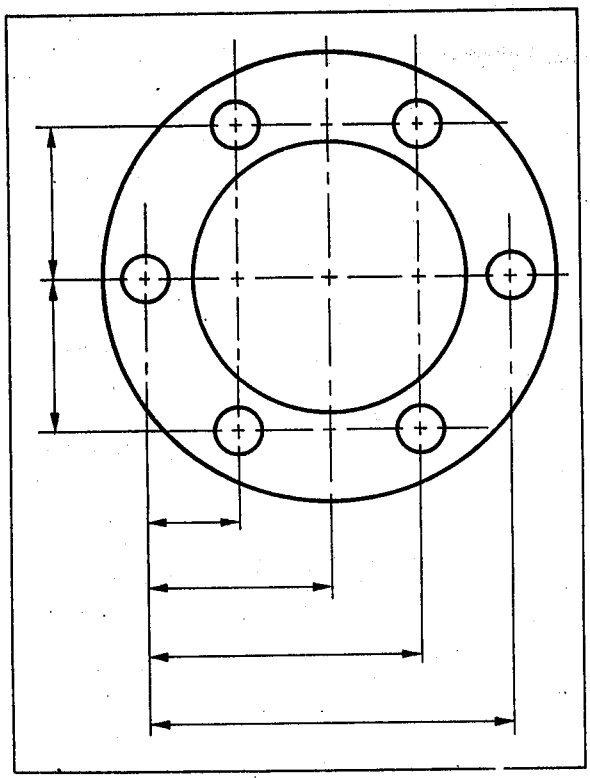


FIGURE 23. *Locating round holes by rectangular coordinates.*

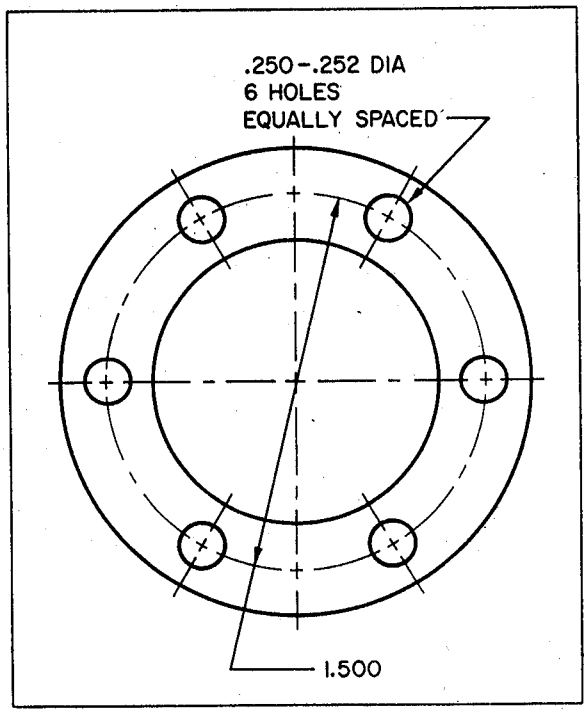


FIGURE 25. *Locating holes on circles by radius or diameter and "EQUALLY SPACED".*

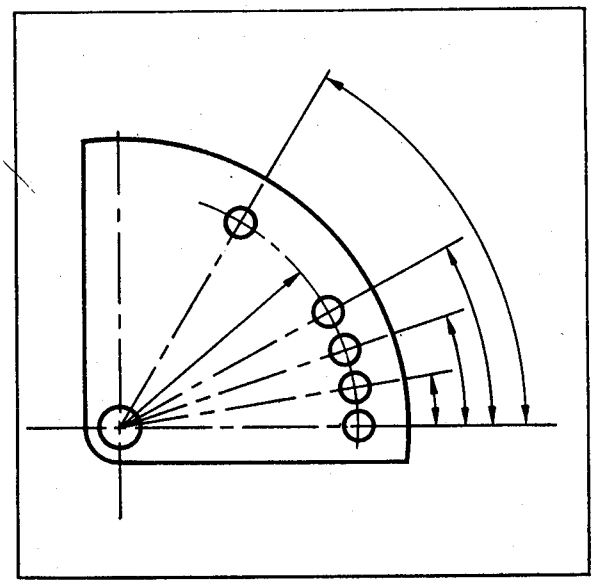


FIGURE 24. *Locating holes on circle by angular dimensioning.*

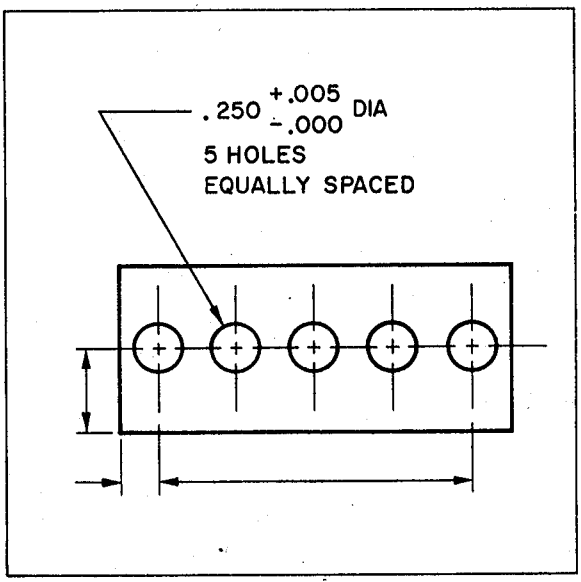


FIGURE 26. *"EQUALLY SPACED" holes in a line.*

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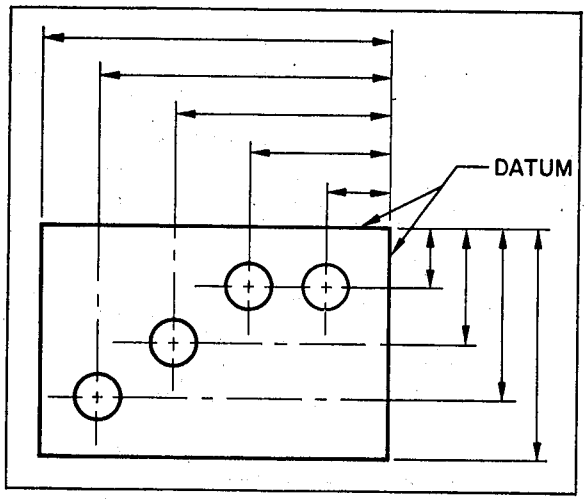


FIGURE 27. *Dimensions from datums.*

tacting points of the actual datum surface with a reference plane (surface plate or other checking device). In establishing datums in relation to defining or measuring a part, at least two and often three datums must be considered in locating a feature.

6.3.1 *Selection of a datum.* Features which are selected to serve as datums must be clearly identified or easily recognizable. To be useful for measuring, a datum on an actual piece must be accessible during manufacture, so that measurements can be referred to it readily. Also, corresponding features on mating parts should be used as datums to insure assembly and facilitate tool and fixture design.

6.3.2 *Accuracy of datum surfaces on actual pieces.* It may be necessary to specify the accuracy of a datum surface by stating form tolerances to assure that locations can be established with the specified accuracy. Recommended expressions for specifying such tolerances are given in 7.6.

6.3.3 *Ordinate dimensioning.* Ordinate dimensioning is a type of rectangular datum dimensioning in which all dimensions are measured from two or three mutually perpendicular datum planes. The datum planes are indicated as zero coordinates and dimensions from them are shown on extension lines without the use of dimension lines or arrowheads (see fig. 28).

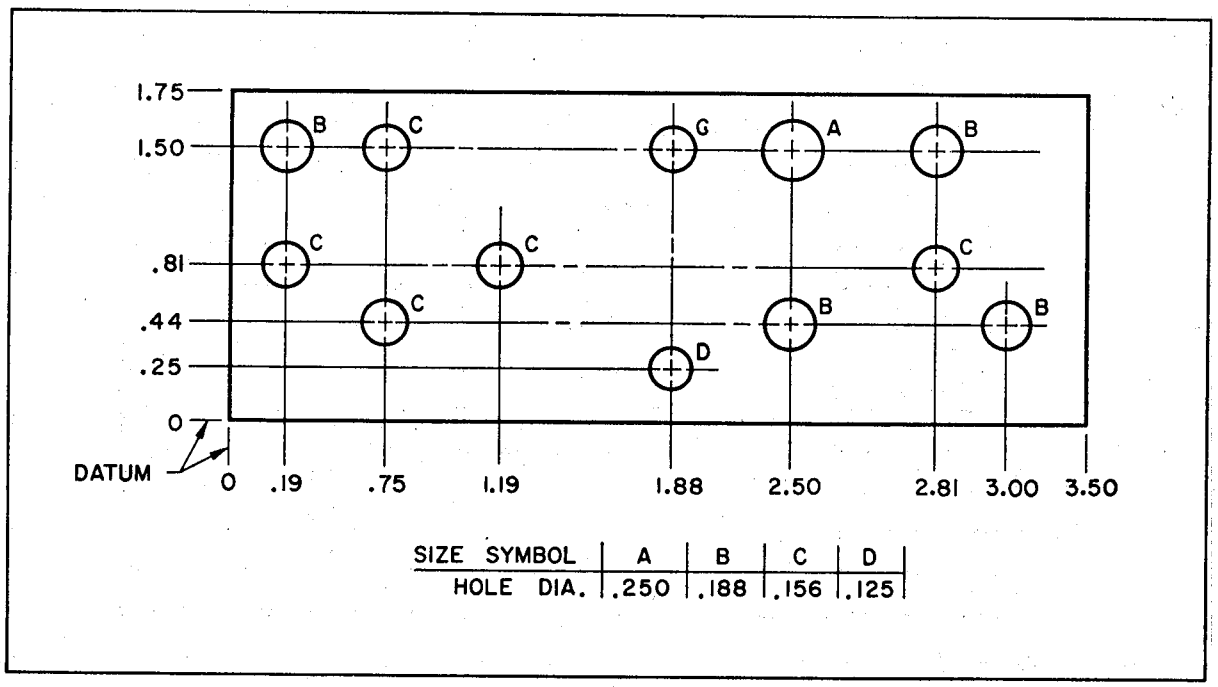
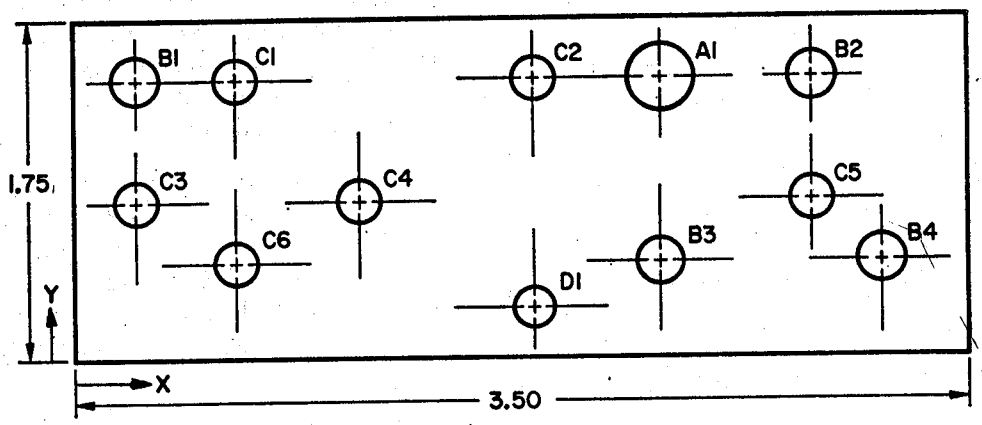


FIGURE 28. *Ordinate dimensioning.*

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	REQD	1	4	6	1
	HOLE DIA	.250	.188	.156	.125
POSITION		HOLE SYMBOL			
X →	Y ↑	A	B	C	D
2.50	1.50	A1			
.19	1.50		B1		
2.81	1.50		B2		
2.50	.44		B3		
3.00	.44		B4		
.75	1.50			C1	
1.88	1.50			C2	
.19	.81			C3	
1.19	.81			C4	
2.81	.81			C5	
.75	.44			C6	
1.88	.25				D1

FIGURE 29. *Tabular dimensioning.*

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6.3.4 *Tabular dimensioning.* Tabular dimensioning is a type of datum dimensioning in which dimensions from mutually perpendicular datum planes are listed in a table on the drawing instead of on the pictorial portion (see fig. 29). This method may be used on drawings which require the location of a large number of similarly shaped features.

6.4 *Reference dimensions.* A reference dimension is a dimension without tolerance used for informational purposes only, and does not govern machining or inspecting operations. A reference dimension shall be indicated on drawings by the abbreviation REF directly following or under the dimension (see fig. 30).

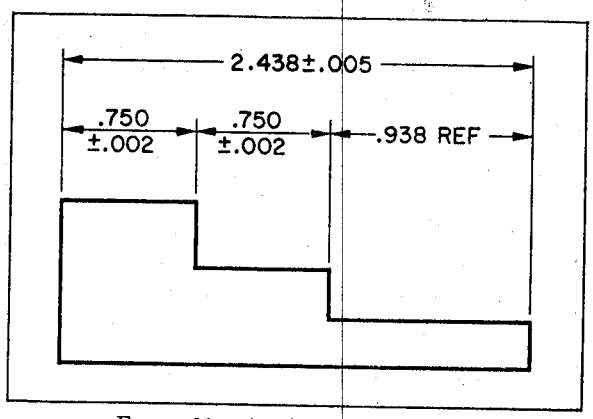


FIGURE 30. *A reference dimension.*

6.5 *Basic dimension.* A dimension specified on a drawing as BASIC is a theoretically exact dimension without tolerance which:

(a) Describes the exact size, shape, or location of a feature from which permissible variations are established by tolerances expressed on other dimensions, in notes or in symbols;

(b) Establishes the exact size of a feature, the location of which may vary within specified limits; or establishes the exact location of a feature, the size of which may vary within specified limits.

6.6 *Rules for dimensioning special features.*

6.6.1 *Cylindrical or spherical surfaces.*

6.6.1.1 *Diameters of concentric elements.* Where the diameters of a number of concentric elements are to be given, it is usually more convenient to show them on a longitudinal view (see figs. 31a and 31b). Where required, a cylindrical hole may be dimensioned in an end view (see fig. 32). Where space is limited, a leader with a note may be used. The numerals shall be followed by the abbreviation DIA to clarify the meaning when the end view is omitted (see fig. 31b).

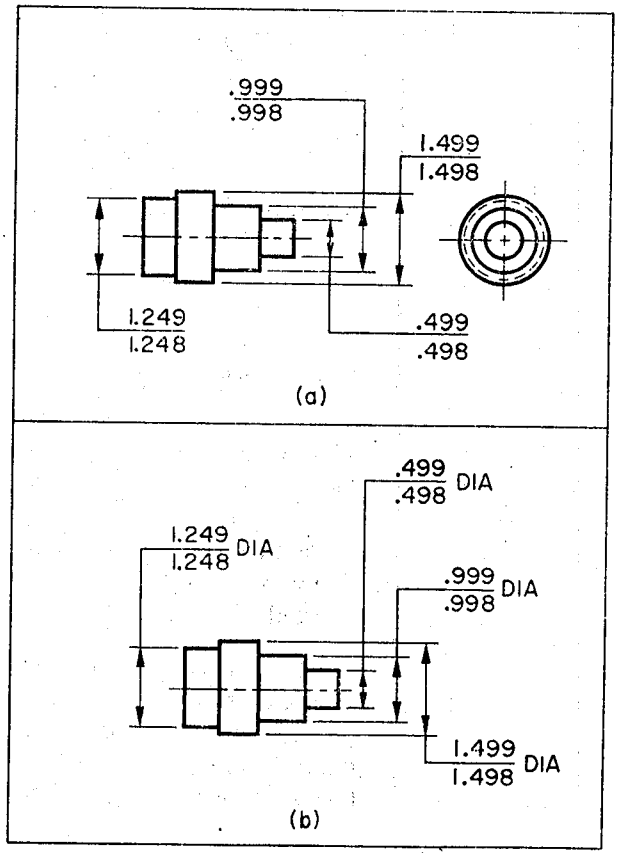


FIGURE 31. *Dimensioning diameters of concentric elements.*

6.6.1.2 *Radii of arcs.* A circular arc is dimensioned by giving its radius. Where space permits, a radius dimension line is drawn from the radius center, with the arrowhead ending at the arc, and with the dimension between the arrowhead and the center. The numerical portion of the dimension shall be followed by the abbreviation "R." Where space is limited, as for a small radius, a leader may be used. Where it is inconvenient to place the arrowhead between the center of the radius and the arc, it may be placed outside the arc. The center of the radius may be indicated

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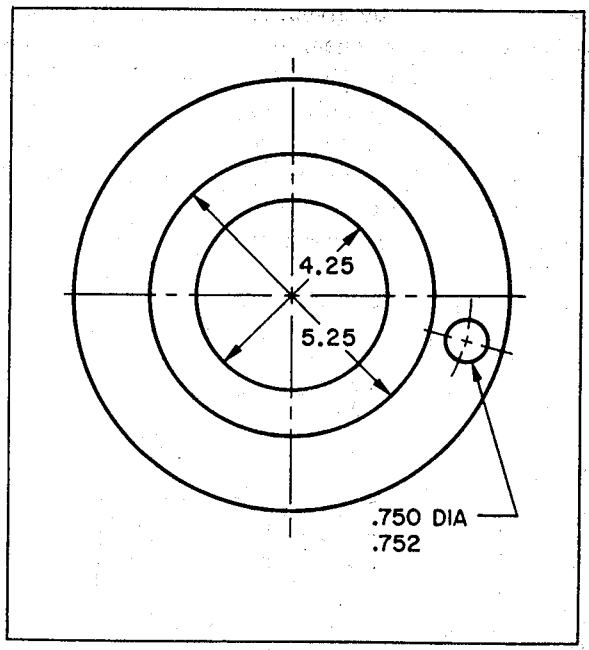


FIGURE 32. *An end view of cylindrical surfaces.*

by a small cross to clarify the drawing (see fig. 33). Where space does not permit showing the complete radius dimension line to scale, the line may be foreshortened, as in figure 33. The portion of the dimension line ending in the arrowhead shall be in the radial direction.

6.6.1.2.1 *True radius indication.* Where a radius is dimensioned in a plane not perpendicular to the plane of projection (see fig. 34), the dimensions may originate from an arbitrary center with the term "TRUE R" specified after the numeral.

6.6.1.3 *Spherical surfaces.* Spherical surfaces may be dimensioned as in figure 35.

6.6.1.4 *Distances along curved surfaces.* Linear distances along curved surfaces may be dimensioned as a chordal distance or as an arc, and should be so indicated on a drawing (see figure 36).

6.6.1.5 *Parts with rounded ends.* A method of dimensioning a part with rounded ends is shown in figure 37a. The radius is indicated, but not dimensioned; it varies with the actual width when specified in this way. Where a hole location is more critical than the location of a radius from the same center, the method of figure 37b is recommended.

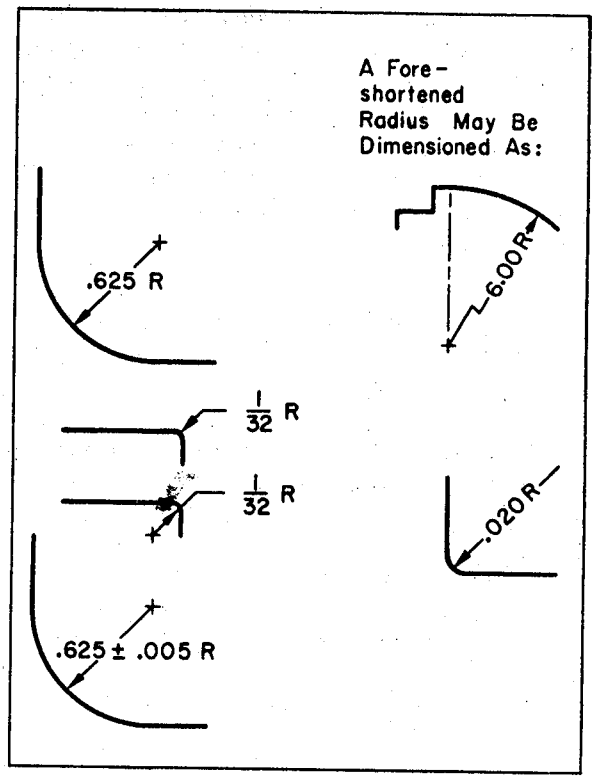


FIGURE 33. *How to dimension a circular arc.*

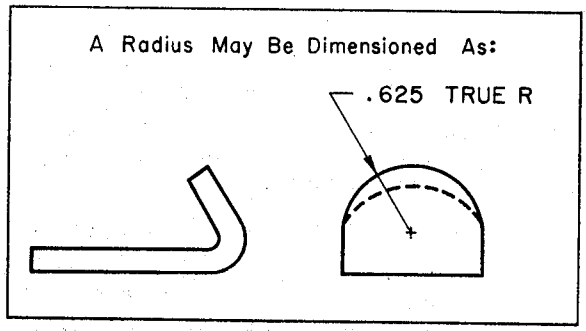


FIGURE 34. *True radius dimensioning.*

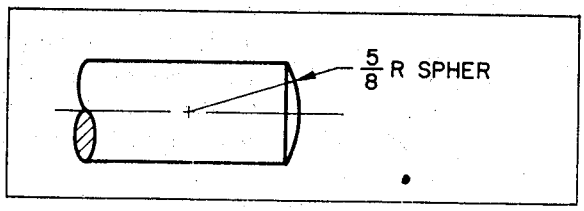


FIGURE 35. *Indicating a spherical surface.*

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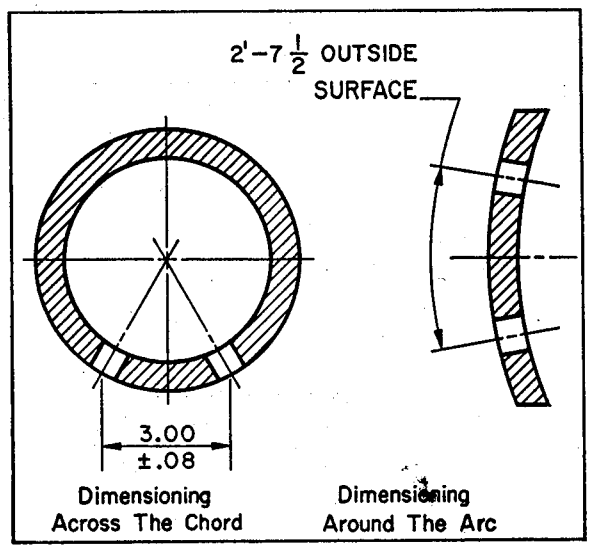


FIGURE 36. *Dimensioning along a curved surface.*

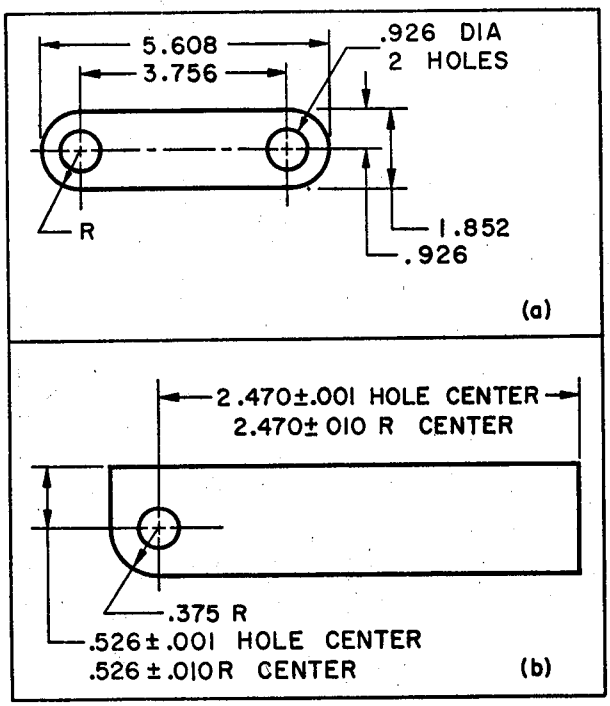


FIGURE 37. *Dimensioning parts with rounded ends.*

6.6.1.6 *Curves consisting of circular arcs; noncircular curves.* A curved line composed of two or more circular arcs should be dimensioned by giving the radii and locating their centers or the points of tangency (see fig. 38). Either a circular or a noncircular curve may be dimensioned by the coordinate (offset) method of figure 39.

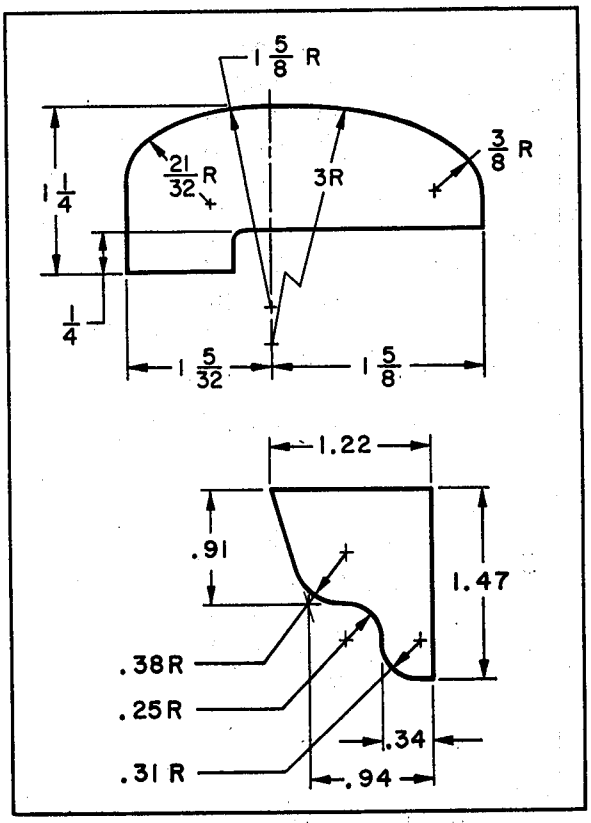


FIGURE 38. *Dimensioning curves made up of circular arcs.*

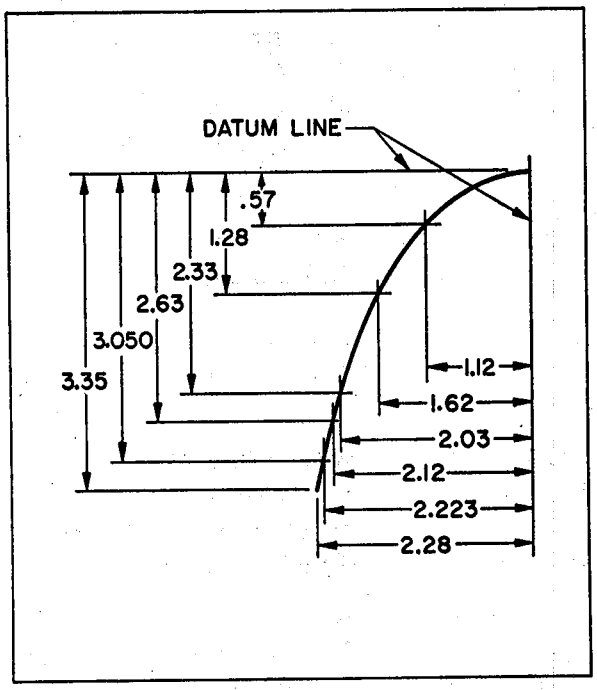


FIGURE 39. *Dimensioning a noncircular curve.*

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6.6.2 *Round holes.*

6.6.2.1 *Plain round holes.* Round holes are dimensioned in various ways, depending on design requirements and manufacturing methods. The expressions of figure 40 are universally understood.

6.6.2.2 *Counterbored holes.* The diameter and depth of the counterbore should be indicated (see fig. 41). In some cases the thickness of the remaining stock may be dimensioned rather than the depth of the counterbore.

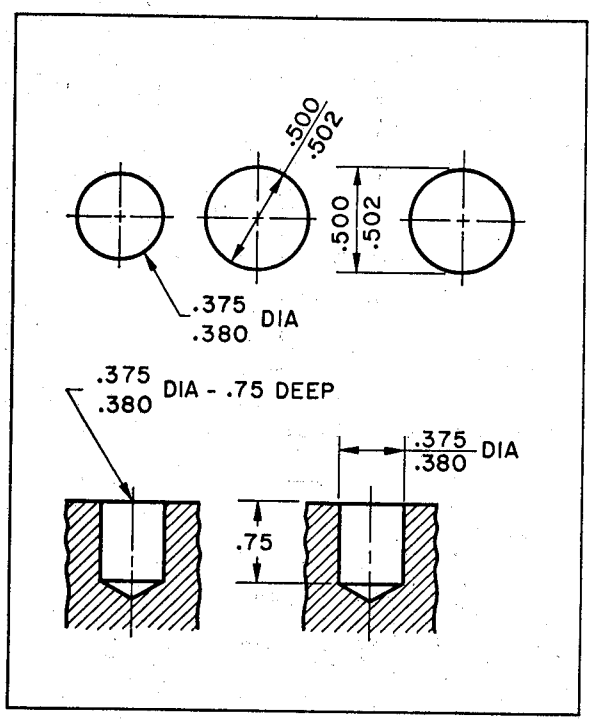


FIGURE 40. *Dimensioning round holes.*

6.6.2.3 *Countersunk holes.* The diameter and the angle of the countersink should be indicated (see fig. 42).

6.6.2.4 *Spot-faced holes.* The diameter of the faced area should be indicated (see fig. 43). Spot face may be specified by a note only, and not delineated on the drawing.

6.6.3 *Shaft centers: Center drills.* Shaft centers, or countersunk center holes, may be required in shafts, spindles, and other cylindrical

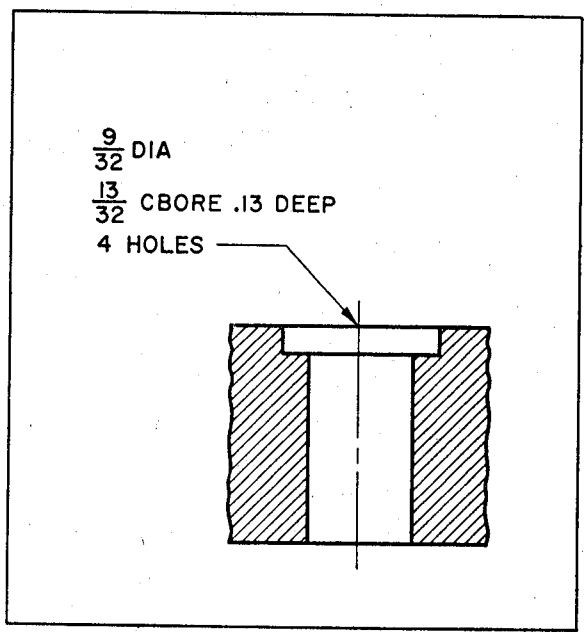


FIGURE 41. *A counterbored hole.*

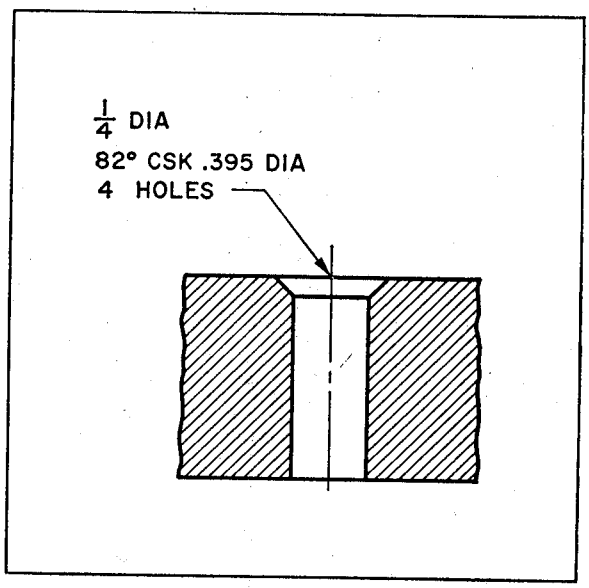


FIGURE 42. *A countersunk hole.*

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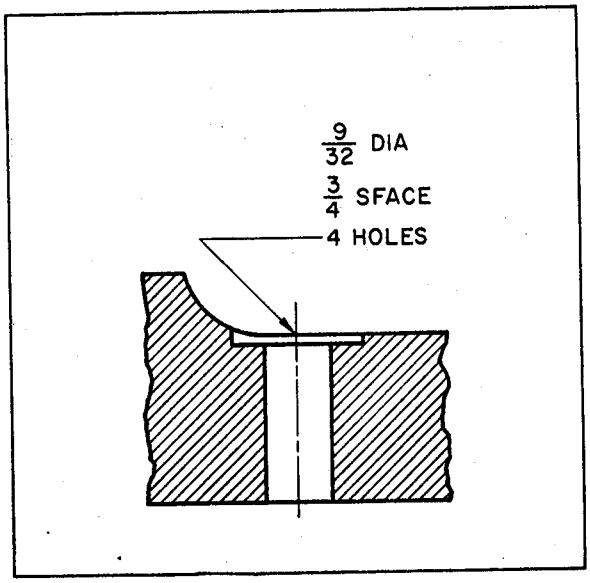


FIGURE 43. *A spot-faced hole.*

or conical parts during manufacture and inspection. The dimensioning of figure 44 specifies what is needed; the work may be done by using a standard combined drill and countersink, or by drilling a straight hole and then countersinking the hole as a separate operation. In figure 44, if the center drill is necessary, omit the word OPTIONAL. A center drill may be specified by a note and not delineated on the drawing.

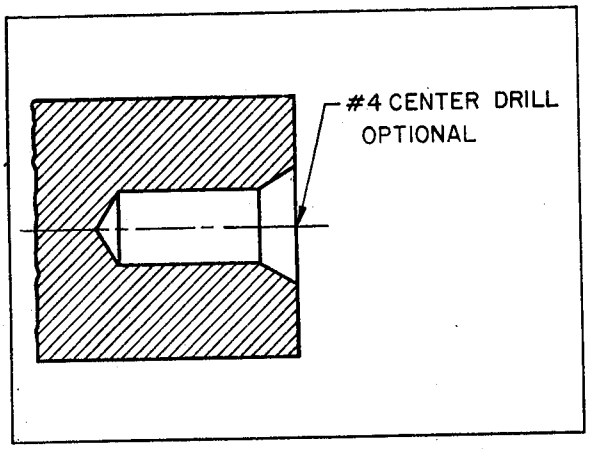


FIGURE 44. *A shaft center.*

6.6.4 *Chamfers.* The recommended method for dimensioning a chamfer is to give an angle and a length, as in figure 45a. For angles of 45°, the method of figure 45b is often used. This

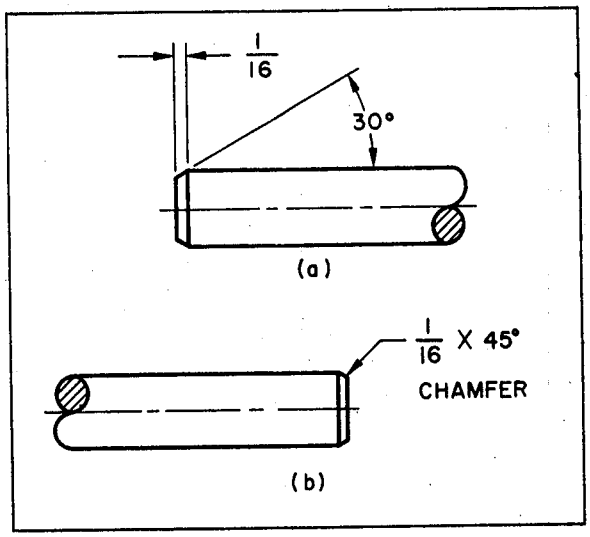


FIGURE 45. *Dimensioning chamfers.*

method is permissible only with 45° angles because the size dimension may apply to either the longitudinal or the radial dimension. Chamfers are never measured along the angle. In the note form of figure 45b, the word "chamfer" may be omitted.

6.6.5 *Keys and keyways.* Figures 46 and 47 illustrate satisfactory methods of expressing dimensions for keyways.

6.6.6 *Knurls.* Close tolerances are not necessary in connection with knurls that provide a rough surface for gripping, or that are used as decoration. For these purposes it is common to specify only the pitch of the knurl, the type of knurling, and the axial length of the knurled area, as in figure 48. To specify knurling for a press fit between two parts, the diameter before knurling should be given, with a tolerance, and the minimum diameter after knurling should be included in the note that specifies the circular or diametrical pitch (CP or DP, respectively) and type of knurl, as in figure 49.

6.6.7 *Conical tapers.* There are two approved methods of dimensioning and tolerancing tapers: Form Dimensioning and Basic Dimensioning. The selection of the method for a particular application shall be determined by the functional characteristics of the part or the manufacturing process.

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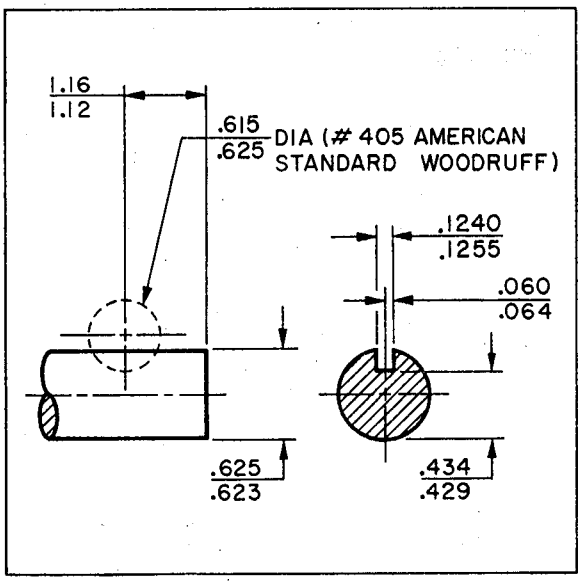


FIGURE 46. *Method of dimensioning Woodruff keyslots.*

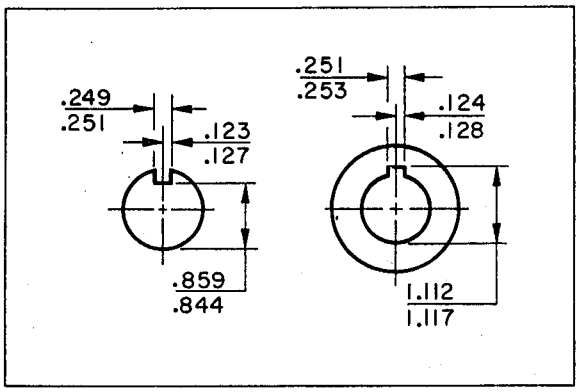


FIGURE 47. *Keyways for stock keys.*

6.6.7.1 *Form dimensioning of tapers.* The four fundamental dimensions which control the form of a conical tapered section are the:

(a) Diameter at the large end;
(b) Diameter at the small end;
(c) Length of the axis of the taper;
(d) Amount of taper for entire length or per unit length, or the included angle.

Three of these four dimensions shall be specifically toleranced. The fourth may be given as a reference dimension (see 6.4). The three dimensions selected for tolerancing shall be those most directly related to the essential functional characteristics of the part. The reference dimension shall have the least influence on function of the part.

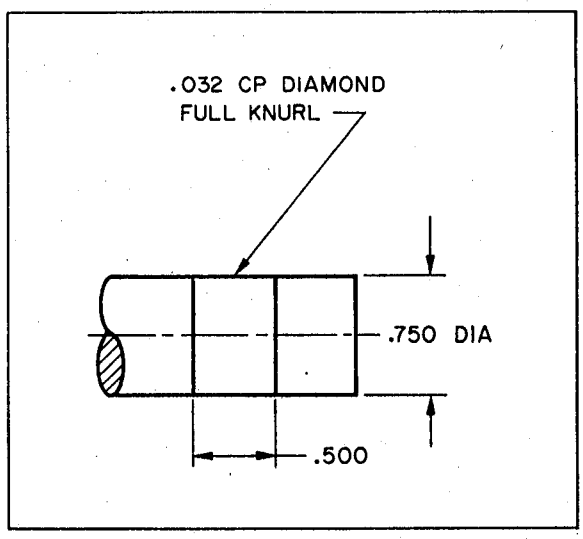


FIGURE 48. *A knurl for decoration or gripping.*

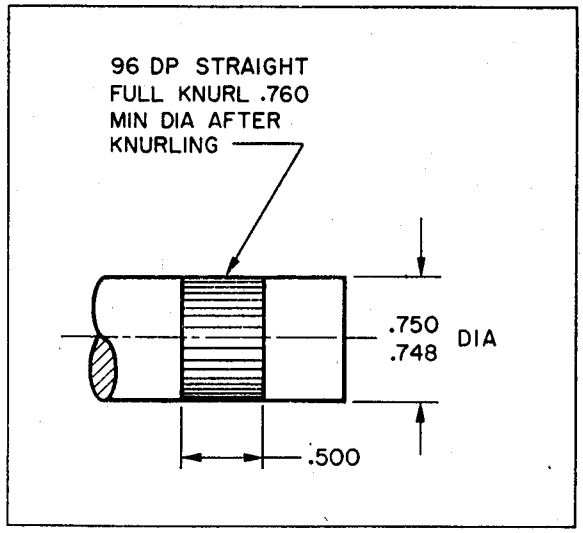


FIGURE 49. *A knurl for a press-fit.*

6.6.7.1.1 Figure 50 is an example of the proper method of dimensioning and tolerancing a tapered section where the most important requirement is the accuracy of the taper. The note on the drawing should always specify the taper on diameter per length or unit length to avoid possible misinterpretation of the taper as being based on the radius.

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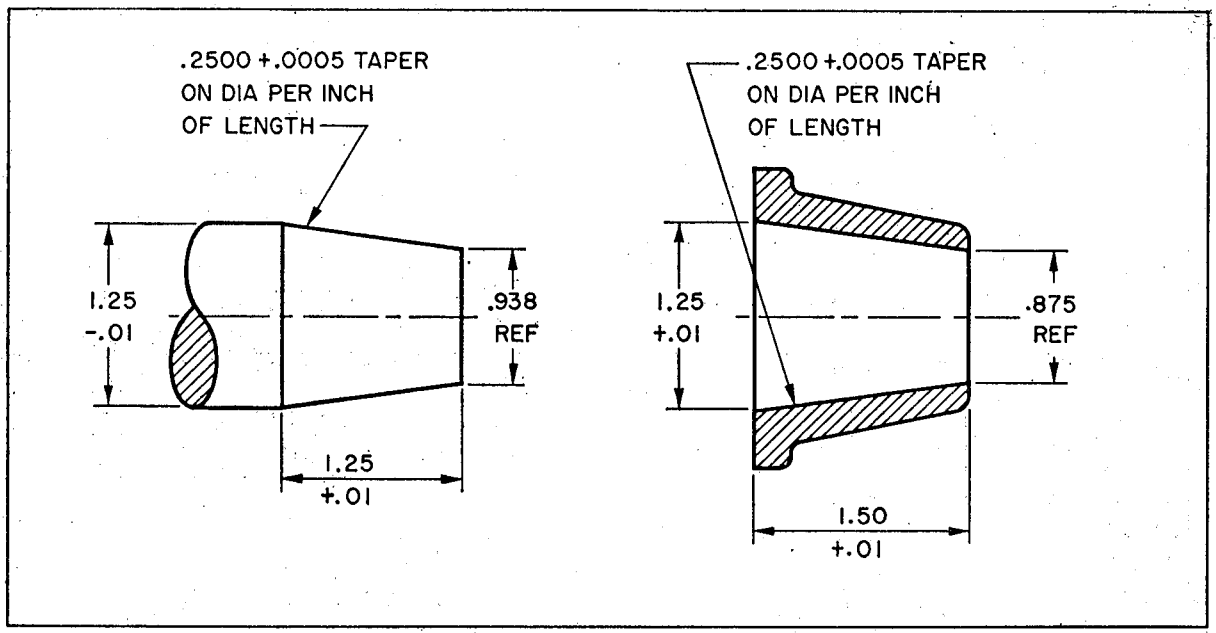


FIGURE 50. *Dimensioning taper, taper per inch method.*

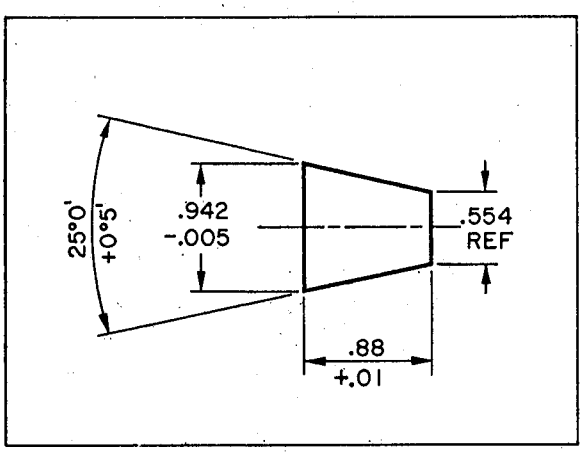


FIGURE 51. *Dimensioning taper, angular method.*

6.6.7.1.2 Figure 51 illustrates a method of dimensioning the same type of tapered section by use of an angular tolerance instead of a tolerance on the taper. This method, though permissible, is not commonly used.

6.6.7.1.3 Figure 53 illustrates the method of dimensioning and tolerancing conical tapers suitable for use when both the diameter and the taper are important, but the length of the piece is not. The word BASIC after the taper means that the taper on the part may vary from the basic value, but any variation must fall within the tolerance zone established by the maximum and minimum limits of the end diameter.

6.6.7.2 *BASIC dimensioning of tapers.*

6.6.7.2.1 *BASIC diameter.* A diameter on a conical taper labeled BASIC is an exact dimension which must be located within specified limits. The BASIC diameter method of dimensioning tapers, illustrated in figure 52, controls the size of the tapered section, as well as its axial position in relation to some other surface. As is shown in the interpretation of figure 52, the tolerance on the location of the BASIC diameter not only controls the axial position of the tapered section, but also sets up a tolerance zone within which the form of the taper must fall. The word BASIC after the taper means that the taper may vary from the value given, but must fall within the zone created by the locational tolerances. The right-hand illustration in the "interpretation" portion of figure 52, a magnified view of the tolerance zone, shows the maximum and minimum permissible variations in taper per inch from the stated BASIC value.

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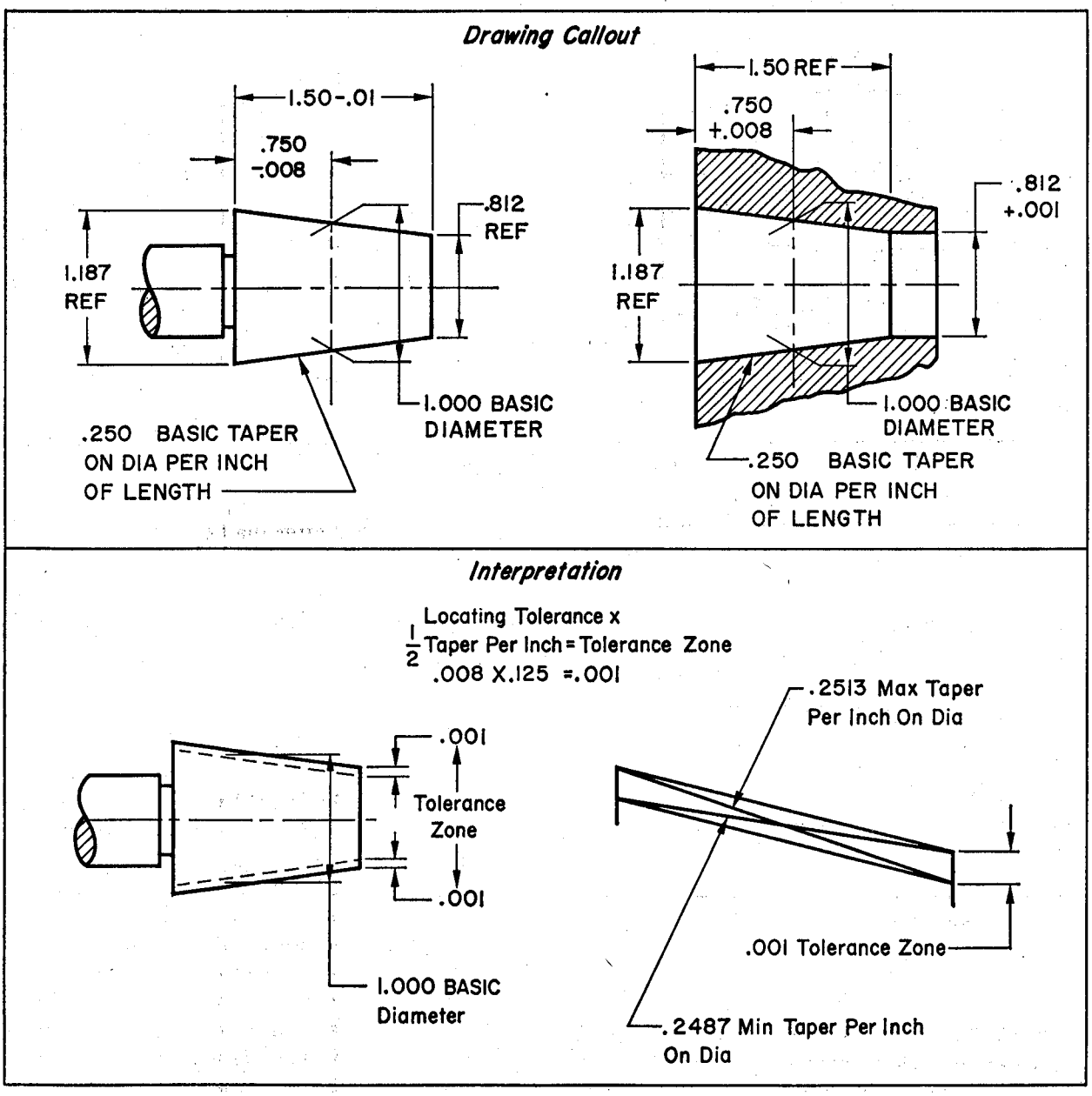


FIGURE 52. *Dimensioning taper, BASIC diameter method.*

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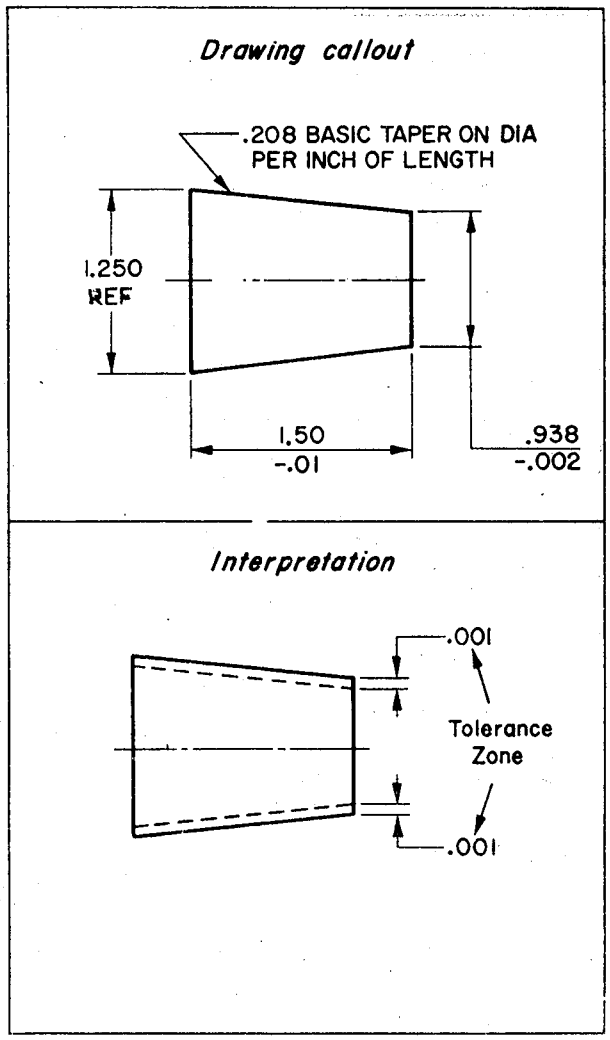


FIGURE 53. *Dimensioning taper where diameter and taper are important.*

6.6.7.2.2 *Basic length.* If the taper on a part is extremely slight, it is sometimes more convenient to label the length as BASIC and apply the tolerance directly to the diameter. In this way the BASIC length controls the diameter as well as the position of the tapered section. This method is illustrated in figure 54. The size of the tolerance zone is directly derived by using one-half the tolerance on the diameter. The lower illustration in the "interpretation" portion of figure 54, a magnified view of the tolerance zone, shows maximum and minimum permissible variations in taper per inch from the stated basic value.

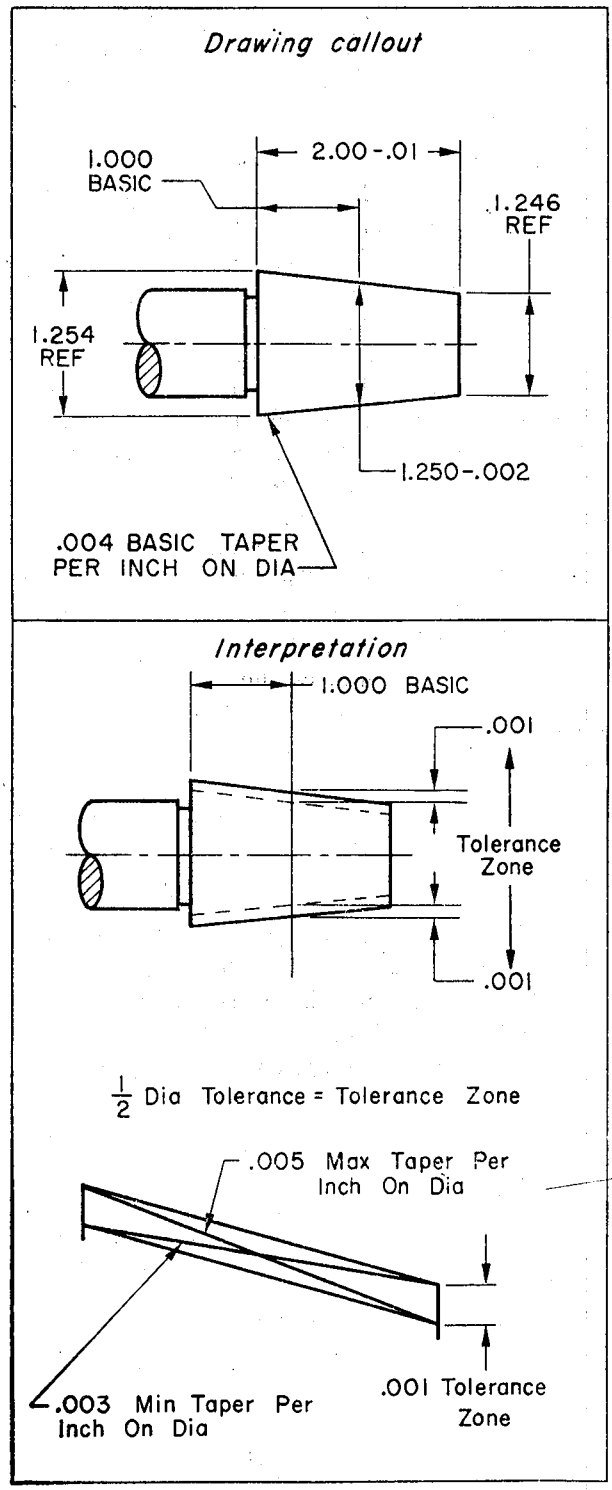


FIGURE 54. *Dimensioning taper, BASIC length method.*

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6.6.8 *Flat tapers.* The methods recommended for dimensioning conical tapers can be adapted to tapered flat pieces. Figure 55 gives an example of the adaptation of one of the methods to dimensioning flat tapers.

6.6.9 *Positioning angular surfaces.* An angular surface may be located by a combination of linear dimensions and an angle, or by linear dimensions alone. Each arrangement of dimensions and tolerances has the effect of specifying a particular tolerance "zone" within which the surface must lie. The shape and extent of the "zone" thus specified depends on the dimensioning method chosen, and on the way tolerances are arranged around the locating dimensions. If an angular surface is located by a combination of linear and polar coordinates, as in figure 56a, the surface must lie between planes represented by the tolerance zone shown in figure 56b. The tolerance "zone" will be wider as the distance from the apex of the angle increases. In order to avoid the accumulation of tolerances, the basic angle tolerancing method illustrated in figure 57a can be used. No specific tolerance is placed on the angle which is specified as "BASIC." This indicates that the actual variation permitted to the angle is defined by the tolerances on the two linear dimensions. These define a tolerance zone within parallel boundaries as shown in figure 57b, and the accuracy of the angles must be such that no part of the resulting surface exceeds these boundaries. If the angular tolerance is expressed by a Tolerance of Form, the angle can be given without loss of precision. (See fig. 95 for an example of a Tolerance of Form on angularity.)

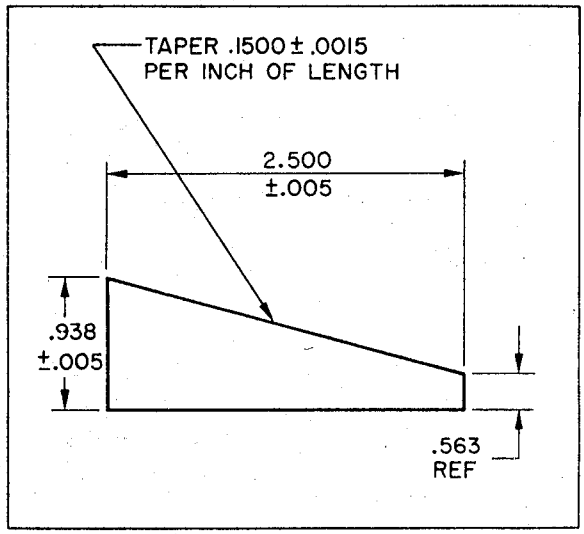


FIGURE 55. *Taper on a flat piece.*

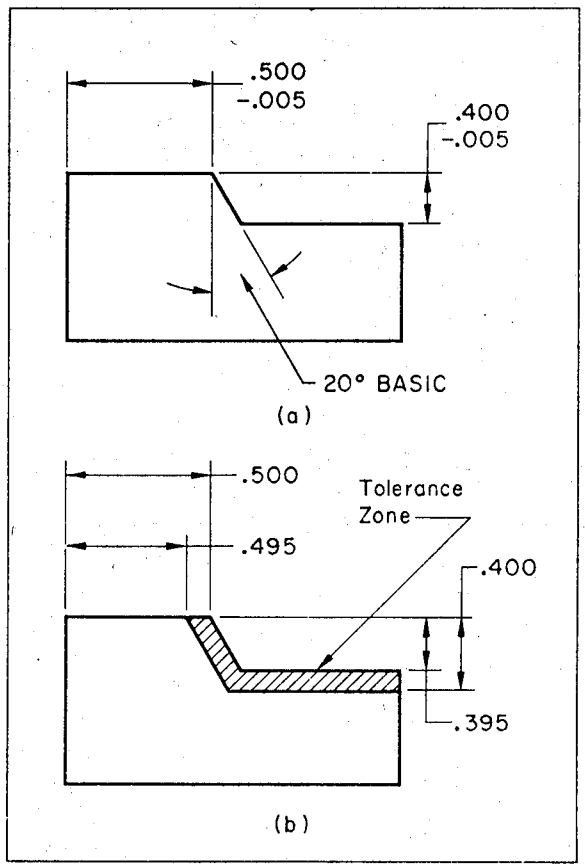


FIGURE 56. *Cumulative angular tolerancing.*

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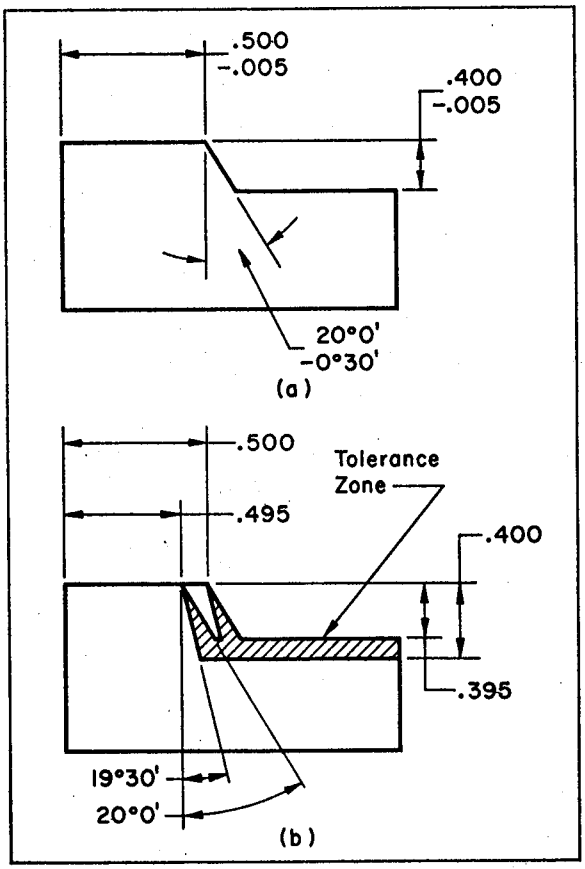


FIGURE 57. *Basic angle tolerancing.*

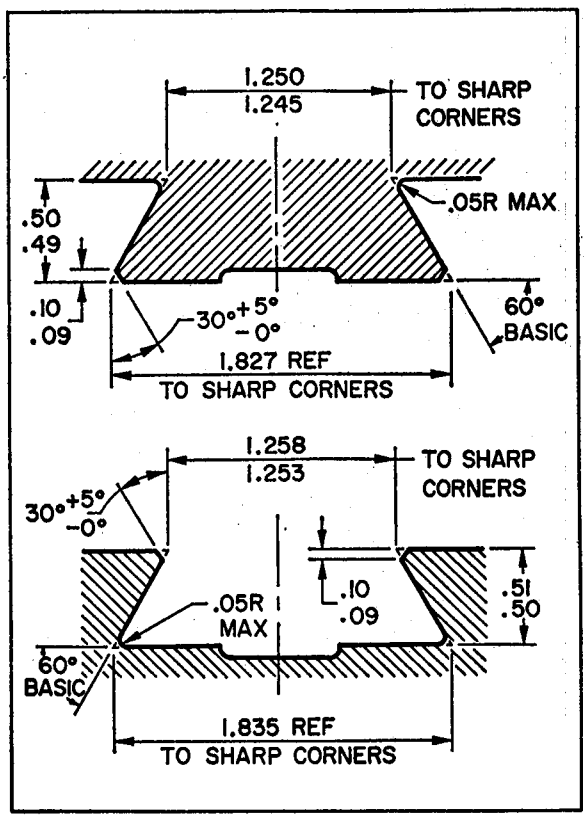


FIGURE 58. *Dovetails: Basic angle tolerancing to control width at contacting surface.*

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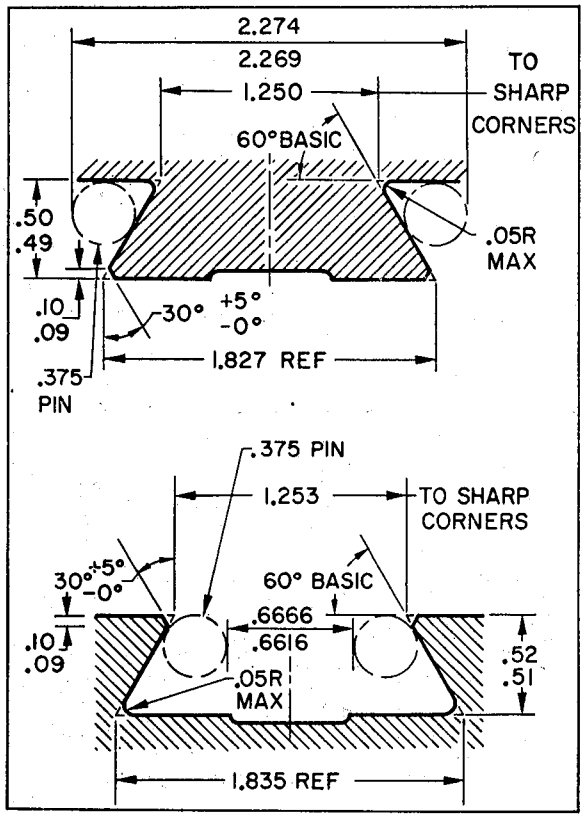


FIGURE 59. *Dovetails: Basic angle tolerancing to show tolerance of pin measurements.*

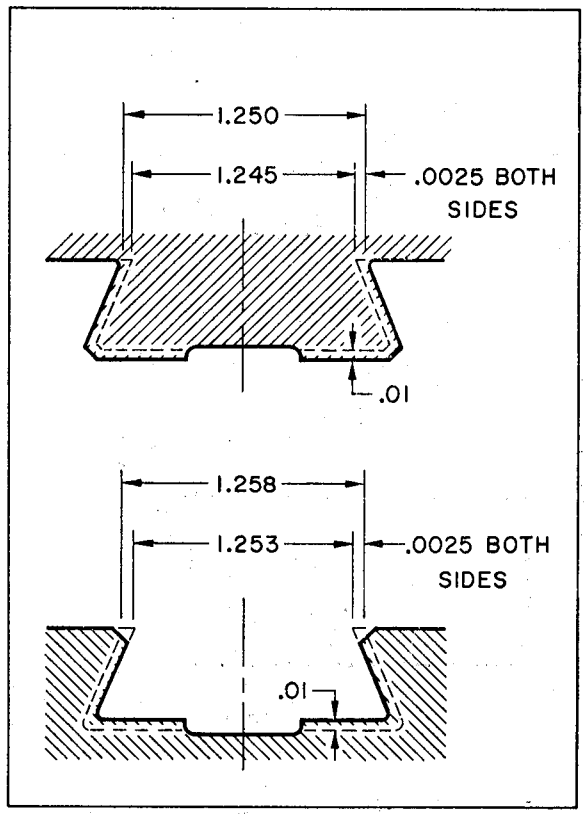


FIGURE 60. *Dovetails: Basic angle tolerancing.*

6.6.9.1 *Dovetails.* Dovetails may be dimensioned by the basic angle tolerancing system, as shown in figures 58 and 59. In both figures, the tolerances are so arranged that the outer face of the internal part will be the contact surface. The tolerancing method used in figure 58 controls the width at the contacting surface. Figure 60 indicates the tolerance zone for each part as defined by the limiting dimensions in figures 58 and 59.

6.6.10 *Dimensioning of sheet metal parts.* Bent or formed sheet metal parts may be dimensioned to the intersections of outside surfaces (see fig. 61) unless other surfaces are critical to provide proper clearance, fit, or functioning of the parts.

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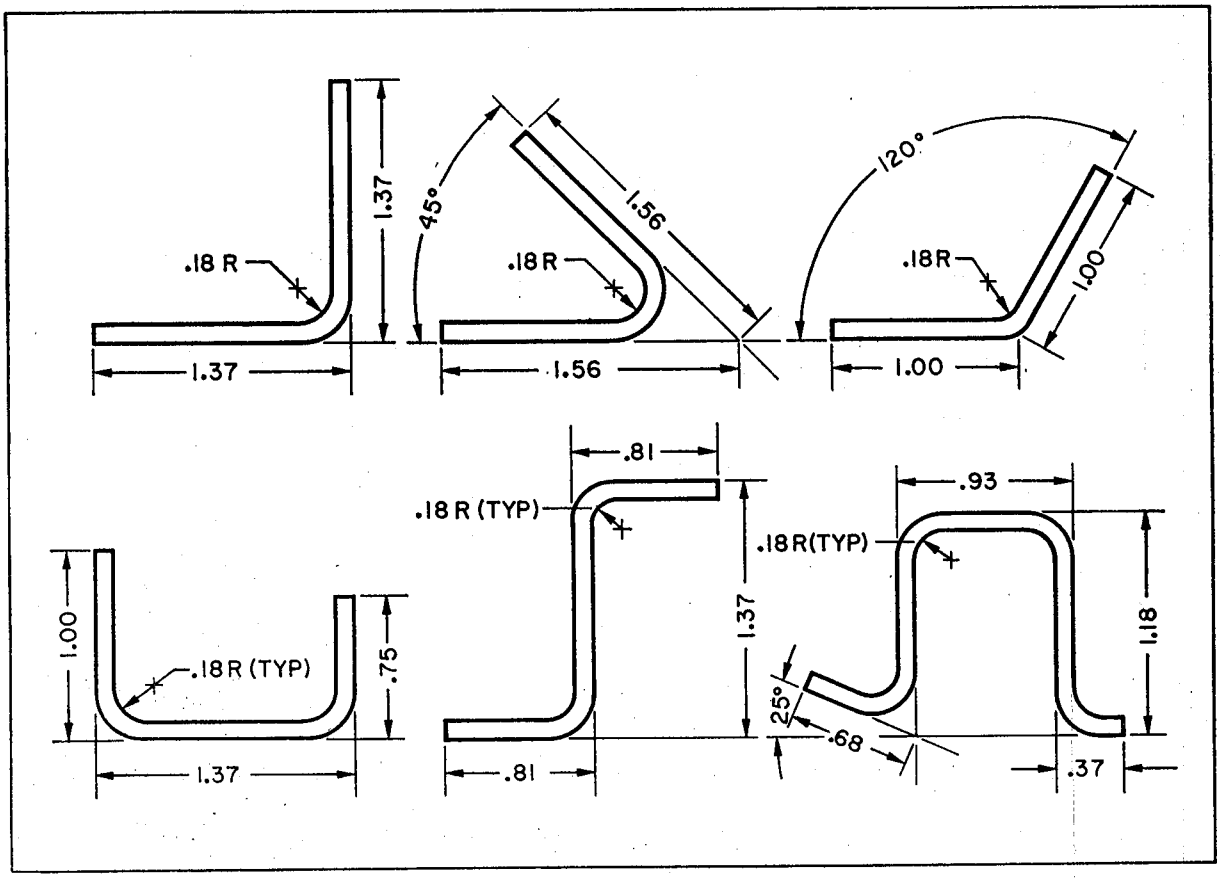


FIGURE 61. *Dimensioning bent or formed sheet metal parts.*

6.6.10.1 *Intersections.* Intersections used for dimensioning may be accented by prominent dots to eliminate all doubt as to the intended points of intersection (see fig. 62).

6.6.10.2 *Stiffening rib sections.* Acceptable methods of dimensioning stiffening rib sections are shown in figure 63.

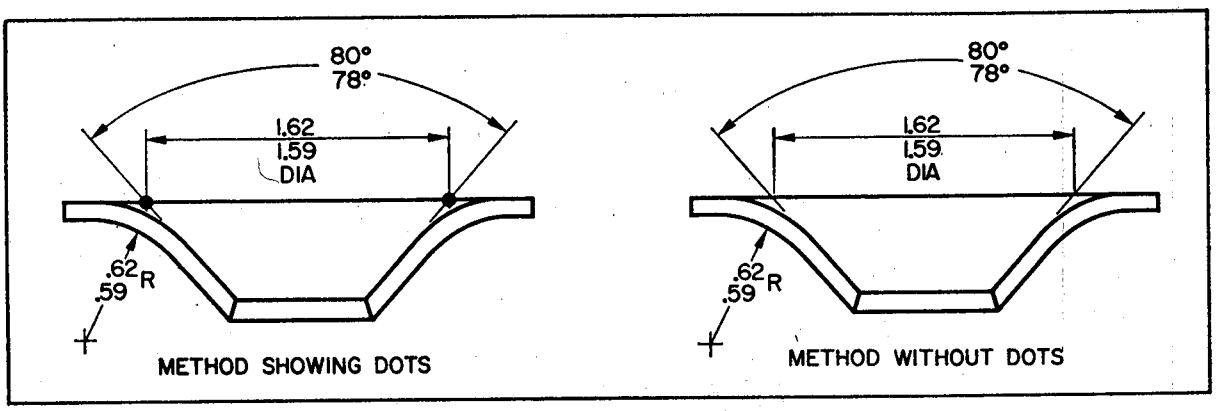


FIGURE 62. *Indicating dimensioning intersections.*

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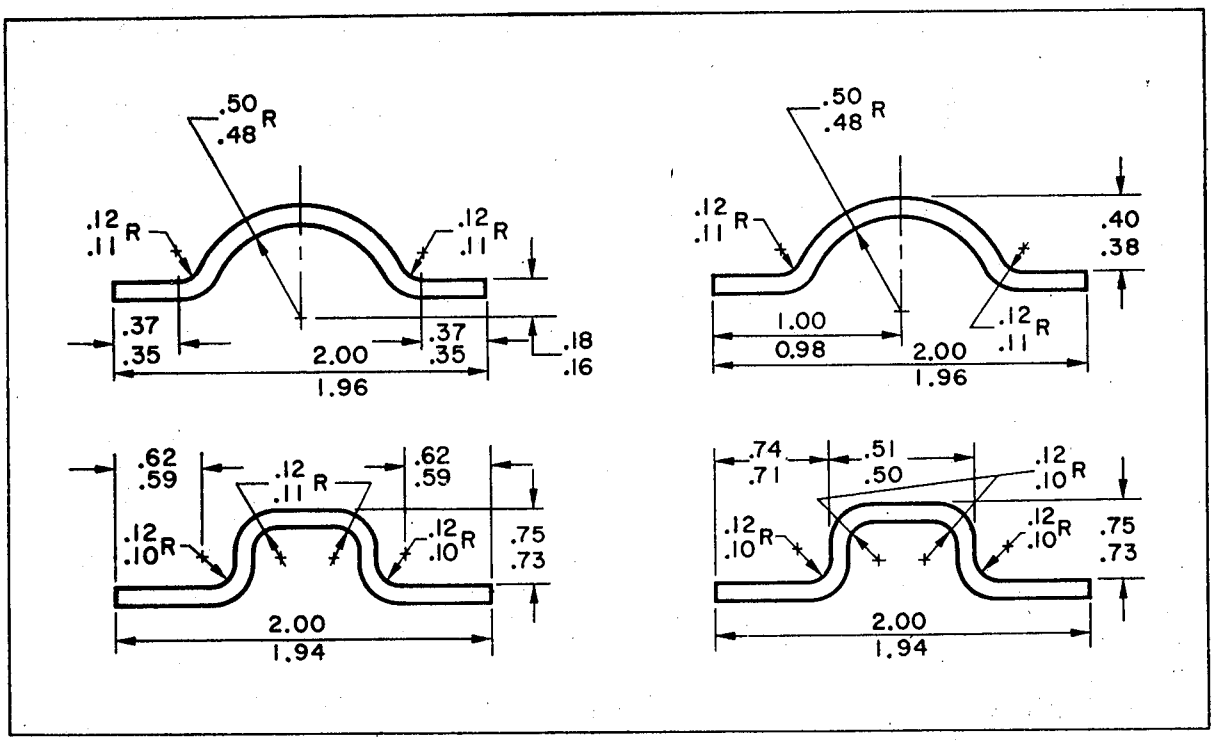


FIGURE 63. *Dimensioning stiffening rib sections.*

6.6.10.3 *Cross sections.* Section lining of sheet metal parts is not normally required but should be done where it will help to define holes, butt joints, or similar features (see fig. 64).

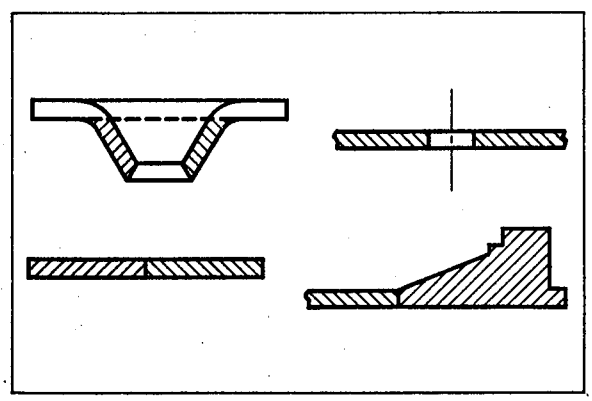


FIGURE 64. *Indicating sheet metal cross sections.*

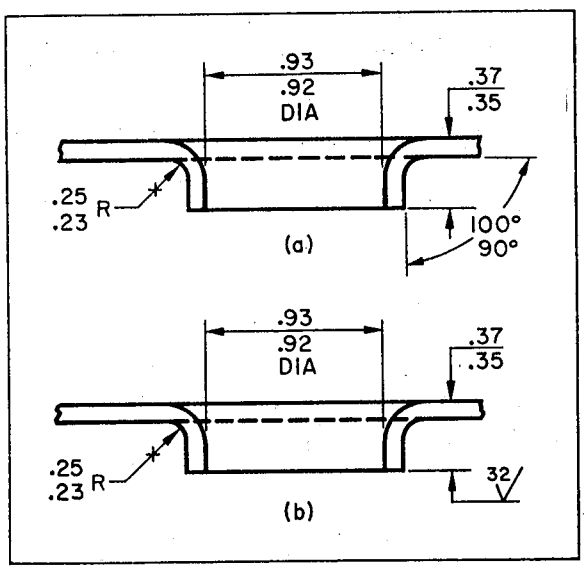


FIGURE 65. *Dimensioning eyelets.*

6.6.10.4 *Eyelets.* Figure 65 shows methods of dimensioning eyelets.

6.6.10.5 *Overlaps for welding.* Where resistance welding is to be used to join sheet metal parts, give dimensions to indicate directly the flat contact width. This is to prevent an accumulation of tolerances which can result in the contact width being reduced below the required minimum (see fig. 66).

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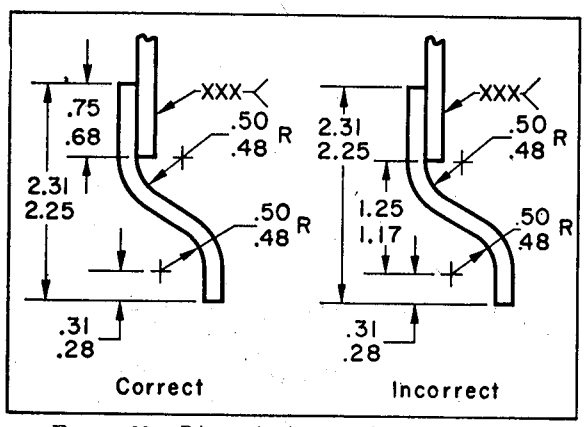


FIGURE 66. *Dimensioning overlaps for welding.*

7. TOLERANCES AND LIMITS

7.1 *General.* Limits are the maximum and minimum values prescribed for a specific dimension. A tolerance represents the total amount by which a specific dimension may vary; thus, the tolerance is the difference between the limits (see 4.2.7). A tolerance shall be expressed in the same form as its dimension; the tolerance on a decimal dimension shall be expressed by a decimal to the same number of places, and the tolerance on a dimension written as a common fraction shall be expressed as a common fraction. An exception to this is a close tolerance on an angle which may be expressed by a decimal representing a linear distance (see figs. 95 and 96). In a "chain" of dimensions with tolerances, overall variations in position that may occur are equal to the sums of the tolerances on the intermediate distances. The datum dimensioning method of 6.3 avoids overall accumulations, but the tolerance on the distance between two features equals the sum of the tolerances on dimensions from the datum. When the distance between the two points must be controlled closely, the distance between the two points should be dimensioned directly, with a tolerance. Figure 67a illustrates a series of "chain" dimensions where tolerances accumulate between points X and Y; figure 67b illustrates the use of datum dimensioning and provides the same tolerance accumulation between X and Y with larger individual tolerances; figure 67c shows a means of providing the same tolerance accumulation between X and Y but with maximum permissible tolerances on individual hole locations.

7.1.1 *Methods of tolerancing.*

7.1.1.1 *Unilateral tolerance method.* In the unilateral tolerance method, permissible variation from the design size is shown in one direction only (see fig. 68a), the design size being one of the limits.

7.1.1.2 *Bilateral tolerance method.* In the bilateral tolerance method, permissible variations from the design size are shown in both directions. Bilateral variations are generally specified for locating (positional) dimensions, or for any dimension that can be allowed to vary in either direction. The plus tolerance is added to the design size and the minus tolerance is subtracted to determine the limits (see fig. 68b).

7.1.1.3 *Limit dimensioning method.* In the limit dimensioning method, only the largest and smallest permissible dimensions are shown (see fig. 70). The tolerance is the difference between the limits.

7.1.1.4 *Expressing allowable variations.* Allowable variations shall be expressed in one of the following ways:

(a) Two tolerance numerals are specified, one plus and one minus (see fig. 68). This form of expression is necessary if the plus variation differs from the minus variation. It may be used in preference to (b) below, where the plus and minus variations are equal. When the variation in one direction is zero, the tolerancing method is unilateral (see fig. 68a). When neither of the variations is zero, the tolerancing method is bilateral (see fig. 68b).

Note: Two variations in the same direction shall not be specified.

(b) Another application of bilateral tolerancing is to use a combined plus and minus sign followed by a single tolerance numeral. This method applies if the plus variation is equal to the minus variation (see fig. 69).

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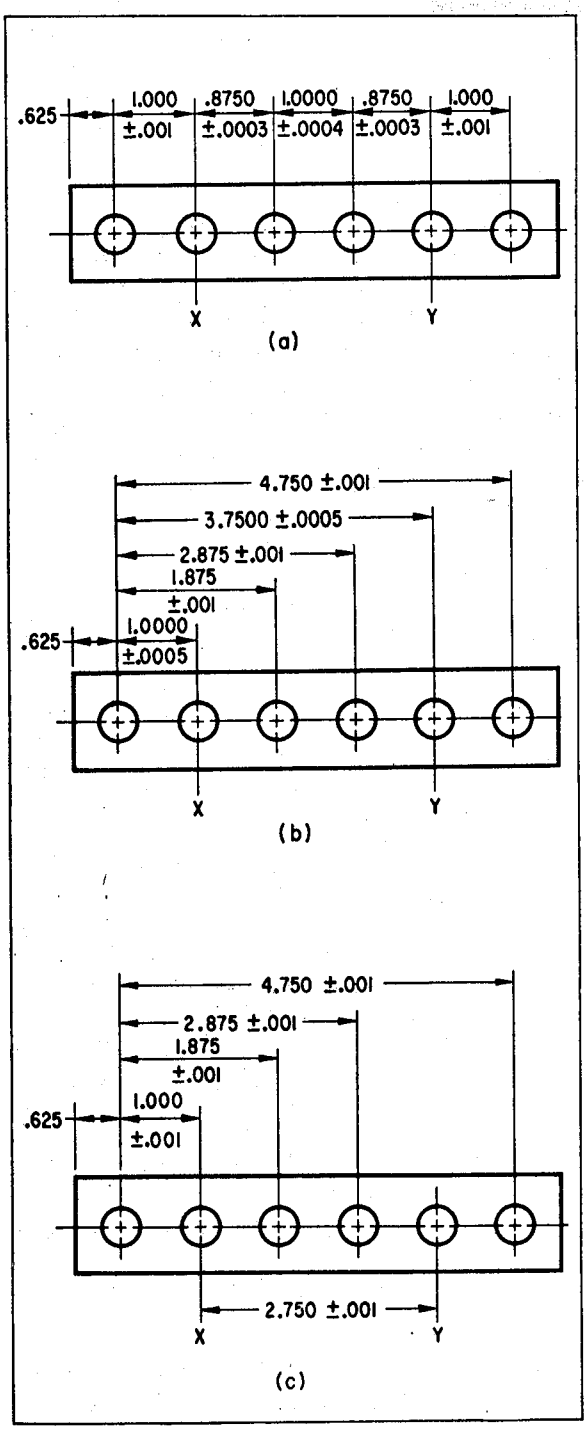


FIGURE 67. *Locating round holes.*

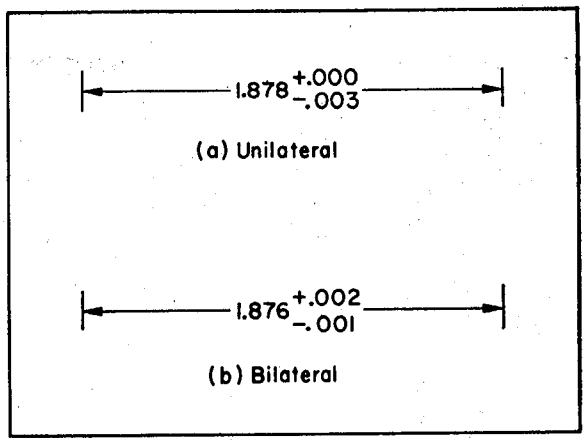


FIGURE 68. *Giving a tolerance by a plus and minus figure.*

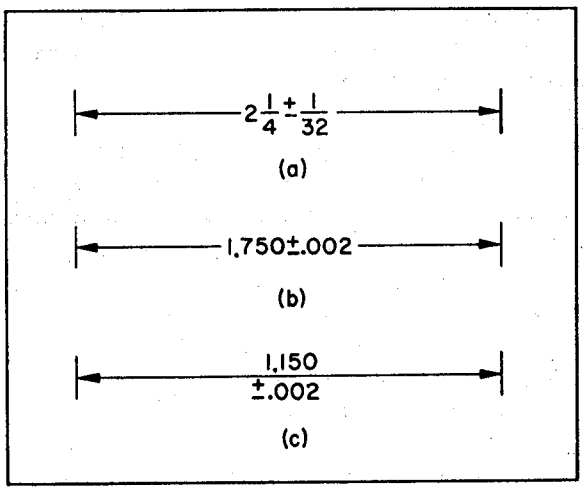


FIGURE 69. *Using a combined plus and minus sign.*

(c) The limit dimensioning method is applied by specifying the maximum and minimum limits of size or position. The numerals shall be arranged by one of two methods; both methods shall not be used on the same drawing.

(1) The high limit is always placed above the low limit when dimensions are given directly, and the low limit always precedes the high limit when dimensions are given in note form (see fig. 70a).

(2) For dimensions of position given directly (not by note), the high limit numeral (maximum dimension) is placed above and the low limit numeral (minimum dimension) is

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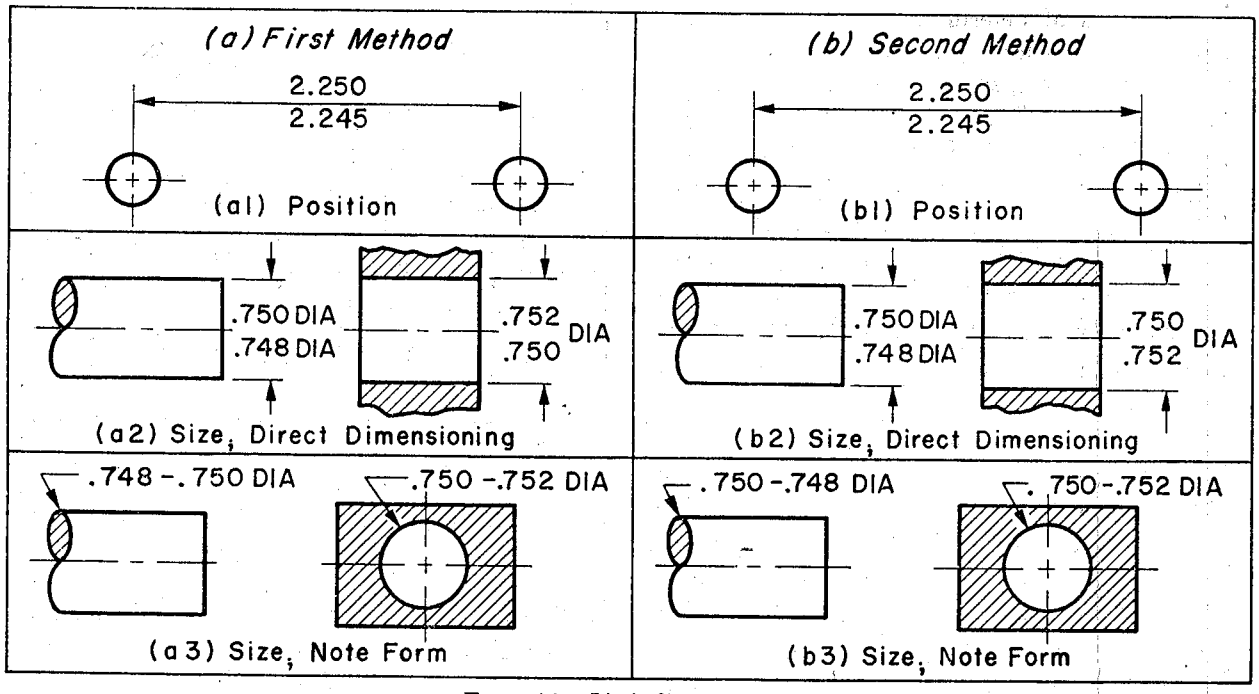


FIGURE 70. *Limit dimensioning.*

placed below. For dimensions of size given directly, the numeral representing the Maximum Material Condition (MMC) is placed above and the numeral representing the minimum material condition is placed below. When the limits are given in note form, the numeral that otherwise would be above precedes the other (see fig. 70b).

(d) It is not always necessary to state both limits when expressing unilateral tolerances.

(1) A unilateral variation is sometimes expressed without stating that the variation in the other direction is zero (see fig. 71a). However, two different methods shall not be used on the same drawing.

(2) MIN or MAX is often placed after a numeral where the other limits are not important. Depths of holes, lengths of threads, chamfers, etc., are often limited in this way (see fig. 71b).

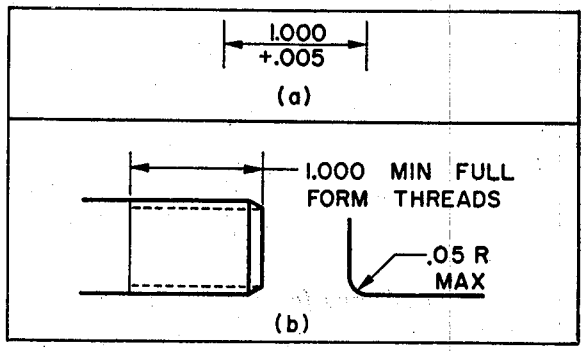


FIGURE 71. *Expressing a single tolerance or limit.*

(e) Other ways of stating allowable variations are described under True Position Tolerances, 7.5, and Tolerances of Form, 7.6.

(f) The above ways of expressing allowable variations apply both to linear tolerances and angular tolerances (see fig. 19). Angular tolerances may be expressed in degrees, minutes, and seconds, or in decimals.

7.1.2 *Interpretation of Limits.* All limits are considered to be absolute. Dimensional limits, regardless of the number of decimal places, are to be used as if they were continued with zeros.

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For purposes of determining conformance with limits, the measured value is to be compared directly with the specified value and any deviation, however small, outside the specified limiting values signifies non-conformance with the limits.

For example:

1.22	means 1.220 - - 0	as read on a suitable measuring device used to determine conformance to the drawing limits.
1.20	means 1.200 - - 0	
1.202	means 1.2020 - - 0	
1.200	means 1.2000 - - 0	

7.1.3 *Placing tolerance numerals.* Numerals indicating tolerances shall be placed in one of the following ways:

(a) To the right and in line with the dimensional numeral (see figs. 69a and 69b).

(b) Below the dimension numeral with the dimension line between (see fig. 69c).

(c) To the right of the dimension numeral with the plus variation above the level of the dimension line and the minus variation below it. In this method the dimension line is not drawn between the numerals indicating tolerances (see fig. 68).

7.1.4 *Placing limit numerals.* Numerals indicating limits shall be placed as illustrated in figure 70.

7.1.5 *Dimensioning for fits.* For interchangeable manufacture, the tolerances on dimensions shall be such that an acceptable fit will result from assembly of parts having any combination of actual sizes that are within tolerances (see fig. 72).

Figure 73 shows a method of dimensioning mating parts that must fit one another, when they do not need to be interchangeable. The size of one part need not be held to a close tolerance, because it is to be modified to the size that is necessary for the desired fit.

7.1.5.1 *"Hole basis" and "shaft basis."* To specify the dimensions and tolerances of an internal and external cylindrical surface so that they will fit together as desired, it is necessary to begin calculations by assuming either the mini-

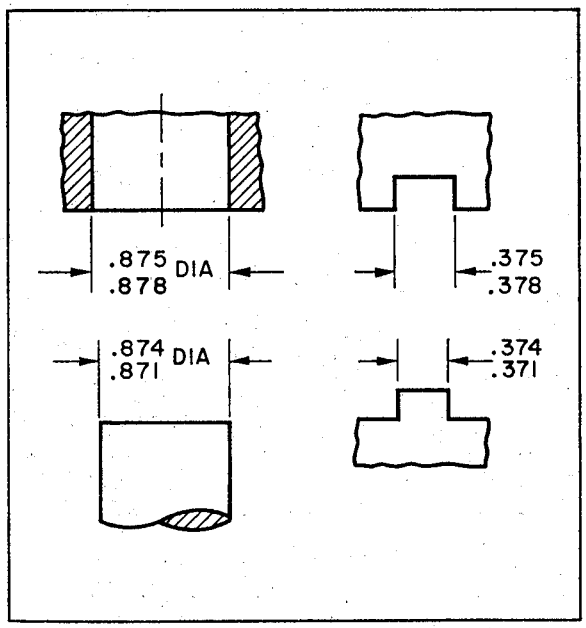


FIGURE 72. *Indicating dimensions of surfaces that are to fit closely.*

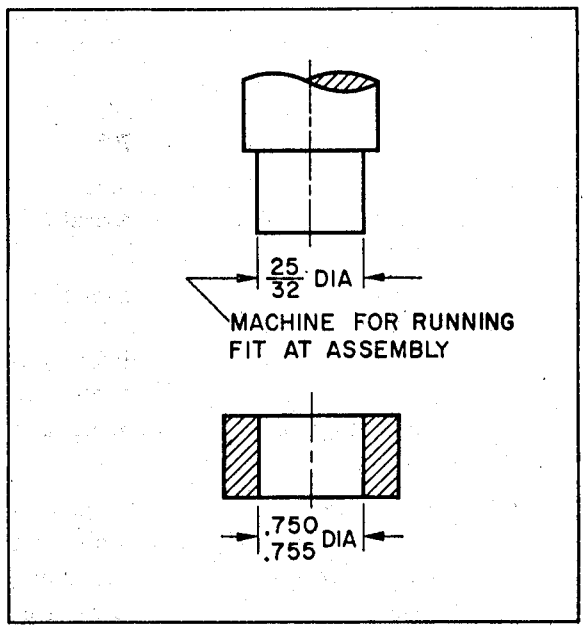


FIGURE 73. *Dimensioning noninterchangeable parts that are to fit closely.*

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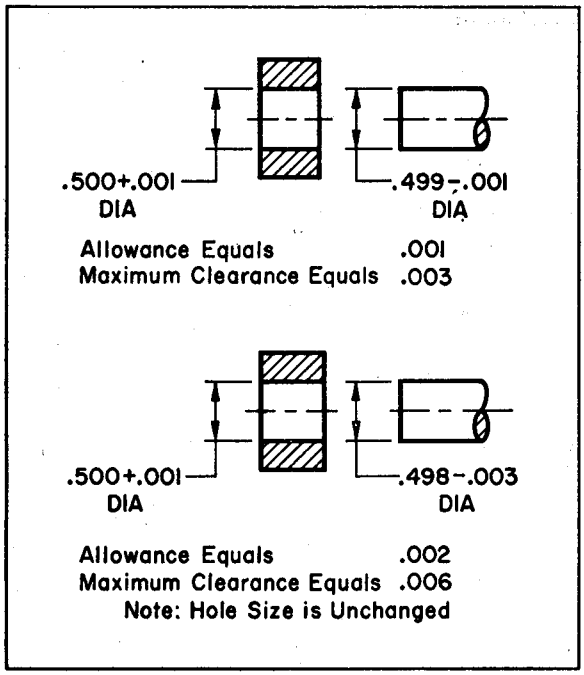


FIGURE 74. *"Basic hole" fits.*

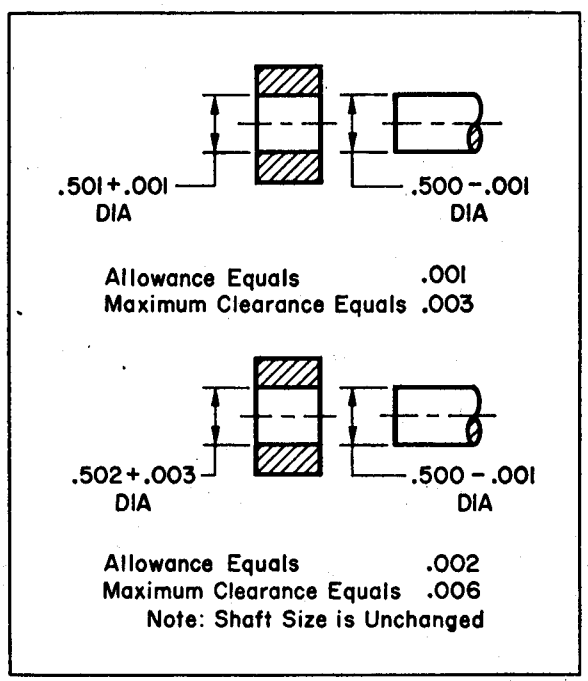


FIGURE 75. *"Basic shaft" fits.*

mum hole size or the maximum shaft size. A basic hole system is a system of fits in which the design size of the hole is the basic size and the allowance is applied to the shaft (see fig. 74).

A basic shaft system is a system of fits in which the design size of the shaft in the basic size and the allowance is applied to the hole (see fig. 75).

7.1.5.2 *General tolerance notes.* When the same tolerance applies to several dimensions on a drawing, repetition of the most common tolerance applicable to decimals, fractions and/or angles should be avoided by entering such general tolerances either in a block supplementary to the title block or as general notes. Such general tolerance notes apply to each dimension on a drawing excepting:

(1) Dimensions having tolerances assigned directly.

(2) Dimensions labeled "REF", "MAX", "MIN", or "BASIC".

(3) Dimensions covered by a general toleracing note relating to a specific category of dimensions.

Examples:

(a) CASTING TOLERANCE .006/IN. OR .03, WHICHEVER IS GREATER.

(b) DIMENSIONS LOCATING THE TRUE POSITION ARE BASIC.

(4) Dimensions which have tolerance controlled by documents invoked by the drawing.

7.2 *Expressing positional and form tolerances.* Either notes or symbols may be used to express positional and form tolerances. A tolerance of form is defined in paragraph 7.6. A positional tolerance is the permissible variation in the specified position of a feature (surface, line or point) in relation to some other feature. A true position tolerance, which is defined in paragraph 7.5.3, is one type of positional tolerance. A positional tolerance also may be expressed directly in relation to a locating (positional) dimension by one of the methods prescribed by paragraph 7.1.1.4.

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7.2.1 *Symbols.* When symbols are used in lieu of notes to express positional or form tolerances, they shall be the symbols shown herein.

7.2.1.1 *Geometric characteristic symbols.* The symbols shown in figure 76 shall be used to state the geometric characteristics being toleranced.

7.2.1.2 *Feature control symbol.* When symbols are used, a positional or form tolerance shall be stated by means of the feature control symbol illustrated by figure 77. The geometric characteristic symbol shall be followed by the permissible tolerance, and in some cases by the modifier Ⓜ or Ⓢ (see 7.3.4). The height of the box outline of the symbol shall be ¼ inch minimum, and the length, 1¼ inches minimum. When characters in addition to the characteristic symbol, datum reference(s), and tolerance must be specified, the length of the box should be increased as necessary to avoid crowding. The feature control symbol shall be associated with the feature(s) being toleranced by one of the following methods:

(a) Adding the symbol to a note pertaining ing to the feature(s);

(b) Running a leader line from the feature(s) to the symbol;

(c) Attaching a side, end, or corner of the control symbol outline box to an extension line from the feature;

(d) Attaching a side or end of the box to the dimension line pertaining to the feature when it is cylindrical.

7.2.1.3 *Datum identifying symbol.* Each datum excepting implied datums (see 7.5.9) on a drawing shall be assigned a different identifying reference letter. Any letter of the alphabet, excepting "I", "O", and "Q" may be used for

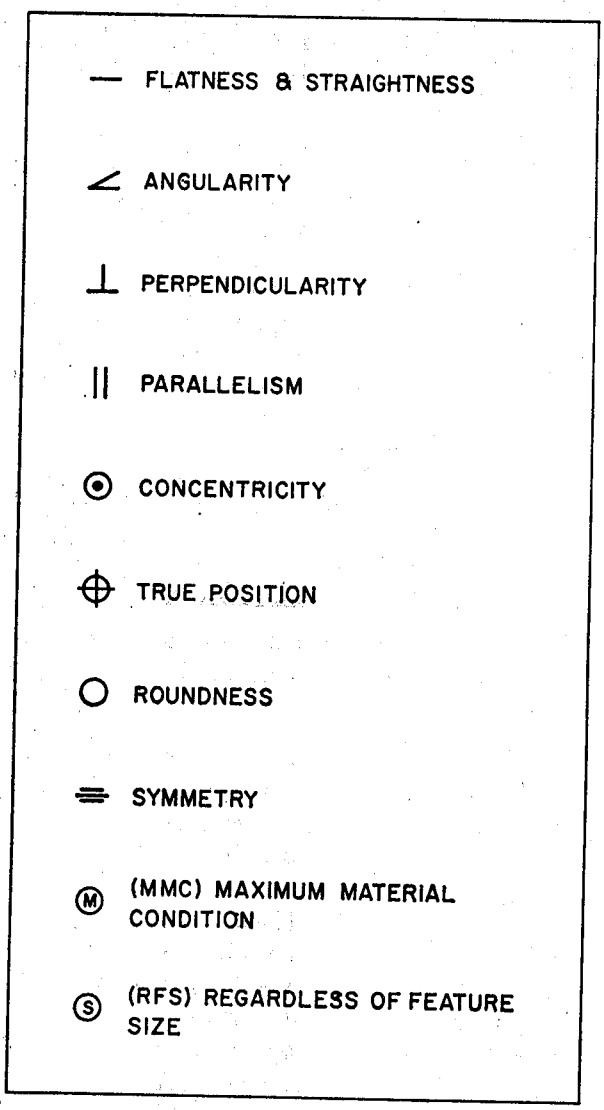


FIGURE 76. *Geometric characteristic symbols and modifiers.*

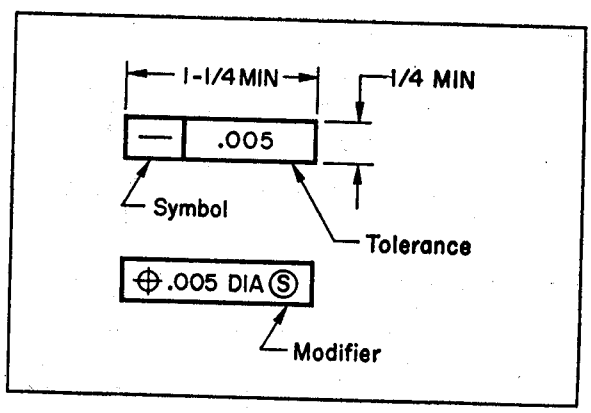


FIGURE 77. *Feature control symbol.*

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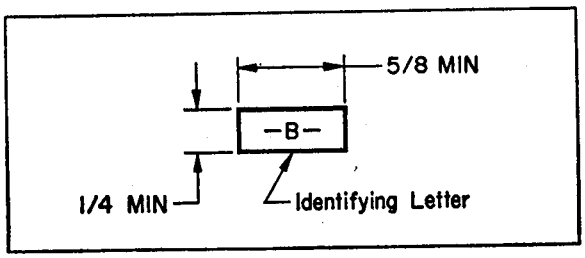


FIGURE 78. *Datum identifying symbol.*

this purpose. The datum identifying symbol shall be formed by a rectangle containing the datum reference letter preceded and followed by a dash (see fig. 78). The symbol shall be associated with the datum feature by the methods prescribed for feature control symbols in 7.2.1.2. Each datum identifying symbol shall be used only once on a drawing.

7.2.1.4 *Reference to datum.* When a positional or form tolerance must be related to a datum, this relationship shall be stated in a feature control symbol by placing the datum reference letter between the geometric characteristic symbol and the tolerance. If there are two or more datums, or if the modifier Ⓜ or Ⓢ is used, and where straightness or roundness symbols are used, vertical lines shall be added to provide clarity (see figs. 77 and 79).

7.2.1.5 *Multiple datums.* Where reference is made to more than one datum used to control the positional or form tolerances of a feature, the first datum reference letter in the sequence shall be the primary datum, the second letter, the secondary datum, etc. The datum reference letters shall be placed in order of importance with respect to establishing the location, form, or position of other features. Thus the datum reference letters will not necessarily be in alphabetical order. In some designs the datums obviously are of equal importance. An example is: a concentricity tolerance related to two equally important bearing surfaces.

7.2.1.6 *Combined symbols.* When a feature serves as a datum and also is controlled by a positional or form tolerance, the feature control symbol and the datum identifying symbol shall be combined as shown in figure 80. In such cases, the length of the box for the datum identifying symbol may be either the same as that of the feature control symbol or 5/8 inch.

7.2.1.7 *Interrelated geometric characteristics.* In some cases it may be necessary to control the geometrical accuracy of a feature by means of different tolerances of form which are:

(a) Both (or all) governed by a single tolerance value;

(b) Each related to a different datum;

(c) Intended for a simultaneous check of both (or all) geometric requirements.

The two (or more) geometric characteristics with their respective datums and the single tolerance shall be shown in the same feature control symbol as illustrated in figure 81 (also see appendix 40).

7.2.1.8 *Former symbols.* The symbol ⌣ formerly was prescribed by this standard for use in expressing a tolerance on flatness. The symbol

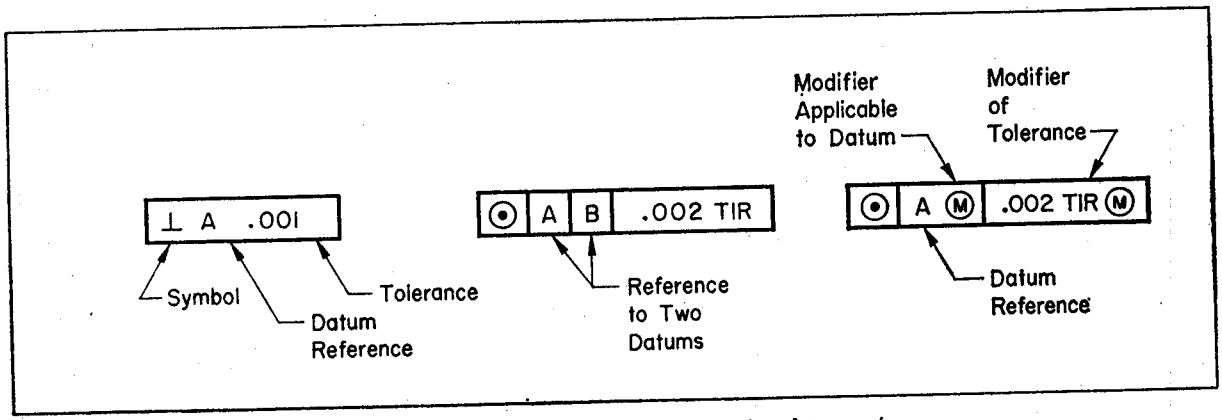


FIGURE 79. *Feature control symbol incorporating datum reference.*

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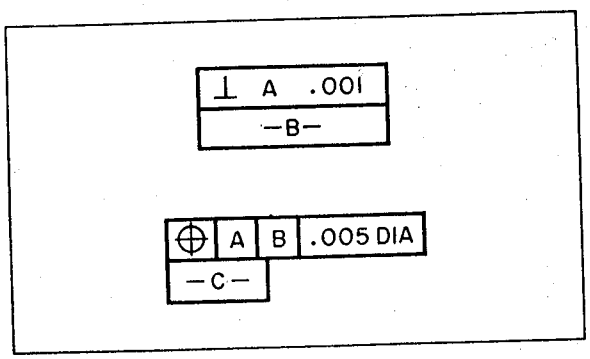


FIGURE 80. *Combined feature control symbol and datum identifying symbol.*

⌓ formerly was prescribed for use in expressing a tolerance on both straightness and flatness. Whenever either of these symbols appears on an existing drawing it shall be interpreted as governing the one of these geometric characteristics that applies to that case.

7.2.2 *Notes.* The majority of figures 88 through 111 illustrate means of expressing positional or form tolerances by notes. Paragraph 7.5.4 shows examples of expressing true position tolerances by notes. Paragraph 7.2.1.3 applies to datum reference letters when notes are used, and datums shall be identified in a manner similar to that shown by the figures herein.

7.3 *Significance of "MMC" and "RFS."* MAXIMUM MATERIAL CONDITION and REGARDLESS OF FEATURE SIZE are used as modifiers of positional and form tolerances and datum references. The modifiers may be abbreviated "MMC" or "RFS". The use of the symbols Ⓜ or Ⓢ is restricted to the feature control symbol (see 7.2.1.2).

7.3.1 The maximum material condition of a feature is that condition wherein the feature contains the maximum amount of material, e.g., minimum hole diameter and maximum shaft diameter.

7.3.2 The application of MMC to a positional or form tolerance means that the tolerance applies at MMC and is increased as the feature(s) depart(s) from MMC by the amount of such departure.

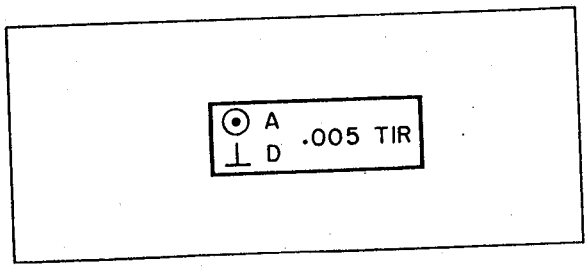


FIGURE 81. *Interrelated geometric characteristics.*

7.3.3 The application of RFS to a positional or form tolerance means that this tolerance must not be exceeded regardless of the actual size to which the feature(s) is produced.

7.3.4 *General rules applicable to MMC or RFS:*

Rule 1. Unless otherwise specified, true position tolerances apply at maximum material condition. MMC also applies to datum references other than plane surfaces.

Rule 2. The second general rule governing positional and form tolerances consists of the following parts:

a. Unless otherwise specified, concentricity, straightness, and symmetry tolerances apply regardless of feature size. RFS also applies to datum references other than planes.

b. Unless otherwise specified, roundness, flatness, parallelism, angularity, and perpendicularity tolerances apply regardless of feature size, provided that the form tolerance zone falls entirely within the tolerance zone for size. RFS also applies to datum references other than planes. (In regard to flatness, see 7.4(c)).

7.3.4.1 Exceptions to General Rules 1 and 2 shall be expressed on drawings in one of the following ways:

(a) By the use of Ⓢ or Ⓜ in feature control symbols;

(b) By the addition of, "REGARDLESS OF FEATURE SIZE" (or "RFS") or "MAXIMUM MATERIAL CON-

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DITION" (or "MMC"), as appropriate, in each applicable drawing note;

(c) By use of a general note.

7.3.4.2 If none of the exceptions to the general rules of 7.3.4.1 is specified, and this standard is referenced on a drawing, the general rules apply.

7.3.4.3 Optionally, the proper modifier may be expressed in connection with each positional or form tolerance and each datum reference, with the following exceptions:

(a) Flatness, perpendicularity, parallelism, or angularity of plane surfaces;

(b) Datums that are planes.

7.3.5 All tolerances specified in connection with positional and form tolerances shall be totals, with the exception of true position and roundness. For true position tolerancing of holes or other cylindrical features, the specification of the diameter of the tolerance zone is preferred; however, the radius of the tolerance zone may be specified. For true position tolerancing of slots or tabs, the specification of the total width of the tolerance zone is preferred; however, the width of the zone either side of true position may be specified. For roundness, either the tolerance on the radius or on the diameter may be specified. Such abbreviations or terms as DIA, TOTAL, WIDE ZONE, TIR, or ON DIA must be used in symbols and notes to indicate that the total tolerance has been specified for true position, concentricity or roundness. Such abbreviations or terms as R, EITHER SIDE, or ON R must be used in symbols or notes to indicate that the radial or half tolerance has been specified for true position or roundness. On existent drawings, if true position tolerances are expressed by symbols which do not specify whether the tolerance is TOTAL, R or EITHER SIDE, such symbols shall be interpreted as specifying the total tolerance ,i.e., DIA, TOTAL or WIDE ZONE, as applicable. On existent drawings, if roundness tolerances are expressed by symbols which do not specify whether the tolerance is "ON R" or "ON DIA", such symbols shall be interpreted as specifying the tolerance on the diameter.

7.4 *Virtual size.* Depending upon its function, a feature may be controlled by (1) one or more of several types of tolerances, such as size, form, and position, (2) certain modifiers such as RFS or MMC, and (3) relationship to specific datums or datum features. Consideration must be given to the collective effect of these specified tolerances and datum relationships in determining the clearance between mating parts and in establishing gage feature sizes. From such consideration a net resultant size is derived, which is termed virtual size. By definition, the virtual size of a feature is a derived size generated from the collective effect of all profile variations permitted by the specified tolerances. It represents the most extreme condition of assembly at MMC.

Examples:

(a) If a straightness tolerance of .005 total at MMC (see fig. 94), or .005 total RFS (see fig. 93) is specified for a 1.000 diameter pin, the virtual size is 1.005 diameter. However, if only a size tolerance of +.000 — .010 for a 1.000 diameter pin is specified, or if this size tolerance is combined with a straightness tolerance of zero at MMC, the virtual size is 1.000 diameter.

(b) Referring to figure 111 and assuming that the size tolerance for the .250 diameter hole is +.001 — .000, the diameter at MMC would be .250. Since a positional tolerance of .010 is specified, the virtual size would be .240. The parallelism tolerance of .003 has no effect on virtual size since the parallelism tolerance must be contained within the positional tolerance zone.

(c) Normally, flatness tolerances have no effect on virtual size; however, for sheet metal, when a specified flatness tolerance exceeds the thickness tolerance, the virtual thickness is increased by the amount of the flatness tolerance.

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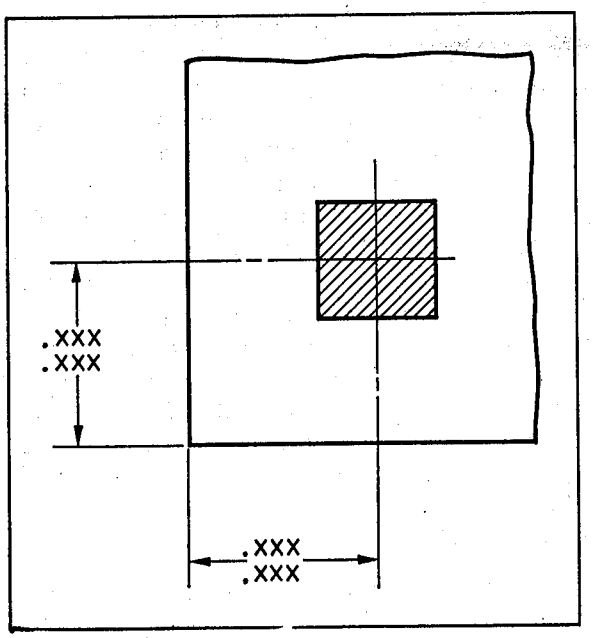


FIGURE 82. *Square tolerance zone resulting from rectangular coordinate dimensioning.*

7.5 *True position tolerancing (see appendix 10).*

7.5.1 *General.* Figures 22 through 27 illustrate the location of holes or other features by means of rectangular coordinates or angular dimensioning. In the past, these coordinates and angular dimensions predominantly have been shown with individual tolerances. This results in a square, rectangular, or wedge-shaped tolerance zone, as shown by figures 82 and 83. The engineering intent can often be expressed more precisely if locations are given as true positions, with tolerances to state how far actual positions can be displaced from true position in any direction. This results in a circular tolerance zone (see fig. 84).

7.5.2 *True position.* True position denotes the basic or theoretically exact position for a feature (see appendix 10). The geometry involved in establishing the true position is perfect. All lines drawn perpendicular to one another and involved in establishing the true position of a pattern shall be interpreted to be at an angle of 90° ±0° (90° basic), even though no angle is specified. All dimensions (rectangular, linear, or angular) that locate the true position with reference to either datums, or other features in the same pattern, shall be interpreted as perfect (true).

7.5.3 *True position tolerance.* A true position tolerance is the total permissible variation in the location of a feature about its true position. For cylindrical features (holes and bosses) the true position tolerance is the diameter of the tolerance zone within which the axis of the feature must lie, the center of the tolerance zone being at the

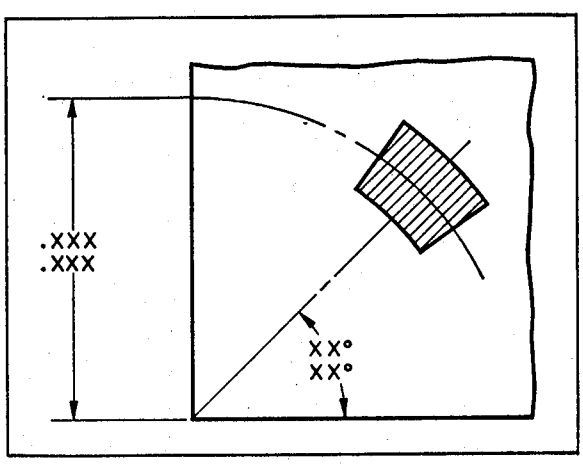


FIGURE 83. *Wedge-shaped tolerance zone resulting from angular dimensioning.*

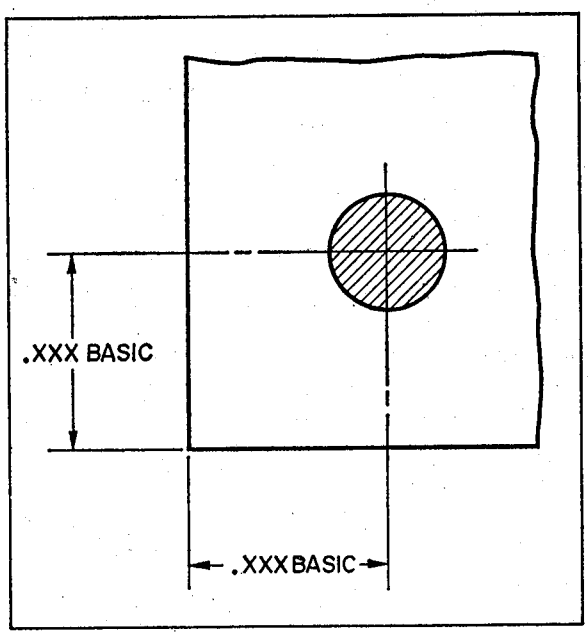


FIGURE 84. *Circular tolerance zone resulting from true position dimensioning.*

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true position. For other features (slots and tabs) the true position tolerance is the total width of the tolerance zone within which the center plane of the feature must lie, the center plane of the zone being at the true position.

7.5.4 *Specifying true position tolerances.* The following methods apply:

7.5.4.1 Specify the true position of each feature in a pattern (groups of holes, slots, tabs, etc.) by untoleranced dimensions labeled BASIC or with the abbreviation BSC (see 7.5.5).

7.5.4.2 When cylindrical features are located by dimensions to their axes and when the location may be allowed to vary in any direction, the zone of tolerance, which in this case is a cylinder, shall be specified. Two acceptable methods are:

(a) 6 HOLES .XXX DIA LOCATED AT TRUE POSITION WITHIN .010 DIA

(b) 6 HOLES .XXX DIA ⌖ .010 DIA

7.5.4.3 When noncylindrical features, such as slots, are located by dimensions to a center plane or to one surface of the feature, true positional tolerancing shall be applied. Two acceptable methods are:

(a) 6 SLOTS LOCATED AT TRUE POSITION WITHIN .010 WIDE ZONE

(b) 6 SLOTS ⌖ .010 TOTAL

7.5.4.4 When a part and its mating part have matching feature patterns, and alignment between such parts depends upon a functional feature such as a pilot, bore, or surface, this feature should be used as a datum. In such cases, the relationship between the true position tolerance and the datum shall be specified. Two acceptable methods are:

(a) 6 HOLES .XXX DIA, LOCATED AT TRUE POSITION WITHIN .010 DIA IN RELATION TO DATUM A

(b) 6 HOLES .XXX DIA ⌖ A .010 DIA

7.5.5 *Identifying dimensions establishing true position.* When true position tolerancing is used on a drawing on which standard tolerances are given in the title block or in a referenced specification, untoleranced dimensions locating true positions shall be identified by one of the following methods:

(a) Add the word BASIC or the abbreviation BSC to the right of, or underneath, each dimension establishing the true position.

(b) Tolerance each dimension not involved in true position dimensioning, or not labeled "REF", "MAX", or "MIN", and use the following general note: DIMENSIONS LOCATING TRUE POSITIONS ARE BASIC.

7.5.6 *Use of projected tolerance zones.* For fixed pins and studs and for threaded holes and inserts, true position tolerances may be specified as projected tolerance zones rather than as tolerances applicable to the depth of hole within the part containing the thread or pin. Only when the mating part is quite thin can the effect of possible error in perpendicularity on projected height be ignored. See figure 85a for an example of the interference that may result even though the axis of both the threaded hole and the clearance hole lies within the tolerance cylinder and the clearance hole is larger than the bolt diameter by the sum of the tolerance cylinders. Note that it is the variation from perpendicularity of the portion of the bolt passing through the mating part that is significant, and that the location and perpendicularity of the tapped hole is only of importance insofar as it affects the projected mating height of the bolt. Figure 85b illustrates that the projected tolerance zone concept realistically specifies this condition affecting assembly of the parts. Figures 86 and 87 illustrate the specification of true position tolerances with projected tolerance zones for tapped holes and fixed pins, respectively.

7.5.7 *True position tolerance formulas for holes.* The following formulas will give a "no interference, zero clearance" fit at maximum material limits. They are based on equal values of Z for mating members:

Let Z = permissible true position tolerance (diametral concept)

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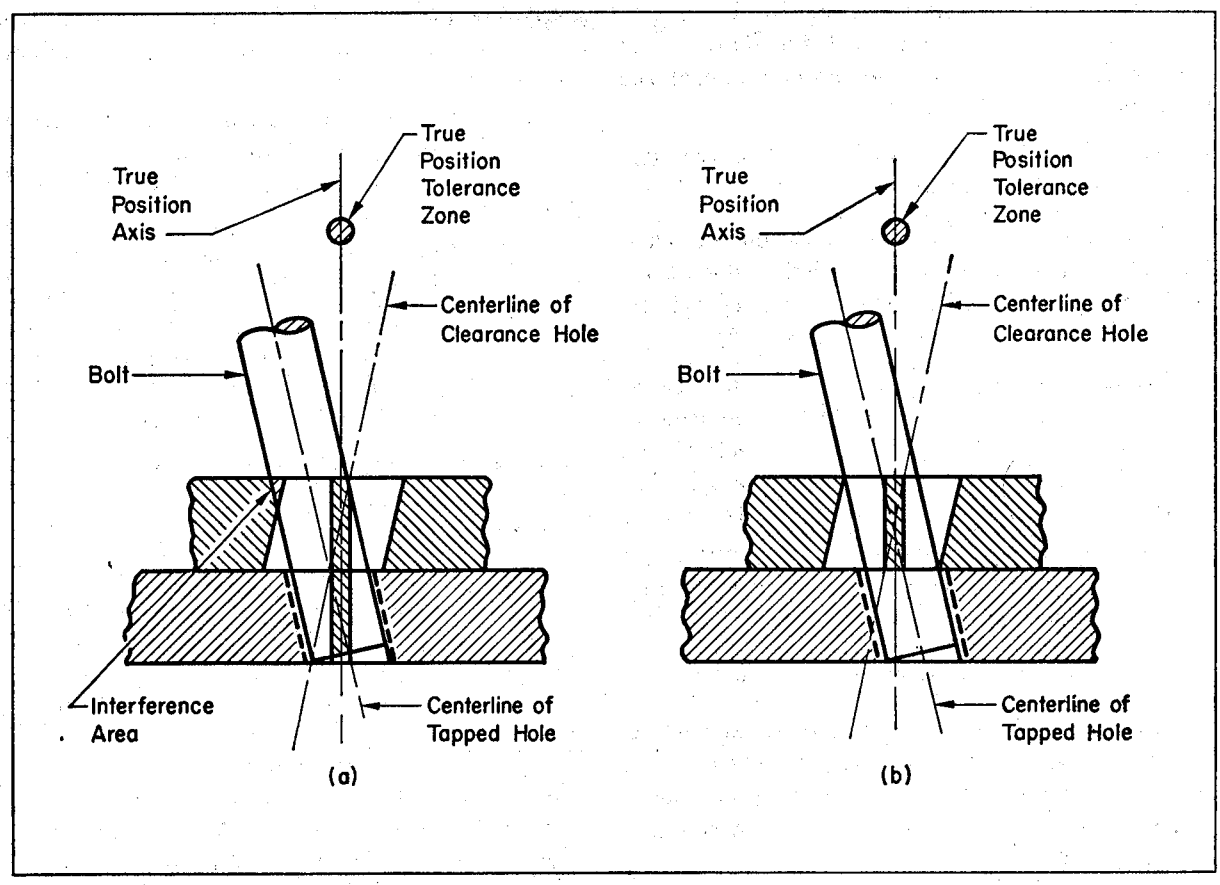


FIGURE 85. *Projected tolerance zone interpretation.*

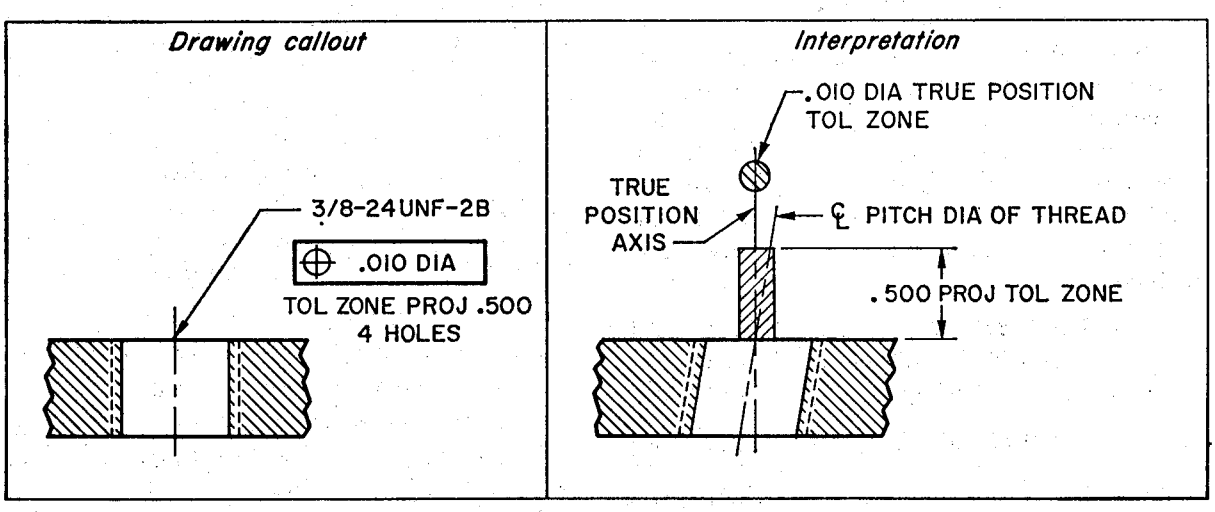


FIGURE 86. *Projected tolerance zone for tapped holes.*

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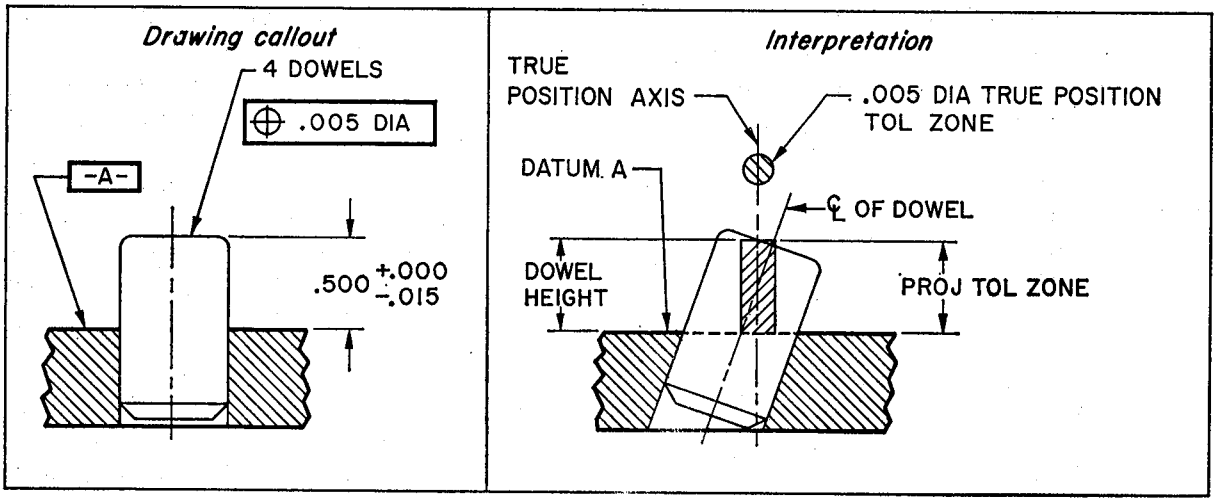


FIGURE 87. *Projected tolerance zone for fixed pins or studs.*

H = minimum diameter of clearance hole
F = maximum diameter of fastener

7.5.7.1 *Bolt clearance holes.* When parts are held by fasteners which have the same basic clearance between holes and fasteners, the following formula applies:

$$Z = H - F$$

7.5.7.2 *Tapped hole and clearance hole.* When one part has a feature, such as a tapped hole, which fixes a clamping bolt or stud, the other part will have clearance holes:

(a) When the true position tolerances are equal, the following formulas apply:

$$Z = \frac{H - F}{2} \quad \text{or} \quad H = F + 2Z$$

(b) When the true position tolerances are not equal for the mating members, the following formula applies:

$$H = F + Z_1 + Z_2$$

in which Z_1 = Permissible true position tolerance in first member.
Z_2 = Permissible true position tolerance in second member.

7.5.8 *True position interpretation.* When applied to a cylindrical feature, such as a hole, the true position tolerance in terms of the axis of the hole means the following: When RFS is the modifier, or when MMC is the modifier and the hole is at its maximum material condition, the axis of the hole must lie within a cylindrical zone having a diameter equal to the true position tolerance and its center at the true position. In cases of "blind" holes, the tolerance zone cylinder for each hole is perpendicular to the surface on which the holes originate and to which they have been shown on the drawing as being perpendicular. In cases of holes passing through a part, the tolerance zone cylinder for each hole is perpendicular to the surface designated on the drawing as a datum and to which the holes are shown as perpendicular; otherwise, either surface of the part may be considered a datum surface. Thus, the tolerance zones under the above conditions define not only the permissible positional variation of the holes, but also the limits of variation in perpendicularity of each hole with respect to a datum surface. For a pattern of holes the cylindrical tolerance zones are parallel, so that the true position tolerance controls both position and parallelism. When no modifier or MMC is specified it is only when the feature is at MMC that the true position tolerance applies (see 7.3.4, Rule 1). When the feature is at other than its maximum material condition (but within its limits of size) an additional tolerance is permitted. Such added tolerance is equal to the difference between MMC and the actual size of the feature.

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7.5.9 *Implied datum surfaces.* Datums shall be identified on drawings whenever failure to do so can result in misinterpretation of dimensions referred to these surfaces. When dimensions are shown from a surface or other feature which is obviously assumed to be at an exact location, that surface or feature is called an implied datum. Figure 88 shows four ways of expressing the true position relationship of a two-hole pattern to three datum surfaces. In views (a) and (b), the left-hand edge and lower edge of the plate are obviously datums, because the basic dimensions locating the holes are measured from them. The top or bottom surface of the plate may be considered a datum, as explained in 7.5.8. In views (c) and (d), the three datum surfaces are not only specified, but also the accuracy of these surfaces and their relationship to one another is controlled by geometric tolerances. Essentially, there is little difference between the requirements of the four drawings in figure 88, with the exception that greater inaccuracies of datum surfaces would be permissible under (a) and (b) in comparison with (c) and (d).

7.5.10 *Avoiding ambiguity of datum surfaces.* It is particularly necessary to specify datums to avoid ambiguity when a pattern of features is located by true position dimensioning with respect to a parallel circular bore or pilot, and is shown perpendicular to two or more plane surfaces. In the case of a part similar to that illustrated by figure 89, for example, it might not be clear whether the left-hand face or the right-hand face of the flange is a mating surface to which certain features must be perpendicular or parallel. Such ambiguity must be avoided by designating which nominally perpendicular plane surface is the datum, as has been done in figure 89.

7.5.11 *Larger tolerance between a datum and a pattern of features.* When the location of a pattern of features is relatively unimportant while greater accuracy within the pattern is required, this can be specified by the use of both directly toleranced dimensioning and true position dimensioning. When such combination dimensioning is used, the directly toleranced dimensions shall be interpreted as referring only to theoretical centerlines (or centerplanes) in each pattern. Each individual feature is then permitted to deviate from its position in the pattern, within the true position tolerance for that feature. Typical examples showing this combined dimensioning are as follows:

(a) Figure 90a shows an example of a circular and a rectangular hole pattern, each of which is located from the edge surfaces of the part by dimensions having the relatively large tolerance of .03, while the true position tolerance governing the accuracy of location of the holes within the pattern is only .005 and .010 diameter, respectively. One means of illustrating the interpretation of this method of dimensioning and tolerancing the rectangular hole pattern of figure 90a is shown by figure 90b. The lower side of the lower .03 squares in figure 90b is located .40 (minimum dimensional limit) from the horizontal datum established by the lower edge of the plate. The lower side of the upper .03 squares is 1.50 from the corresponding side of the lower squares. The left side of the squares is similarly located from the vertical datum, which is perpendicular to the horizontal datum and touches the high point of the left edge of the plate. The true position pattern as an entity (consisting of four .010 diameter zones about the true positions) is limited in location and orientation (in an angular relationship to the datums) by the requirement that the centerline of each of the true position tolerance cylinders must lie within, or on the periphery of, its respective .03 square tolerance zone. This interpretation is in accord with the requirement stated above that the directly tolerance dimensions shall be interpreted to apply to pattern centerlines, and hence to apply to patterns as entities. Although the dimensions of .53/.50 and .43/.40 are drawn to the centerlines of the nearer row of holes, they locate the complete

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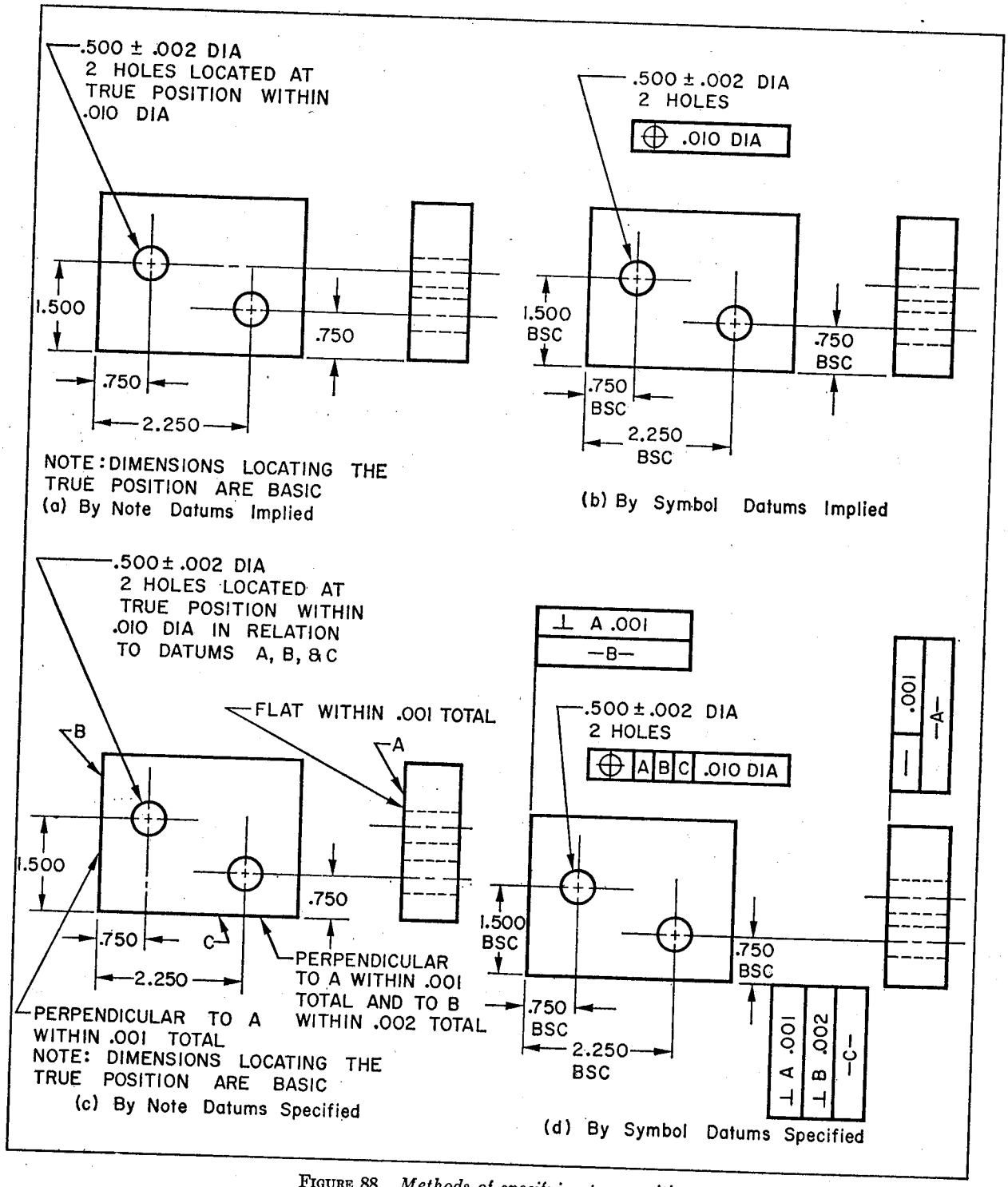


FIGURE 88. *Methods of specifying true position.*

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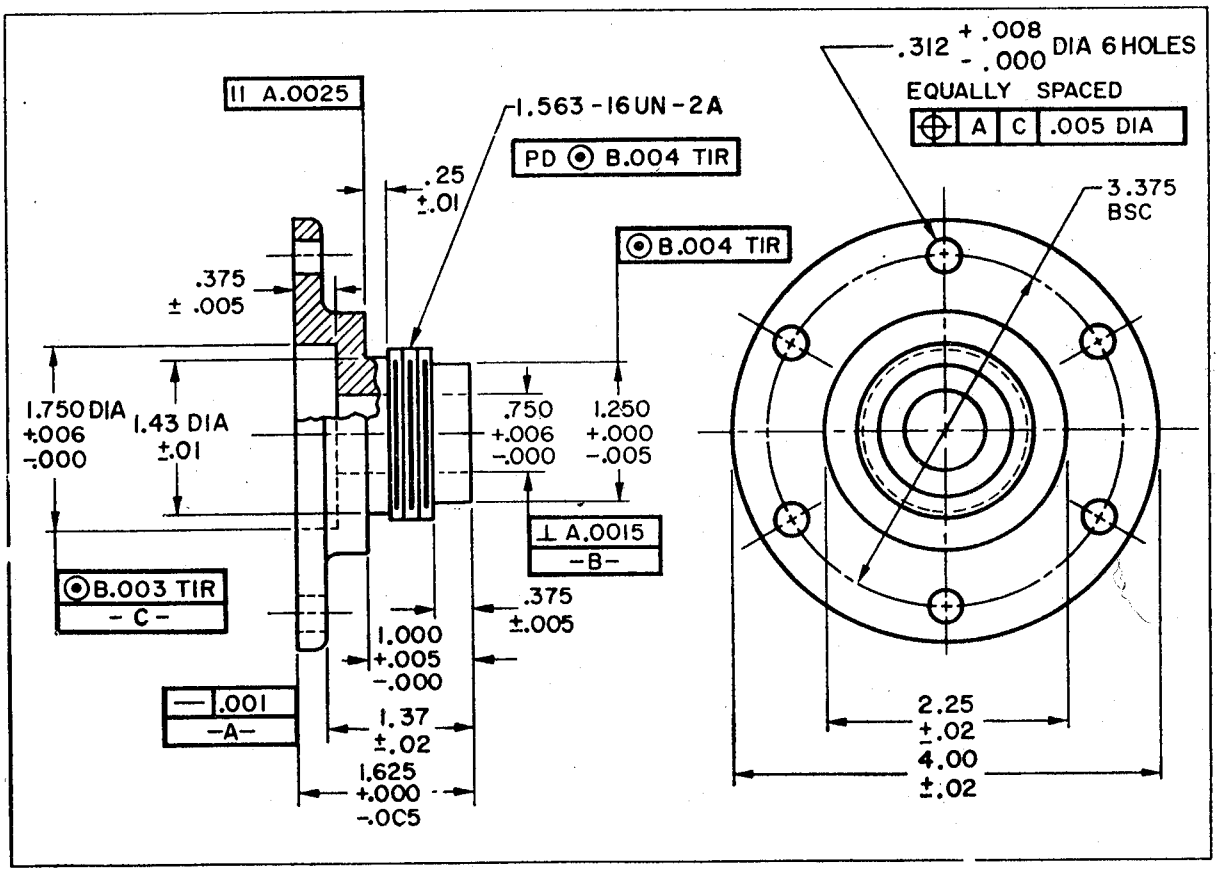


FIGURE 89. *Application of positional tolerances.*

pattern; and each hole may vary within the pattern by an additional amount as permitted by the true position tolerance zone.

(b) Figure 90c shows an example of radial hole pattern in which the pattern is located from a datum face of the part by a dimension which has a larger tolerance (.06) than the true position tolerance (.02 dia) applicable to the location of the holes within the pattern. In accordance with the rule of 7.5.11, the tolerance of .06 shall be interpreted to apply to the location of the center plane of the pattern. The axis of the holes in the pattern can vary from being coplanar only by the .02 diameter tolerance zones located at the true angular positions. Thus, the axis of one or more holes can be as much as 10.07 (10.06 + .02/2) from the datum face (at maximum material condition of the holes). However, as explained in (a), the hole-to-hole relationship is confined to a smaller deviaton by the true position tolerance.

(c) Figure 90d shows an example of a hole pattern which is basically concentric with a spline on the same part. The concentricity note in this figure shows that the hole pattern may deviate a greater amount than each individual hole is permitted to deviate within the pattern. Any or all holes may deviate from the spline PD by the sum of ½ the concentricity tolerance and ½ the true position tolerance, i. e., .005/2 + .002/2 equals a total permissible deviation of .0035 for each hole from its

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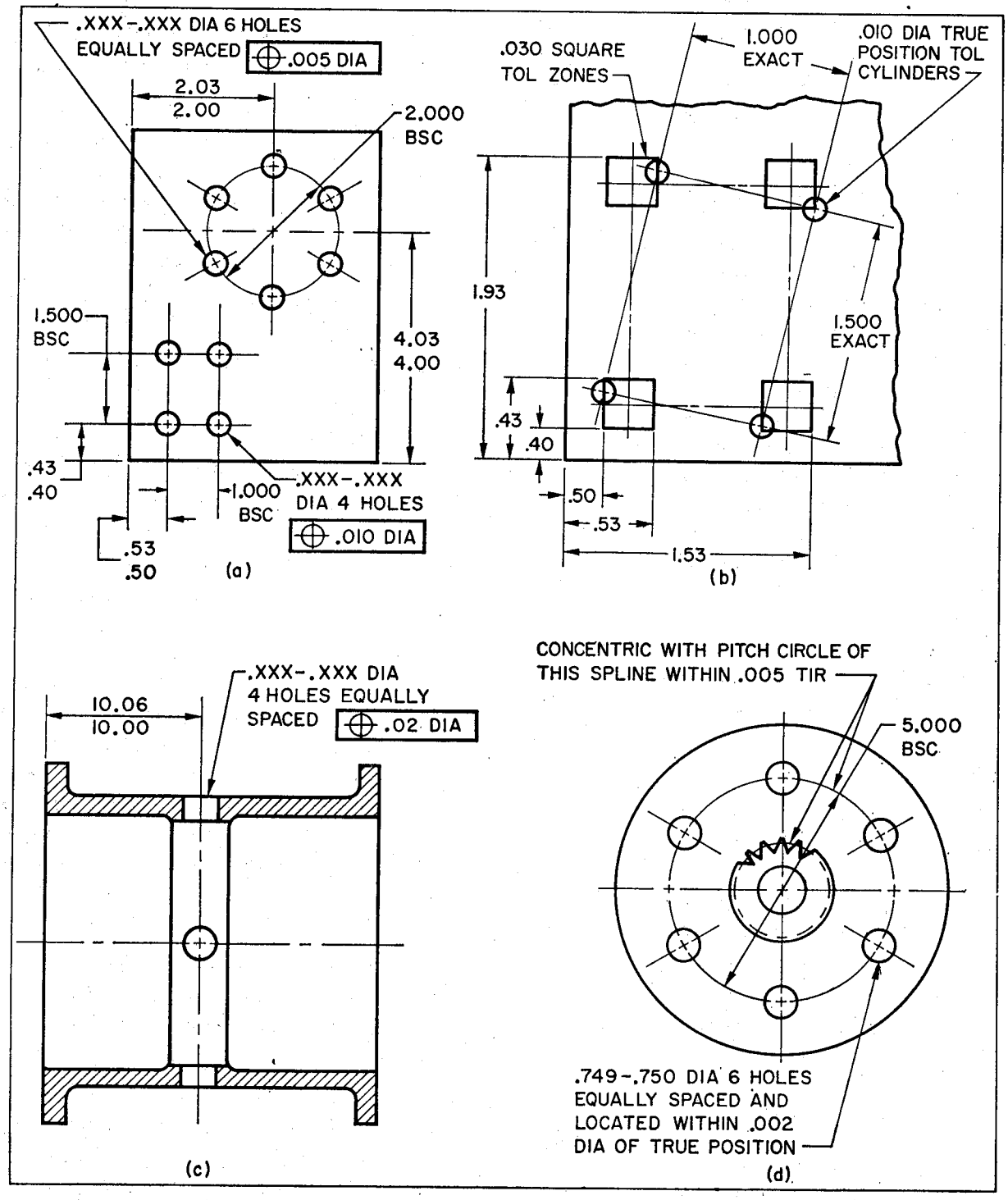


FIGURE 90. *Hole patterns at true position.*

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datum (at maximum material condition). However, greater accuracy than this is required within the pattern, thus the hole-to-hole relationship must conform to the provisions of the true position tolerance given in the figure.

7.6 *Tolerances of form.* Tolerances of form state how far actual surfaces are permitted to vary from the perfect geometry implied by drawings. Expressions of these tolerances refer to straightness, flatness, parallelism, perpendicularity, angular displacement, symmetry, concentricity, roundness, etc. Figures 91 through 106 illustrate recommended expressions for specifying tolerances of form on drawings. The meanings of recommended expressions are illustrated in figures 91 through 106 using axes planes, cylinders, etc. Tolerances of form often affect one another; parallelism includes flatness or straightness, etc. Tolerances of form are also interrelated with tolerances that limit size or position. Figures 91 through 106 consider tolerances of form individually, ignoring the effects of combination with other tolerances of form, size, or position. Paragraph 7.5.13 and figures 110 and 111 illustrate combinations of tolerances of form and position.

7.6.1 Tolerances of form should be specified for all features critical to function and interchangeability:

(a) When established workshop practices cannot be relied upon to provide the required accuracy;

(b) When documents establishing suitable standards of workmanship cannot be prescribed;

(c) When the boundary of perfect form for each feature at MMC does not provide the necessary control (see 7.6.15).

Also, a straightness tolerance or a flatness tolerance for sheet metal may be specified when it is desired to permit a greater variation in a feature than is permitted by the boundary of perfect form for an individual feature at MMC.

7.6.2 *Specifying and interpreting flatness.* Flatness is the condition of a surface which does not deviate from a plane.

7.6.2.1 *Flatness tolerance.* A flatness tolerance is the total deviation permitted from a plane, and consists of the distance between two parallel planes between which the entire surface so toleranced must lie (see fig. 91).

7.6.3 *Specifying straightness of axis.* Straightness of axis of a figure of revolution is the condition in which the axis does not deviate from a straight line.

7.6.3.1 *Axis straightness tolerance.* A tolerance governing the straightness of an axis is the diameter, or width, of the tolerance zone within which the axis must lie. For methods of specifying straightness of axis see figure 92.

7.6.4 *Straightness of elements.* Straightness of the elements of a cylindrical or conical surface is the condition in which no element deviates from a straight line. It is possible for a nominal cylinder or cone to have a straight axis and yet have elements that are slightly curved. Often straightness of the elements of nominally cylindrical or conical surfaces is of primary importance and straightness of the axis is secondary. Therefore, this standard provides a means to distinguish between straightness tolerances pertaining to the elements and those pertaining to the axis.

7.6.4.1 *Element straightness tolerance.* A tolerance governing the straightness of the elements of a cylindrical or conical surface is the maximum permissible deviation of any element from a straight line.

7.6.4.2 For the method of specifying the straightness of the elements of a cylinder or cone regardless of feature size see figure 93. A leader shall run from the symbol or note to the pertinent element.

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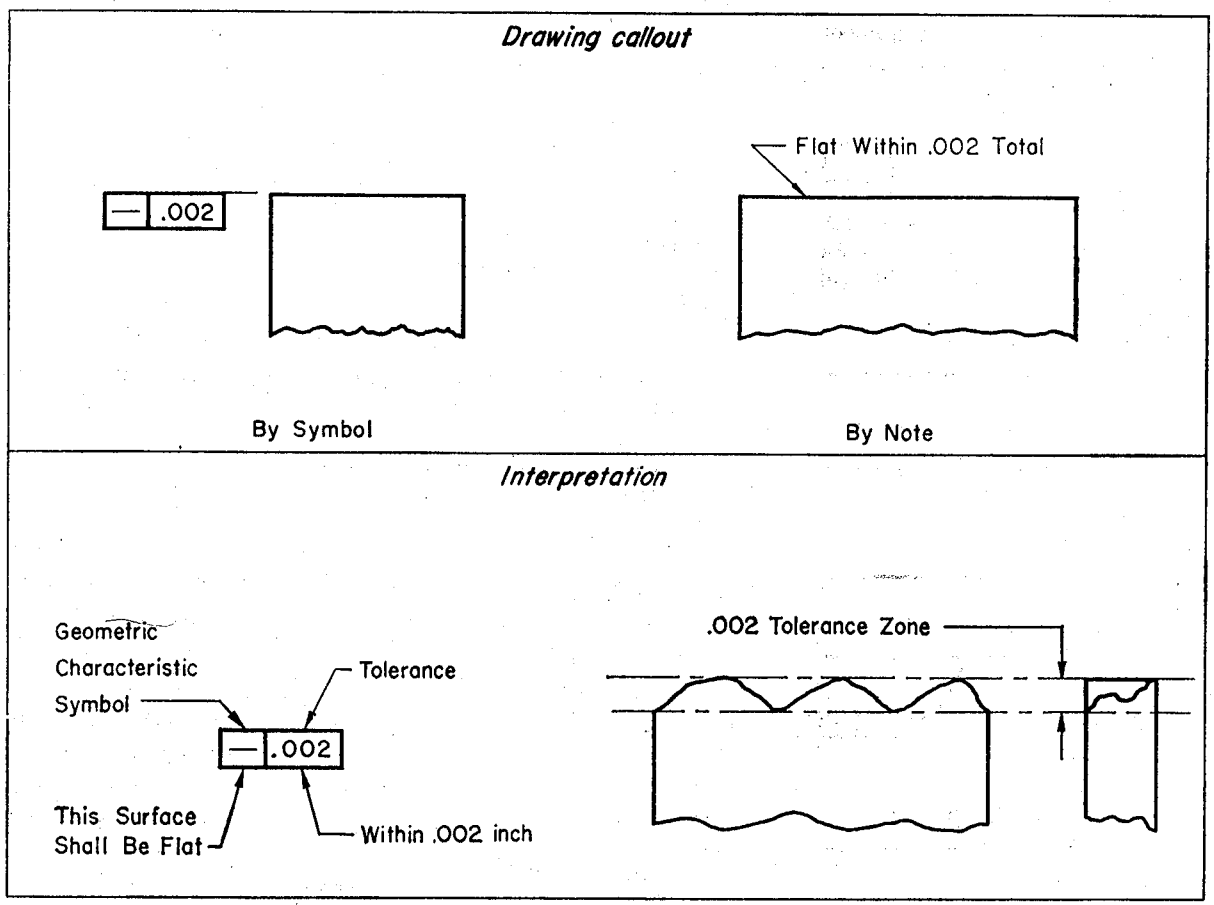


FIGURE 91. *Flatness.*

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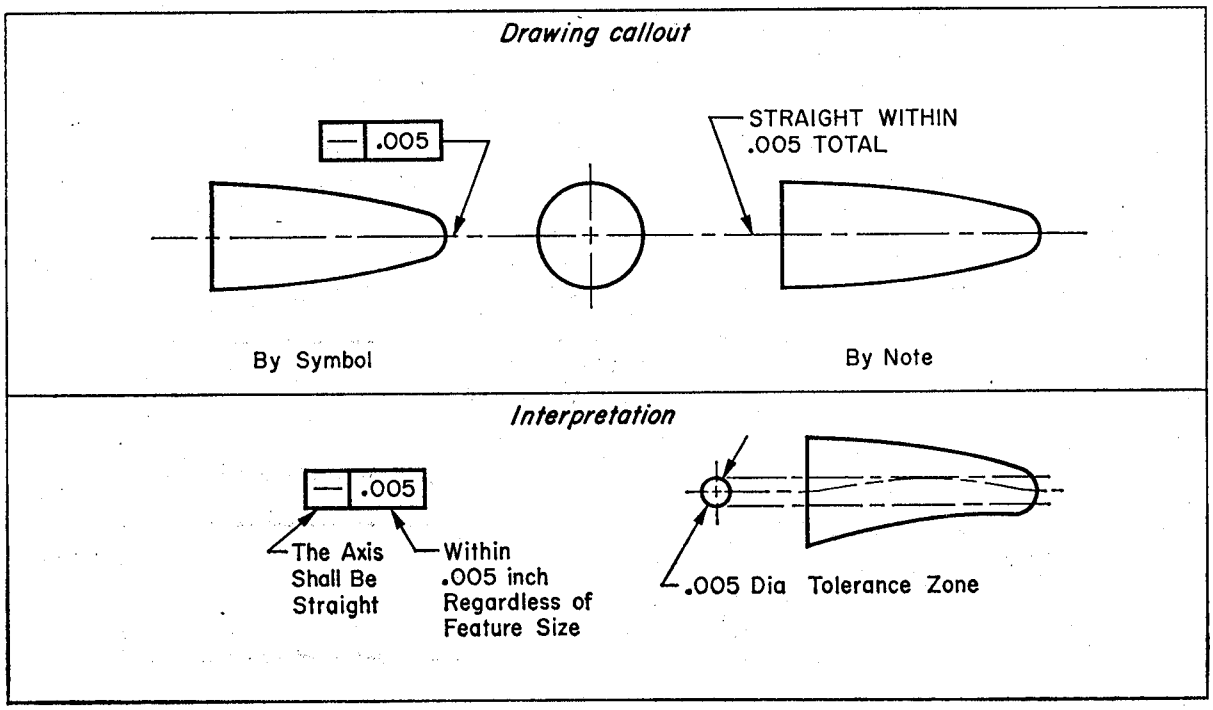


FIGURE 92. *Straightness of axis.*

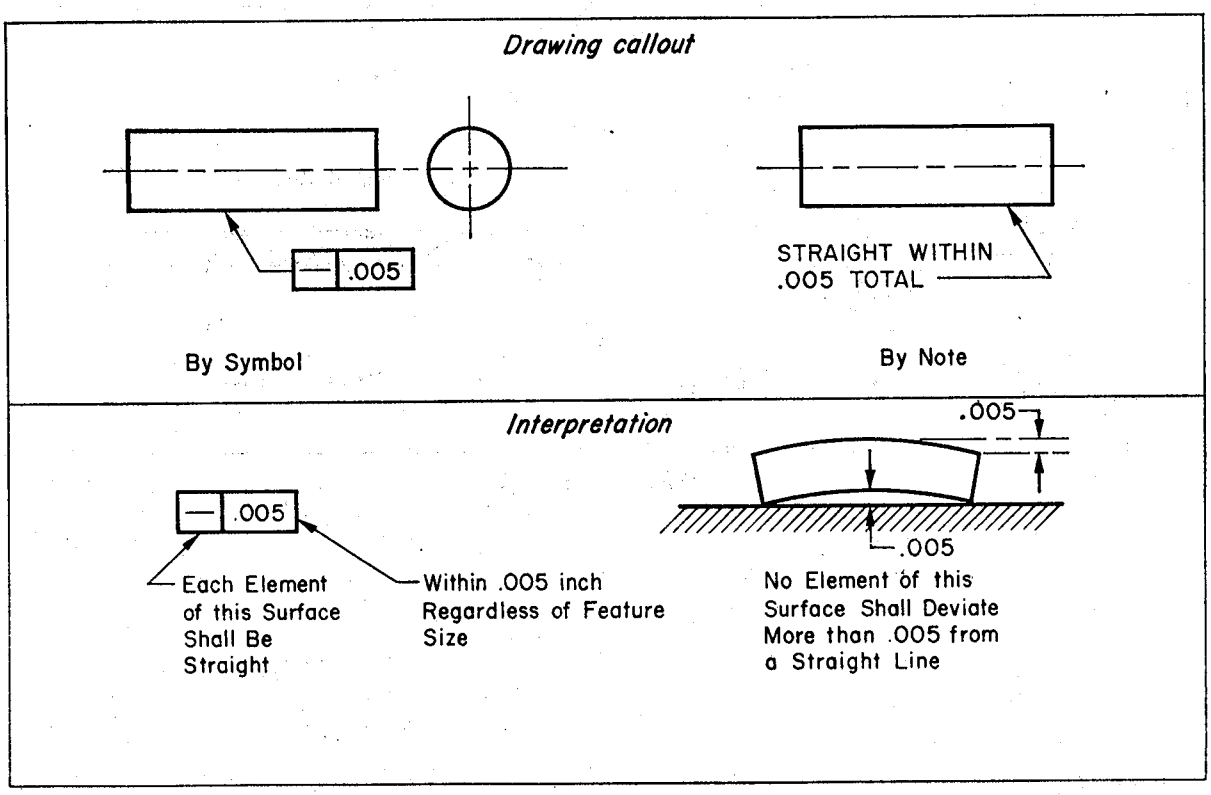


FIGURE 93. *Straightness of elements RFS.*

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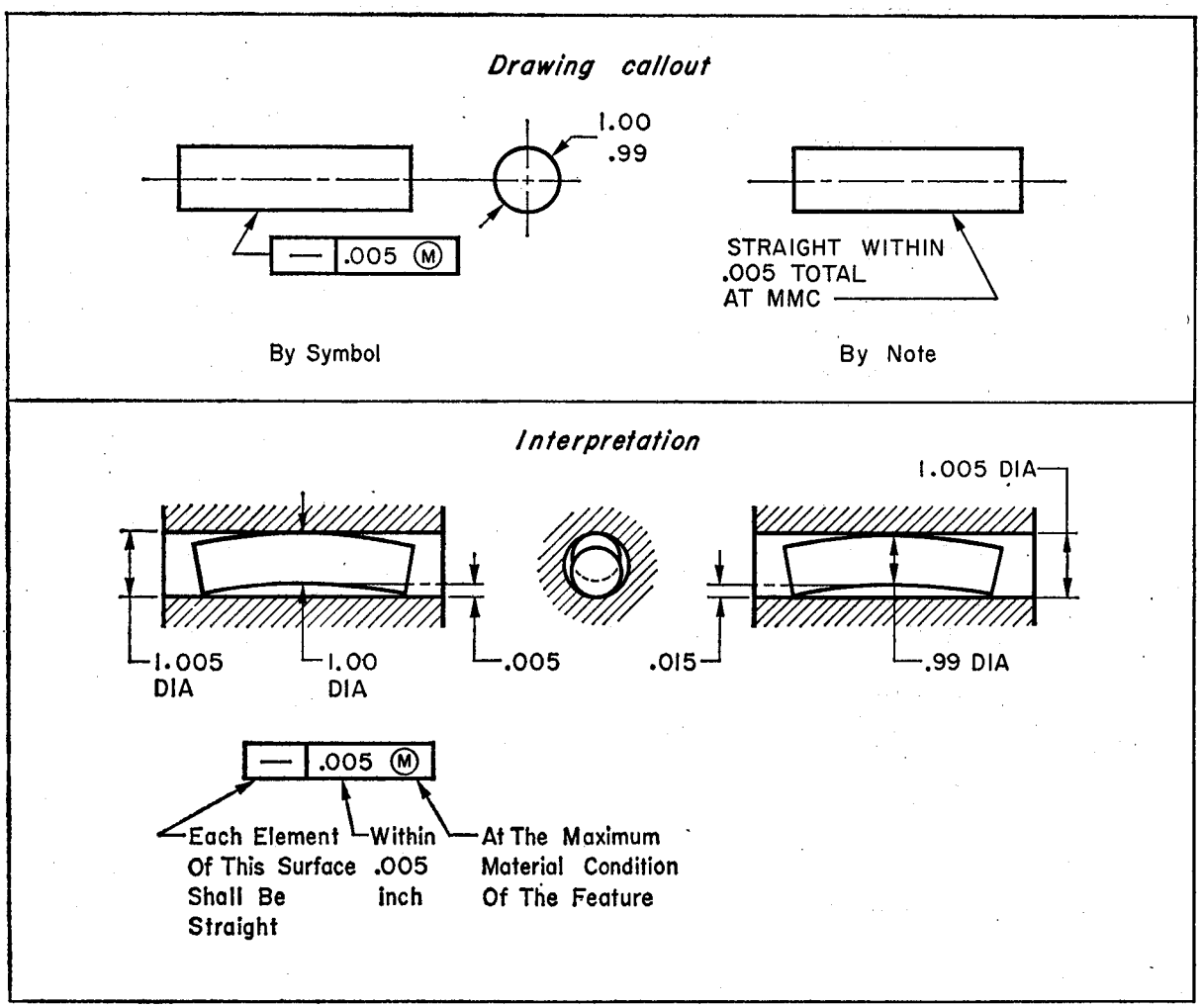


FIGURE 94. *Straightness of elements MMC.*

7.6.4.3 For the method of specifying the straightness of the elements of a cylinder or cone at its maximum material condition see figure 94. A leader shall run from the symbol or note to the pertinent element. The meaning of such a tolerance is that each element of the surface shall be straight within the tolerance when the cylinder is at its maximum material condition, and when the cylinder departs from MMC the straightness tolerance is increased by the difference between MMC and actual size.

7.6.5 *Angularity.* Angularity is the condition of a surface or axis which is at exactly the specified angle (other than 90°) from a datum plane or datum axis.

7.6.5.1 *Angularity tolerance.* An angularity tolerance is the distance between two parallel planes between which the entire surface so toleranced must lie, the planes being inclined from the datum by exactly the specified basic angle (see fig. 95). When angularity is expressed in relation to a horizontal or vertical plane through the axis of a solid of revolution (such as a cylinder) the tolerance zone is equally disposed about the logitudinal axis (see fig. 96).

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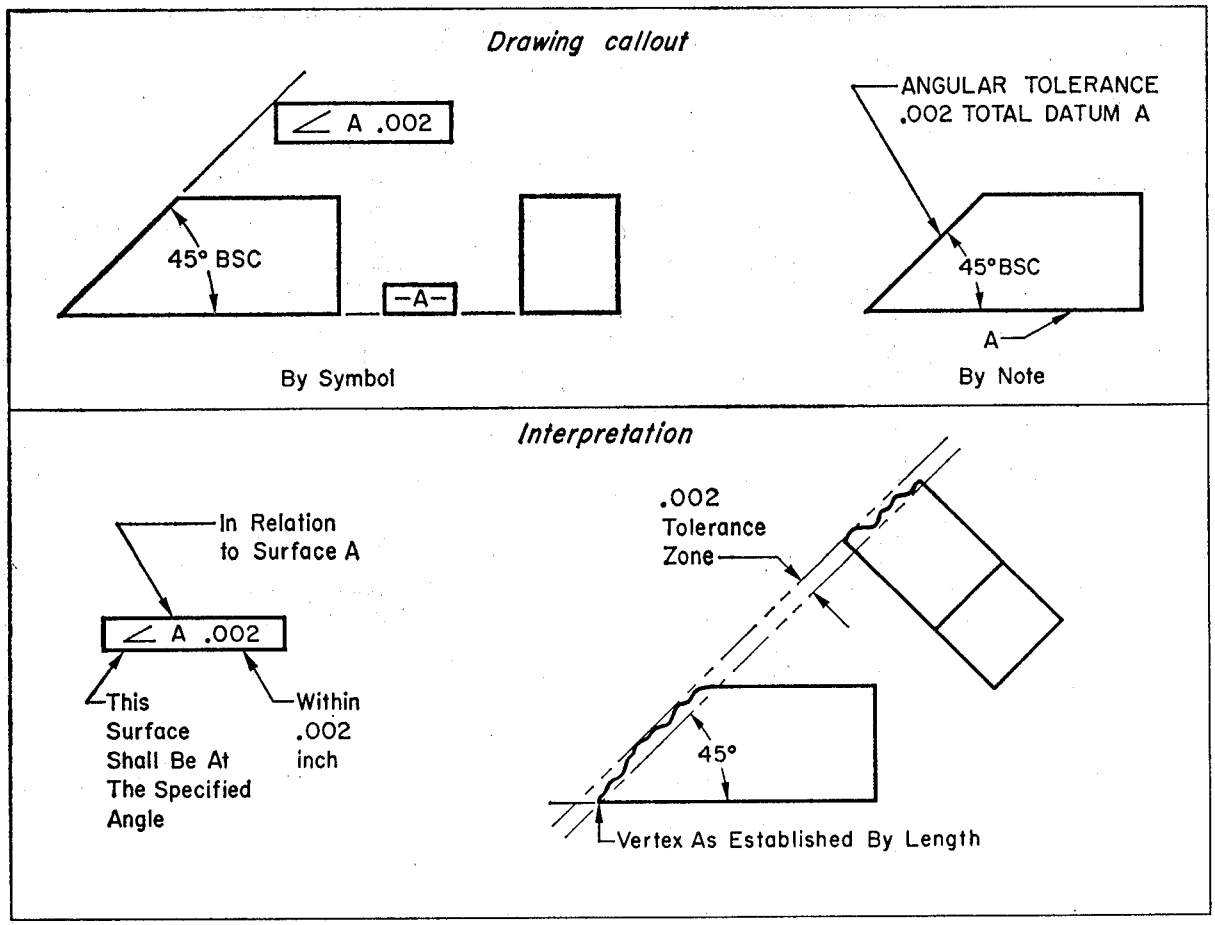


FIGURE 95. *Angularity.*

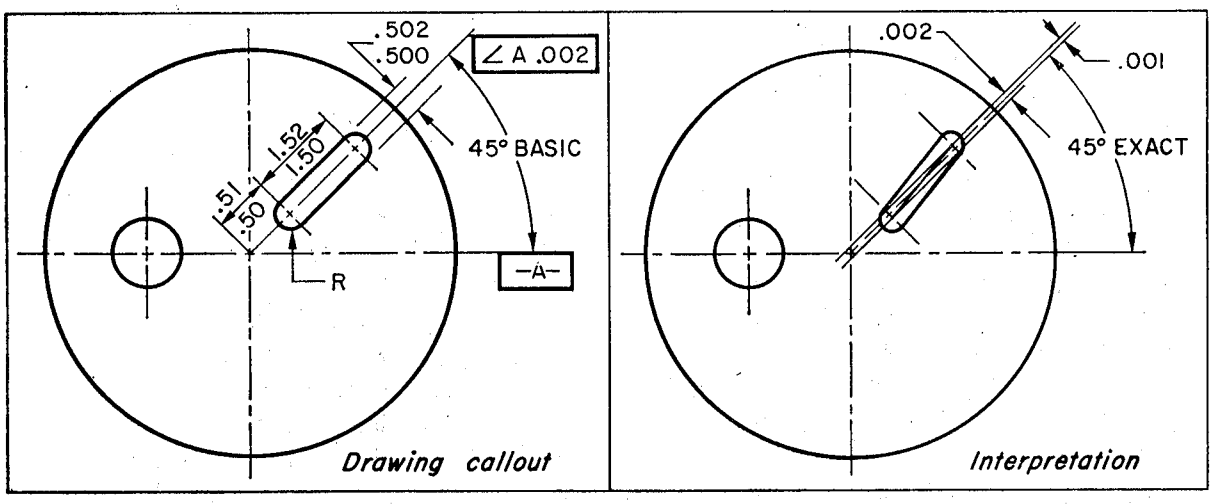


FIGURE 96. *Angularity tolerance for solids of revolution.*

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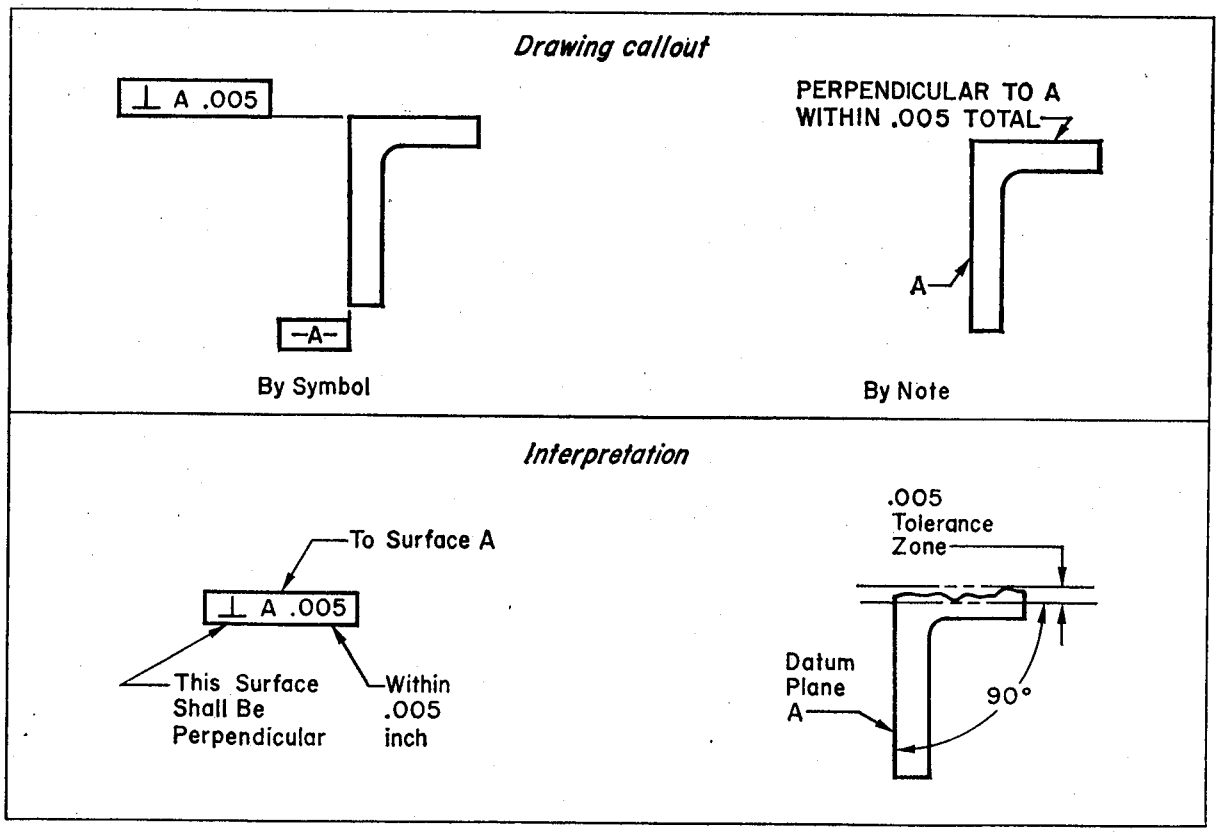


FIGURE 97. *Perpendicularity.*

7.6.6 *Perpendicularity.* Perpendicularity is the condition of a surface or axis which is at exactly 90° from a datum plane or datum axis.

7.6.6.1 *Perpendicularity tolerance.* A perpendicularity tolerance applicable to a plane surface is the distance between two parallel planes between which the entire surface so toleranced must lie, the planes being at exactly 90° from the datum plane (see fig. 97). If the feature is a cylinder (or other figure of revolution) and the datum is a plane, the perpendicularity tolerance is the diameter of the tolerance zone within which the axis of the cylinder must lie, the axis of the tolerance zone being at exactly 90° from the datum plane (see fig. 98). If both the feature and the datum are cylinders, the perpendicularity tolerance is the distance between two parallel planes between which the axis of the cylinder so toleranced must lie, the planes being at exactly 90° from the datum axis (see fig. 99).

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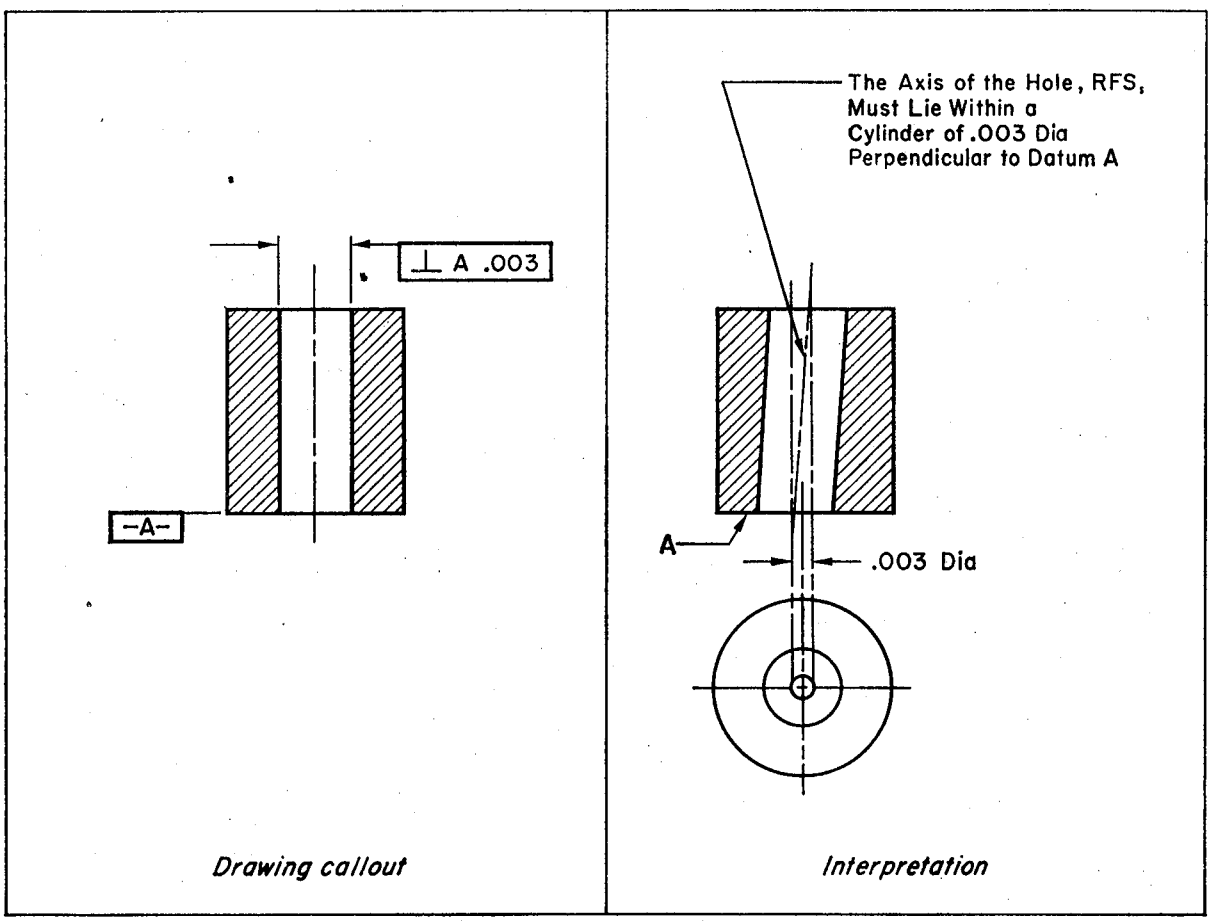


FIGURE 98. *Perpendicularity of cylinder to datum plane.*

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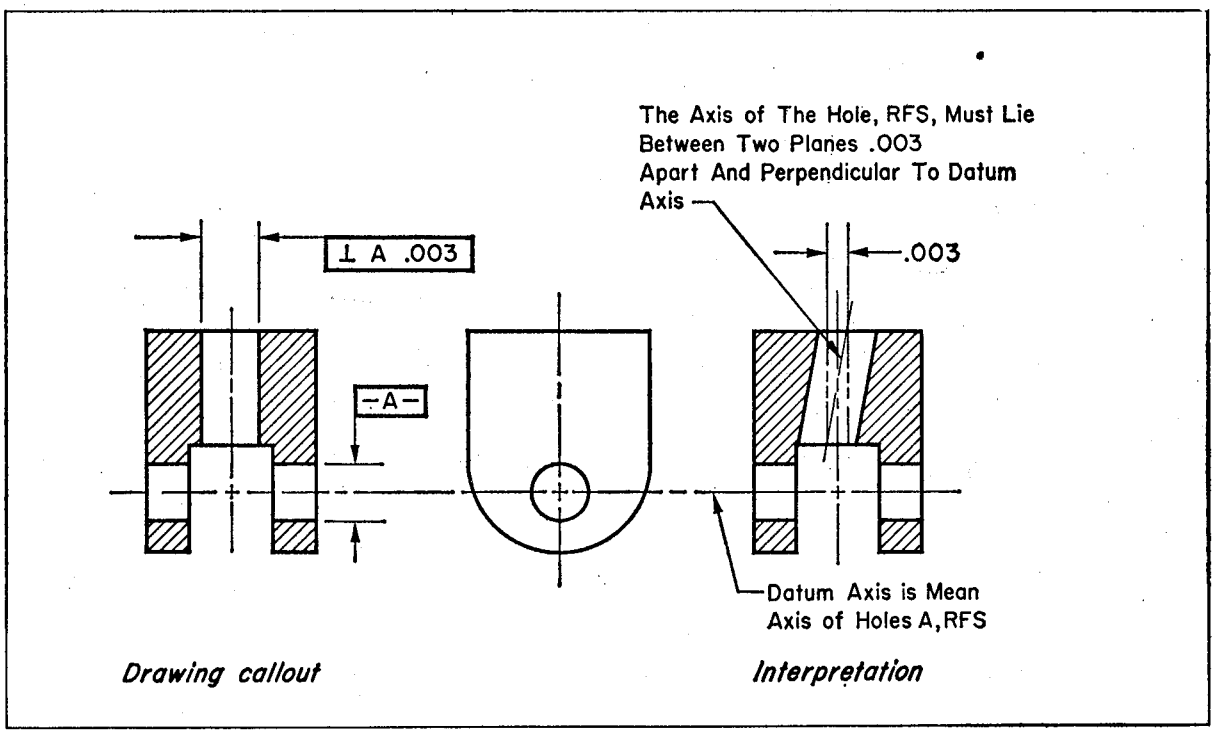


FIGURE 99. *Perpendicularity of cylinder to datum cylinder.*

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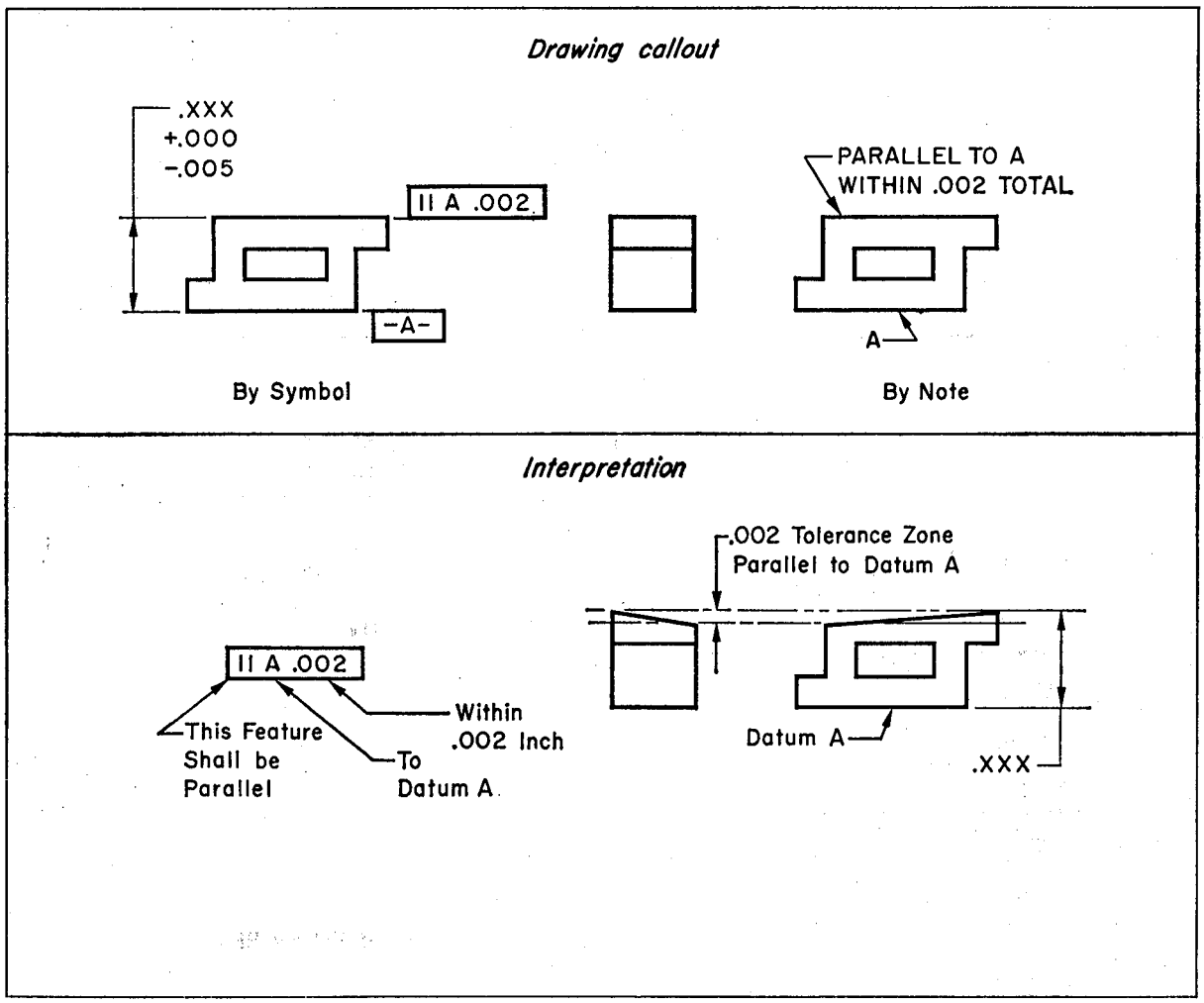


FIGURE 100. *Parallelism.*

7.6.7 *Parallelism.* Parallelism is the condition of a plane or axis which is equidistant at all points from a datum plane, or the condition of an axis which is equidistant at all points from a datum axis.

7.6.7.1 *Parallelism tolerance.* A parallelism tolerance applicable to a plane surface is the distance between two planes between which the surface so toleranced must lie, the planes being parallel with the datum plane. If the feature is a cylinder and the datum is a plane, the parallelism tolerance is the distance between two planes between which the axis of the cylinder must lie, the planes being parallel with the datum plane. If both the feature and datum are cylinders, the parallelism tolerance is the diameter of the tolerance zone within which the axis of the cylinder so toleranced must lie, the axis of the tolerance zone being parallel with the axis of the datum cylinder (see fig. 100).

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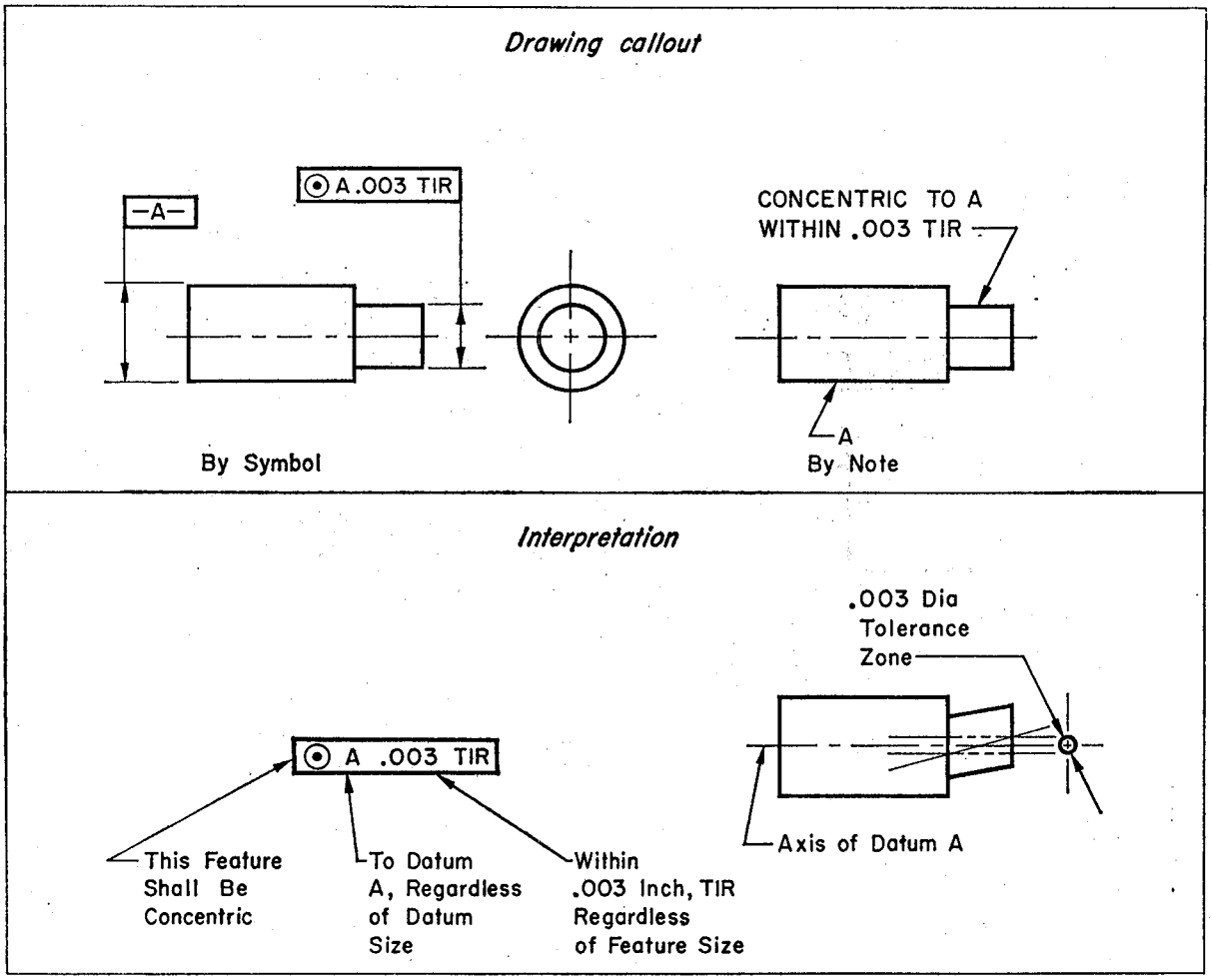


FIGURE 101. *Concentricity.*

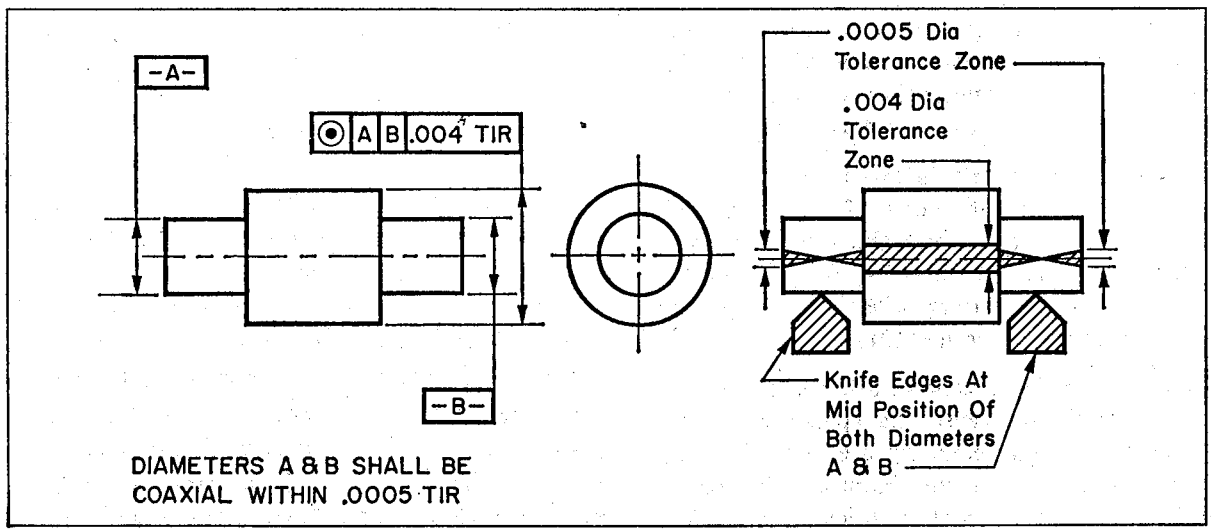


FIGURE 102. *Concentricity (Example 1).*

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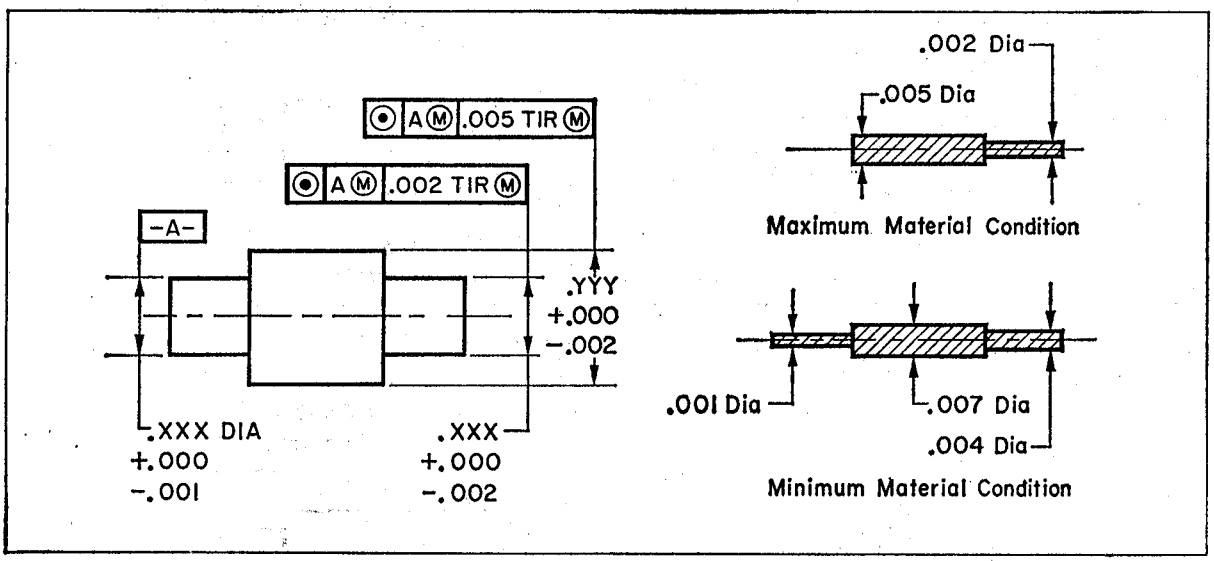


FIGURE 103. *Concentricity (Example 2).*

7.6.8 *Concentricity.* Concentricity is a condition in which two or more regular features (such as cylinders, cones, spheres, hexagons, etc.) in any combination have a common axis.

7.6.8.1 *Concentricity tolerance.* A concentricity tolerance is the diameter of the cylindrical tolerance zone within which the axis of the feature(s) so toleranced must lie, the axis of the tolerance zone coinciding with the axis of the datum feature(s), (see fig. 101).

7.6.8.2 It is important to select datums that are functional, otherwise the specification and measurement of concentricity are of little value. Figure 102 shows datum diameters A and B specified as coaxial, with a tolerance of .0005 total indicator reading (TIR) and the larger cylinder related to the common axis with a concentricity tolerance to both datums of .004TIR. It is assumed that diameters A and B are journals. Hence, it is a functional requirement that the concentricity of the larger cylinder be maintained with both datums A and B and that the coaxiality of A and B be closely toleranced. If the concentricity is specified only in relation to datum A, as illustrated by figure 103, a minor misalignment between datum A and the mean axis will result in a major difference in indicator reading. However, the method of specifying concentricity shown by figure 103 is proper if only diameter A is the bearing surface and the other diameters must be held concentric thereto for proper functioning.

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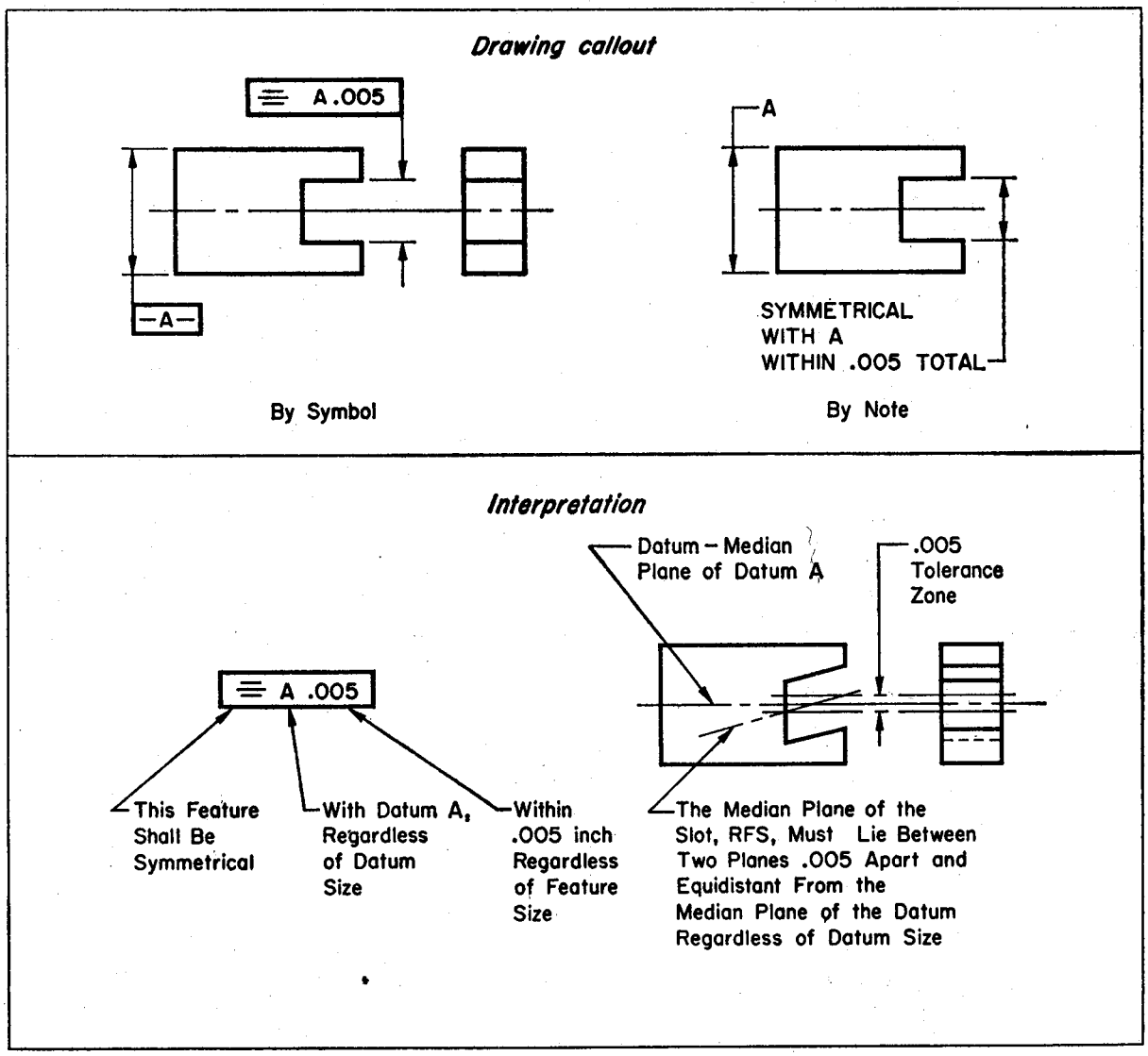


FIGURE 104. *Symmetry.*

7.6.9 *Symmetry.* Symmetry is a condition in which a part or feature has the same contour on opposite sides of a central plane.

7.6.9.1 *Symmetry tolerance.* The tolerance governing the symmetry of a feature with respect to a datum is the distance between two parallel planes between which the central plane of the feature must lie, the parallel planes being parallel to, and equally disposed about, the central plane, or axis, of the datum feature (see fig. 104).

7.6.10 *Roundness.* Roundness is the condition of a figure of revolution (such as a cylinder or cone) in which all points of the surface intersected by any plane normal to the axis are equidistant from the axis.

7.6.10.1 *Roundness tolerance.* The tolerance governing the roundness of a feature is either the difference in the diameters of two concentric circles (in planes normal to the axis) between which the surface so toleranced must lie (diametral method), or the width of the annular zone

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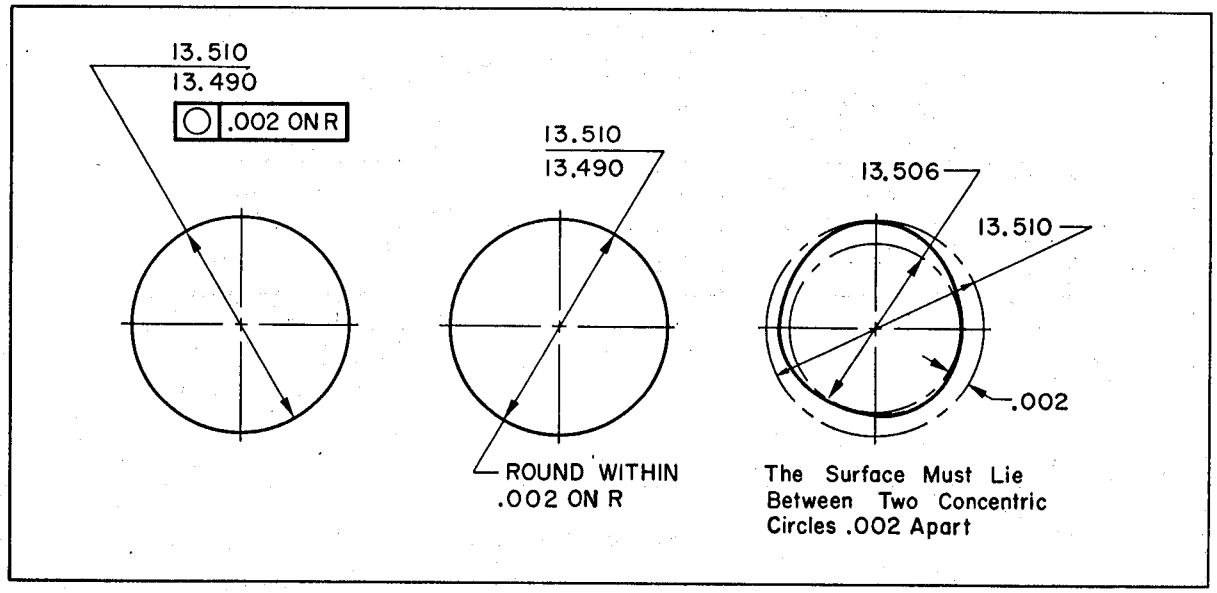


FIGURE 105. *Roundness of rigid parts.*

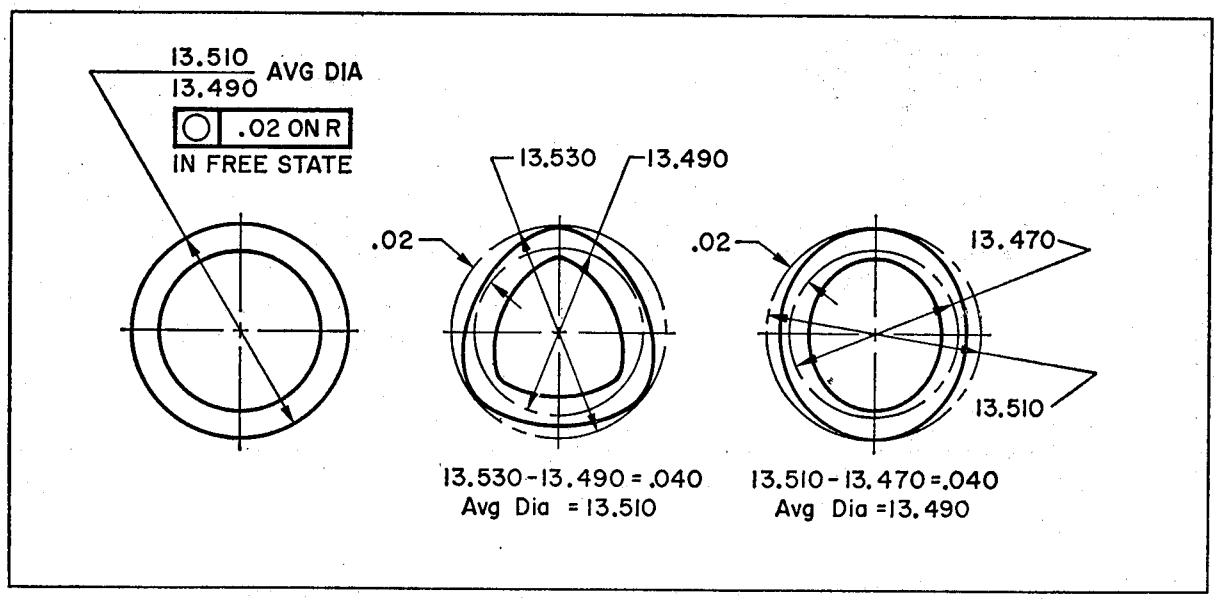


FIGURE 106. *Roundness of non-rigid parts (free-state variation).*

between such concentric circles (radial method) (see last sentence in 7.3.5). Methods of specifying roundness are illustrated in figure 105. If it is desired that roundness be measured by means of a Vee block, the angle of the V should be specified. When a Vee block is used for checking roundness, the number and arrangement of lobes on the surface and the angle of the Vee will have an appreciable influence on the reading. When roundness is checked by rotating the part about its nominal center, the indicator reading will be the out-of-roundness on the radius.

7.6.10.2 The concept of free-state variation as discussed in 7.6.11 affects the method of specifying a roundness tolerance for nonrigid parts (see fig. 106).

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7.6.11 *Free-state variation.* Free-state variation is the amount a part distorts after removal of external forces applied during manufacturing. For parts consisting essentially of shells with a thin wall thickness in proportion to the diameter, geometric tolerances (such as roundness and concentricity) cannot be properly applied without controlling free-state variation.

7.6.11.1 *Regidity.* A part shall be considered rigid (not subject to free-state variation) under the following conditions:

(a) When it does not distort due to its own weight or flexibility;

(b) When forces greater than those that can be exerted employing the expected method of assembly are required to return the part to drawing tolerance.

7.6.11.2 *Allowable free-state variation.* Variations in the free state can exist in two ways:

(a) Distortion due to the weight or flexibility of the part;

(b) Distortion due to internal stresses set up in fabricating.

These variations to be allowable must be within an elastic range that will allow the part to be brought within drawing tolerance by forces equivalent to those that can be exerted by employing the expected method of assembly. The word RESTRAINED in a drawing note stating an allowance for free-state variation shall be interpreted to mean restraint within the limitations of the preceding sentence when the note does not specify the nature of the restraint.

7.6.11.3 *Control of free-state variation.* Free-state variation shall be controlled as follows (fig. 107):

(a) Determine if the part is rigid;

(b) If rigid, dimensions and geometric tolerances shall be stated without reference to free-state variation;

(c) If the part is not rigid, free-state variation shall be limited in the following manner:

1. Select and identify the features (rabbet diameter, bosses, flanges, etc.) to be used as datum surfaces. Since these surfaces may vary from drawing tolerance when they are in the free state, it is necessary to specify either their maximum allowable free-state variation, or the maximum force necessary to restrain each of them to drawing tolerance.
2. Determine the amount of the restraining and holding forces necessary to simulate expected assembly conditions. Specify on the drawing that when restrained to this condition, the remainder of the part, or certain features thereof, shall be within stated tolerances.

(d) It should be noted that if the dimensions and tolerances are met in the free state it is not necessary to restrain the part.

7.6.11.4 *Average diameter.* An average diameter is the mean of several diameters (not less than four) across a circular or spherical part. It is used to determine conformance to diameter tolerance only. When practicable, the average diameter may be determined by using a periphery tape. Only when a diameter is allowed a maximum roundness tolerance in the free state should it be labeled "AVG DIA". Free-state variation for simple parts may be controlled by specifying the roundness tolerance (by symbol or note) and by labeling the diameter "AVG DIA" (see fig. 106).

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DIA X SHALL BE ROUND WITHIN .10 ON DIA IN FREE STATE.
WHEN DIA X IS RESTRAINED TO DRAWING TOLERANCE
AND SURFACE Y IS HELD FLAT WITH THE EQUIVALENT OF
80 TO 130 IN. POUND TORQUE AT EACH OF 64 BOLT HOLES,
DIA Z SHALL BE CONCENTRIC TO DIA X WITHIN .06.
DIA Z SHALL BE ROUND WITHIN .04 ON DIA AND ALL
BOSSES SHALL BE WITHIN DRAWING TOLERANCE.

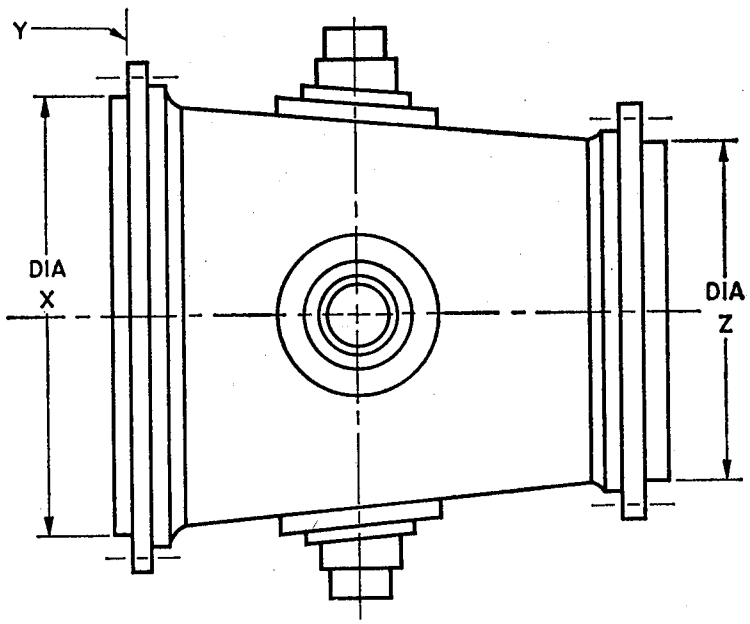


FIGURE 107. *Free-state variation.*

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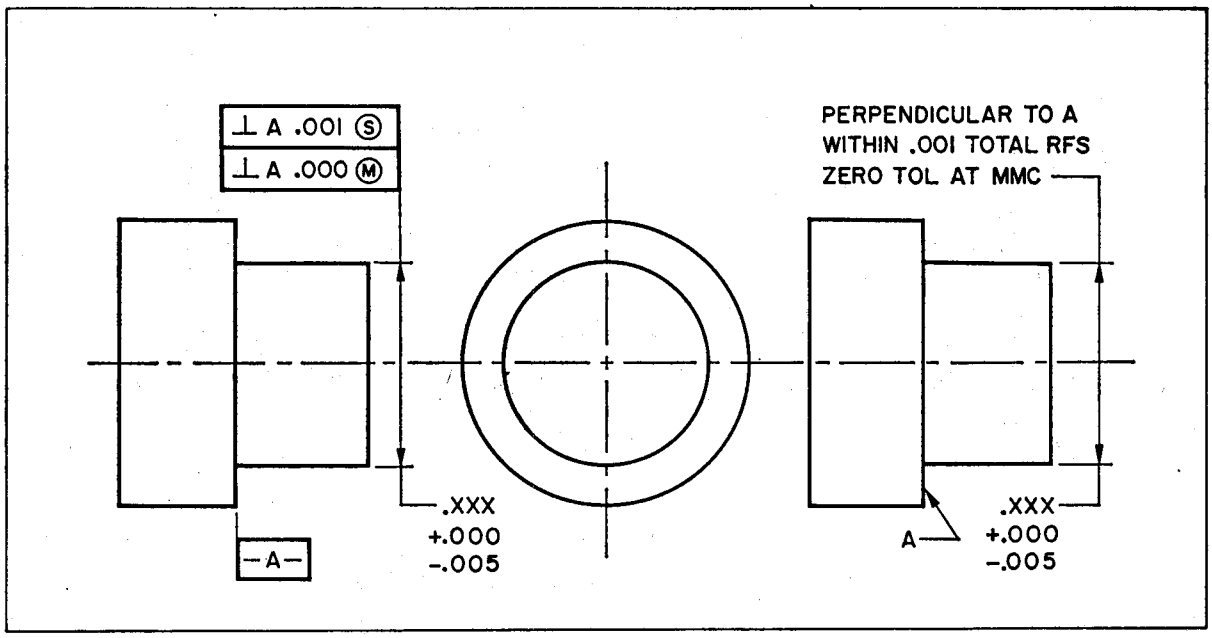


FIGURE 108. *Perfect form at MMC (Example 1).*

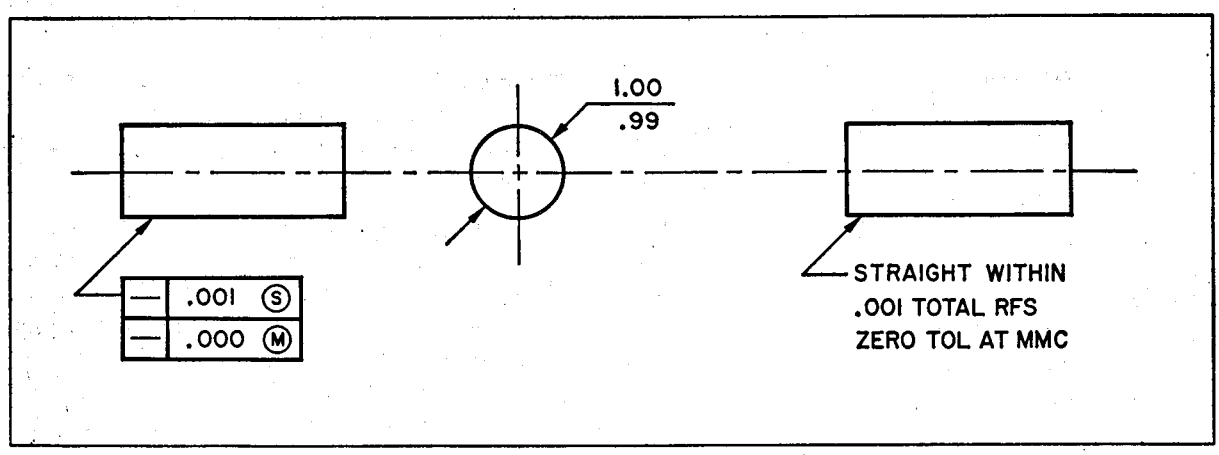


FIGURE 109. *Perfect form at MMC (Example 2).*

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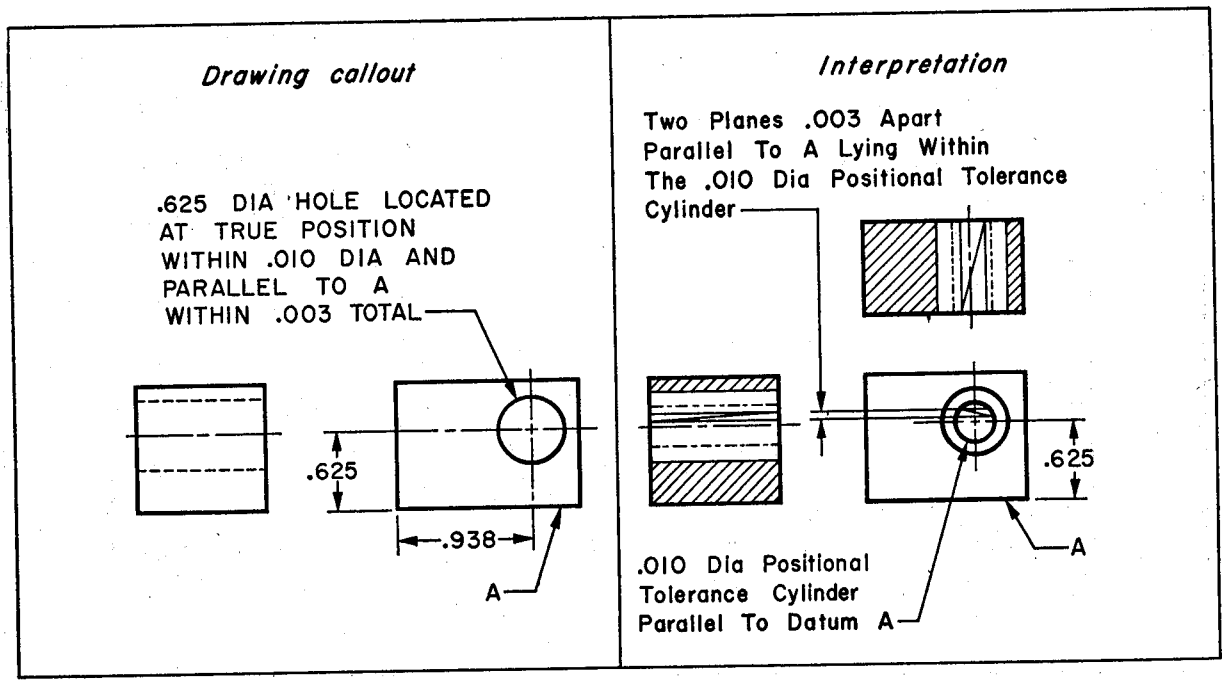


FIGURE 110. *Specifying that a hole must be parallel with a flat surface.*

7.6.12 *Perfect form at MMC.* As explained in 7.4, certain geometric tolerances, whether applied at maximum material condition or regardless of feature size, have an effect upon the virtual size of a part. When it is desired to maintain a profile of perfect form at MMC so that the virtual size can be no greater than that at MMC, yet it also is desired to specify a geometric tolerance to control the form as the part departs from MMC, this may be achieved in the manner illustrated by figures 108 and 109. Whereas in figures 93 and 94 the virtual size of the pin is 1.005, in figure 109 the virtual size is 1.000. The meaning of the tolerancing symbol in figure 109 is:

(a) The tolerance on straightness at MMC is zero.

(b) Between MMC and a diameter .001 less, the tolerance on straightness is equal to the difference between MMC and the actual diameter.

(c) From diameters .001 below MMC to .010 below, the tolerance on straightness remains constant at .001.

The symbol, or note, of figure 108 requires that if diameter .XXX is at its maximum material condition it must be exactly perpendicular to datum A.

7.6.13 *Relation of positional tolerances to other tolerances of form.* Figure 110 illustrates an expression for specifying that a hole must be held parallel with a flat surface more closely than its positional tolerance alone would indicate. The meaning of the combined expression is given by the "interpretation" portion of figure 110. In a similar manner, figure 111 illustrates expressions to require that a hole be located from and parallel with another hole more closely than the positional tolerance alone indicates. The "interpretation" portion of figure 111 illustrates the intended meaning.

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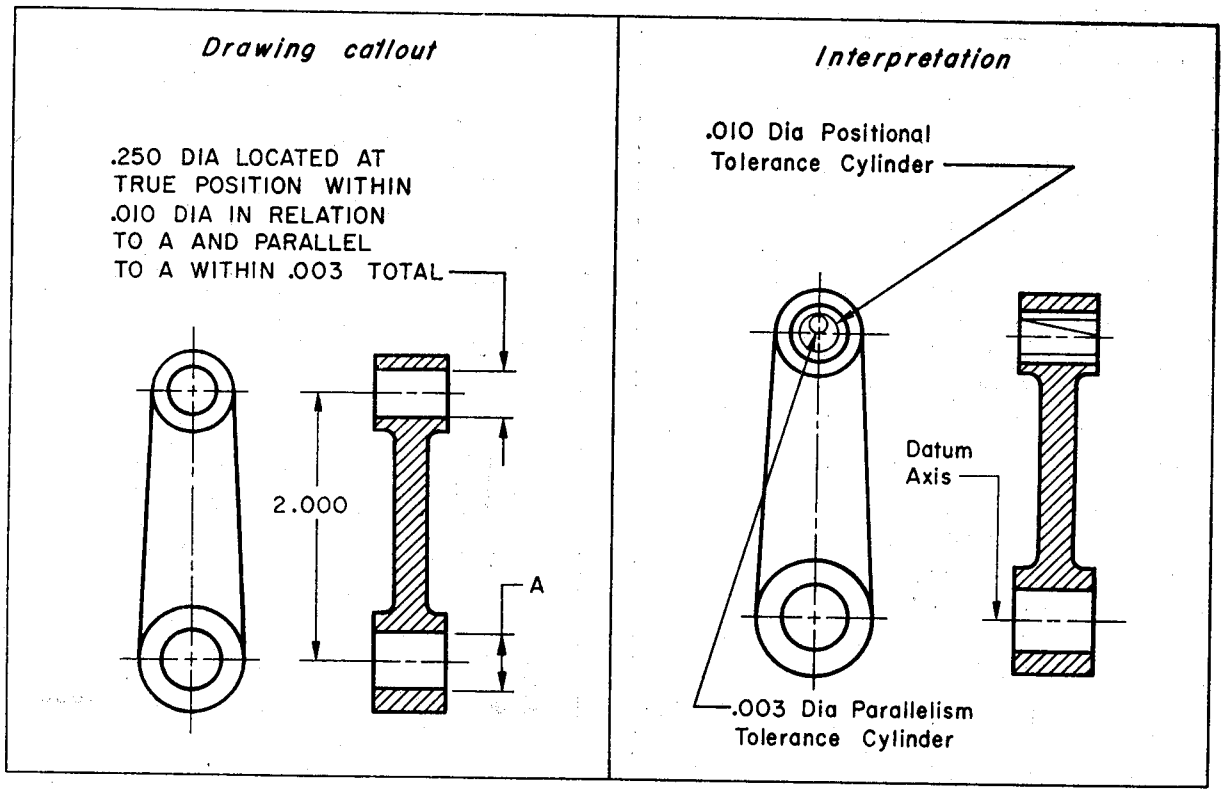


FIGURE 111. *Specifying that a hole must be located from and is parallel to another hole*

7.6.14 *"Zone" tolerances for contours.* A "zone" tolerance may be given where a uniform amount of variation can be permitted along a contour. The drawing is constructed to show the desired contour fully defined by dimensions without tolerances. Information must be included specifying that the general tolerances do not apply to the dimensions establishing the desired contour. At a conspicuous place along the contour, one or two phantom lines are drawn with dimension lines and arrowheads to indicate the location of the tolerance zone. The value of the tolerance is given by note (see figs. 112a and 112b). (For drawing clarity, the distance between the contour and the phantom lines may be greater than its scale value). If some limits on a drawing are expressed by a zone tolerance and others by a general tolerance, the extent of the zone tolerance must be clearly indicated, as in figure 112b. The "zone" may be equally disposed about the desired contour (see fig. 112c).

7.6.15 *Meaning if no tolerance of form is specified.* When tolerances of form are not specified on a part drawing, the toleranced dimensions of size effect a partial control of the form of individual features. The drawing shall be interpreted to require the following control of the form of simple or regular features (cylindrical surfaces, widths, lengths, or thicknesses):

(a) No portion of each such individual feature shall cross the boundary of perfect form at maximum material condition.

(b) Errors of form are permitted within that boundary provided the measured dimensions of the feature at all cross sections and in any appropriate direction are not less than the low limit of size of an external feature or greater than the high limit of size of an internal feature. Figure 113 illus-

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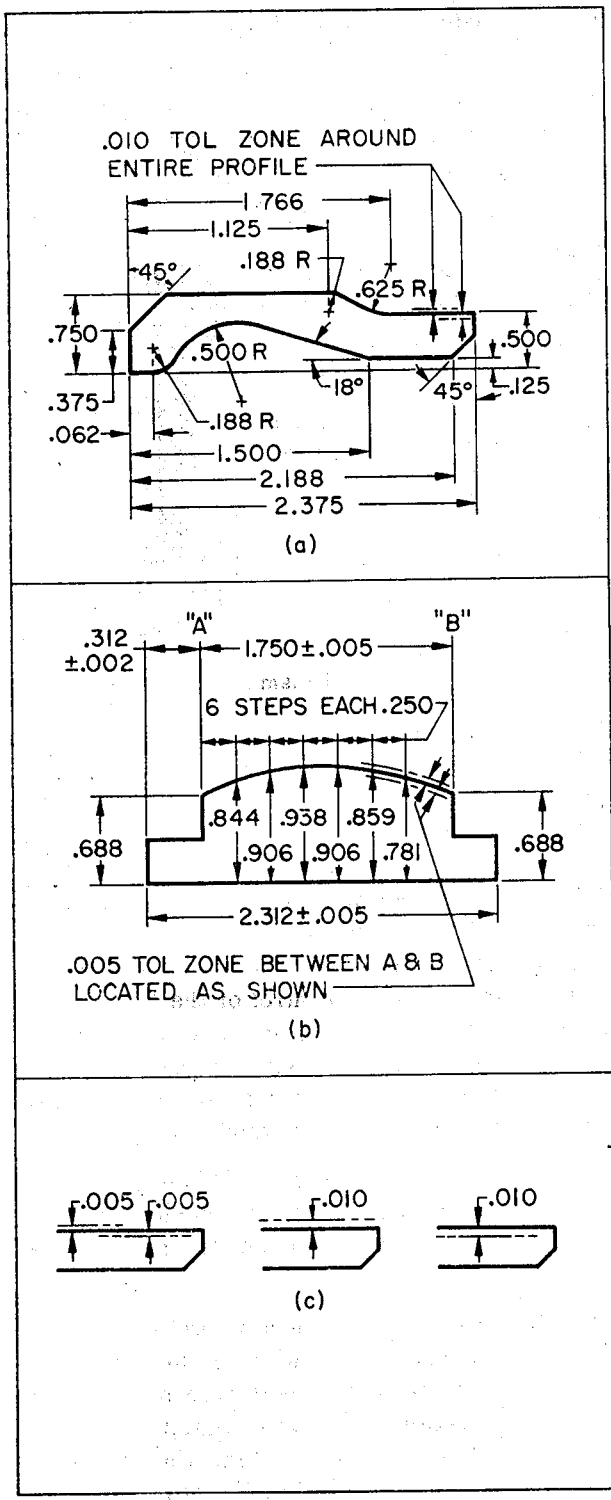


FIGURE 112. *Methods of zone tolerancing.*

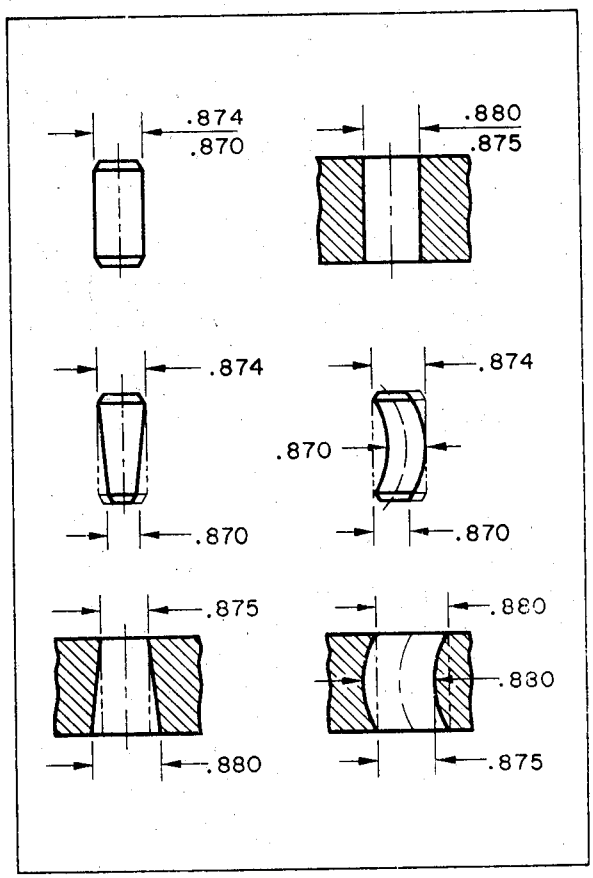


FIGURE 113. *Tolerance zone or boundary within which form may vary when no tolerance of form is given.*

trates typical extreme errors of form that would be acceptable under this interpretation.

Note: The above stated interpretation prescribing a boundary of perfect form at "MMC" does not apply to the complete envelope of a part, but only to individual features. The interpretation also does not apply to fabricated materials, such as bars, sheets, tubing, etc. There, established industry standards prescribe straightness, flatness, and other conditions. When tolerances of form are not given on drawings of parts made from these materials, industry standards for the materials govern the surfaces that are not machined.

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8. NOTES

8.1 Provisions of this standard are the subject of international standardization agreements, as indicated by ABC-ARMY-STD-65 "Dimensioning and Tolerancing" (ILS 9.2.5) and ABC-Navy-STD-29, "Basic Engineering Drawing Practices." When amendment, revision, or cancellation of this standard is proposed, the departmental custodians will inform their respective Departmental Standardization Office (DepSO) so that appropriate action may be taken respecting the international agreement concerned.

8.2 Parts of this document have been taken from the American Standards Association, Standard Y14.5, with permission of the Association.

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(Copies of this standard for military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.)

(Copies of this standard may be obtained for other than official use by individuals, firms, and contractors from the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D. C.)

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Custodians:

Army—Munitions Command
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Air Force

International interest (see sec. 8.1)

Preparing activity:

Navy—Bureau of Naval Weapons

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APPENDIX

10. TRUE POSITION. This section supplements the information on true position tolerancing contained in paragraph 7.5. Further comparisons are given with other methods of tolerancing.

10.1 When features are located by means of rectangular coordinates or angular dimensioning, each dimension having its own tolerances, square, rectangular or wedge-shaped tolerance zones result (see figs. 82 and 83). Figure 10–1 is an example of such direct tolerancing, in which allowable variation in position is specified in the direction indicated by the dimension lines. The resultant tolerance zone is the .010 square shown in figure 10–2. Since the actual position of the feature may be anywhere within the square, the maximum allowable variation from the desired (or nominal) position occurs at 45° from the horizontal or vertical direction in which the tolerances are indicated. As illustrated by figure 10–2, the allowable variation on the diagonal is 1.4 times the specified tolerance.

10.2 In true position tolerancing of the location of cylindrical features the tolerance zone also is cylindrical. The diameter of this cylinder may be 1.4 times the tolerance that would be used in direct tolerancing of rectangular coordinate dimensions without any increase in the maximum allowable variation, as explained in paragraph 10.1. Thus, the permissible tolerance in all directions other than at 45° from the horizontal is increased without any detrimental effect upon the accuracy of location of the features. Such increase in tolerance often results in some reduction in cost of manufacture.

10.3 True position tolerancing often is applied to groups or patterns of holes in two mating parts. For complicated patterns, true position tolerancing is subject to more uniform interpretation than direct tolerancing by rectangular coordinates or angular dimensioning. Also the permissible inaccuracies are easier to analyze in the case of true position tolerancing; hence, there is greater assurance that mating parts will assemble without interference.

10.3.1 Figure 10–3 is an example of a 4-hole pattern located by rectangular coordinate dimensions with directly applied tolerances. Figures 10–4a, 10–4b, and 10–4c are three tolerance patterns that are acceptable interpretations of the tolerances set forth in figure 10–3. Other variations of these tolerance zones also would meet the requirements of figure 10–3. Since so many interpretations are possible, analysis of the probability of assembly of mating patterns is difficult.

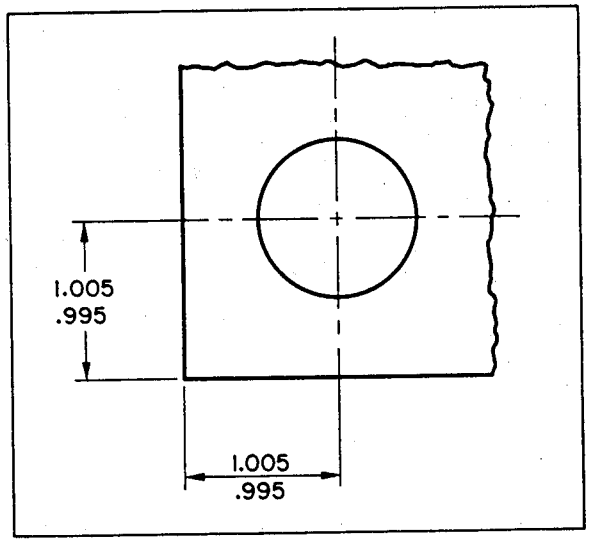


FIGURE 10–1. *True position.*

10.3.2 The design intent often can be expressed more simply and precisely if locations are given as true positions with tolerances to state how far actual positions may be displaced from true positions. Figure 10–5 shows the same pattern of holes as figure 10–3, but with a .007 dia true position tolerance substituted for the comparable ±.005 direct tolerance on the dimensions between hole centers. Figure 10–6 shows the tolerance zones corresponding with figure 10–5. Although the location of the four tolerance cylinders as a group may vary from that shown, only the one interpretation is possible as to the size of the tolerance cylinders and the relation-

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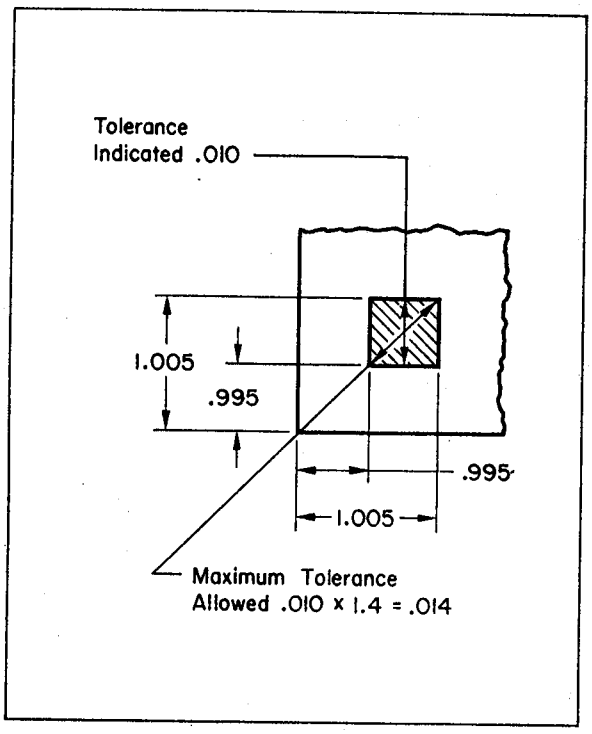


FIGURE 10–2. *True position.*

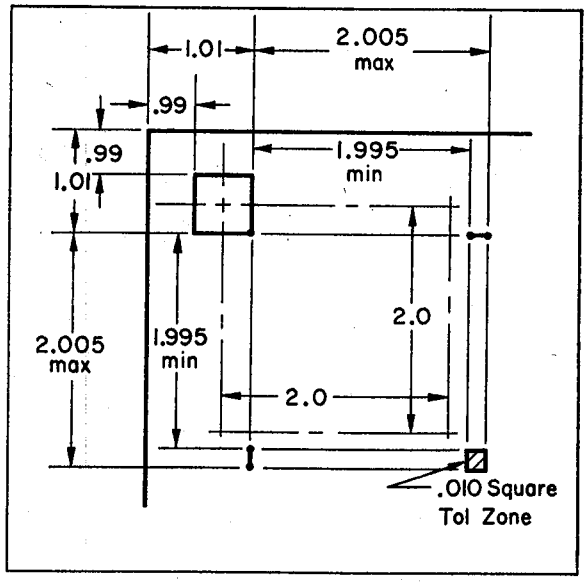


FIGURE 10–4a. *True position.*

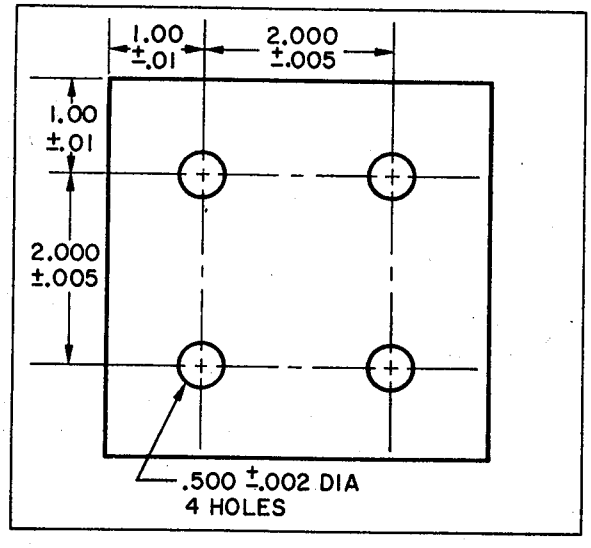


FIGURE 10–3. *True position.*

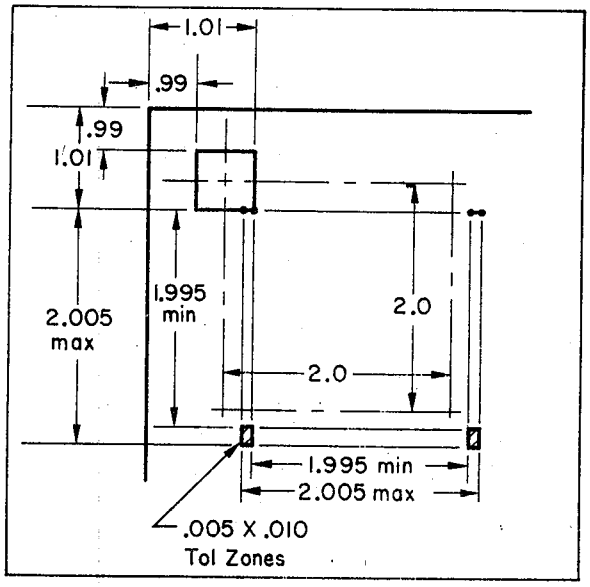


FIGURE 10–4b. *True position.*

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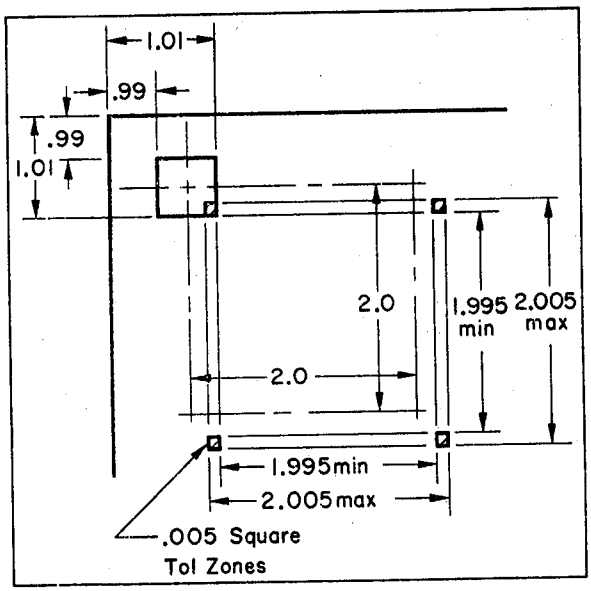


FIGURE 10–4c. *True position.*

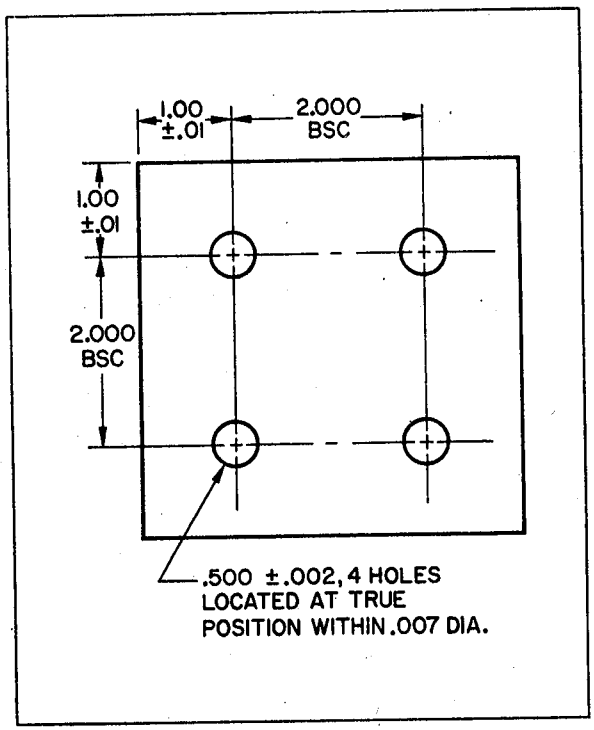


FIGURE 10–5. *True position.*

ship between their axes. Analysis of mating patterns obviously is simplified by the existence of only one permissible interpretation.

10.3.3 An example of angular dimensioning with direct tolerancing is shown by figure 10–7. The angular dimensions are specified from a slot and the holes are located on a toleranced bolt circle. The resultant tolerance zones are shown in figure 10–8. The difficulty of analyzing the probability of assembly of bolts, studs, or dowels with the holes in two such mating parts is obvious, since the tolerance zones are an altogether different shape and are expressed in different units of measure as compared with the cylindrical features to which they apply.

10.4 In true position dimensioning, the basic location of the feature concerned is given by the use of untoleranced locating dimensions in any suitable form, and a note specifying the positional tolerance for that feature. A complete true position expression consists of two, or three, parts. First, the basic locating dimensions are given with some indication that general drawing tolerances do not apply (see 7.5.5). Second, expressions such as "LOCATED AT TRUE POSITION WITHIN .XXX DIA" are added to the note concerning the feature to be located (see

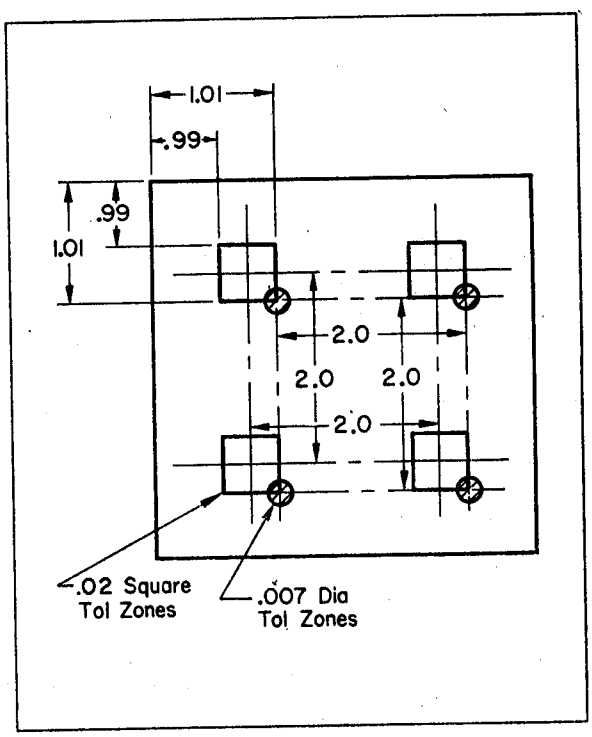


FIGURE 10–6. *True position.*

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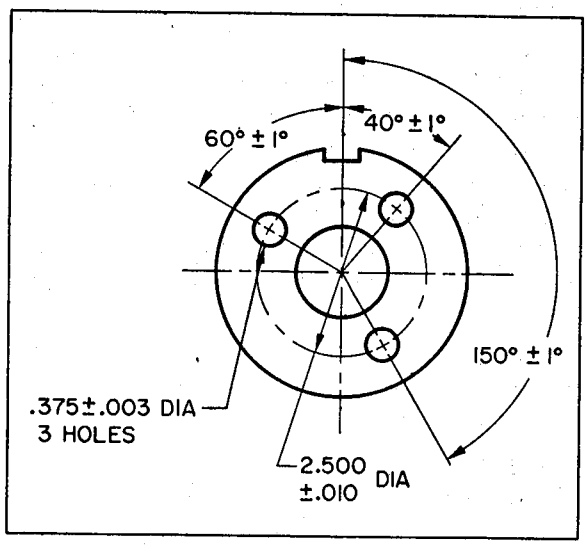


FIGURE 10–7. *True position.*

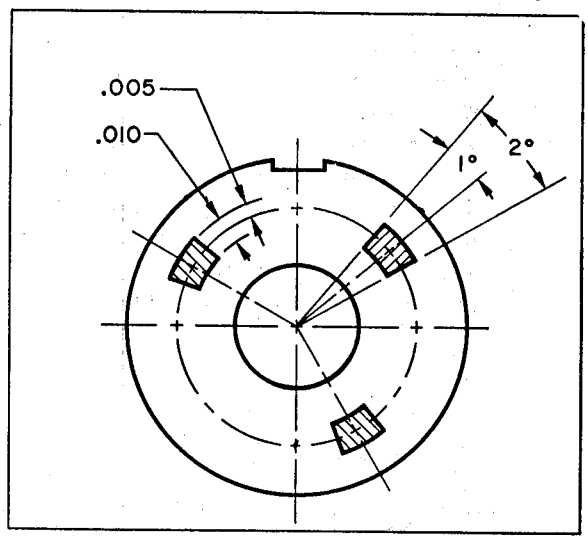


FIGURE 10–8. *True position.*

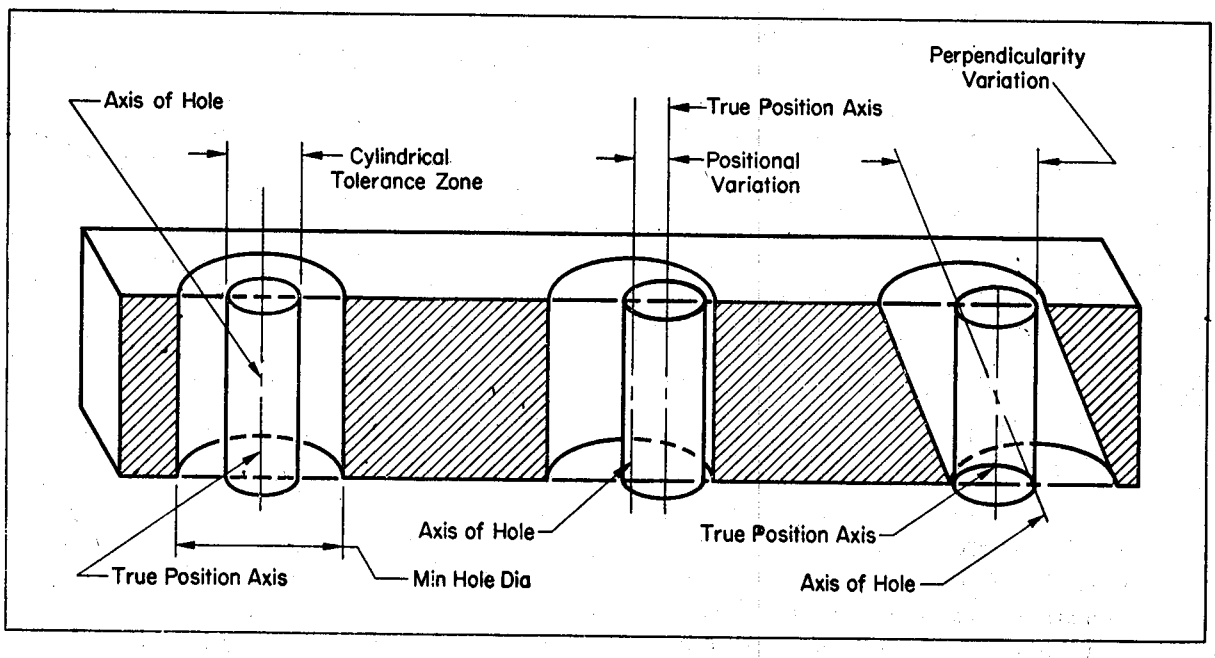


FIGURE 10–9. *True position.*

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fig. 88). And third, if necessary, a phrase identifying the datum is added.

10.5 Figure 10–9 illustrates the relationship of the axis of an actual hole and the true position tolerance zone. In terms of the axis of the hole, it means that the axis of each hole at all sections along its length must fall within the specified cylindrical tolerance zone whose center is located at true position. It is important to note that this tolerance zone also defines the limits within which variations in the perpendicularity of the axis of the hole must be confined with respect to the surface.

10.6 Separate patterns of holes on a part which are related by common basic dimensions are considered to be a single pattern. If it is desired to gage the patterns separately, it should be so indicated by a note on the drawing (see fig. 10–10). Separate hole patterns on a part which are separated by rectangular coordinates are considered to be separate patterns (see fig. 90a).

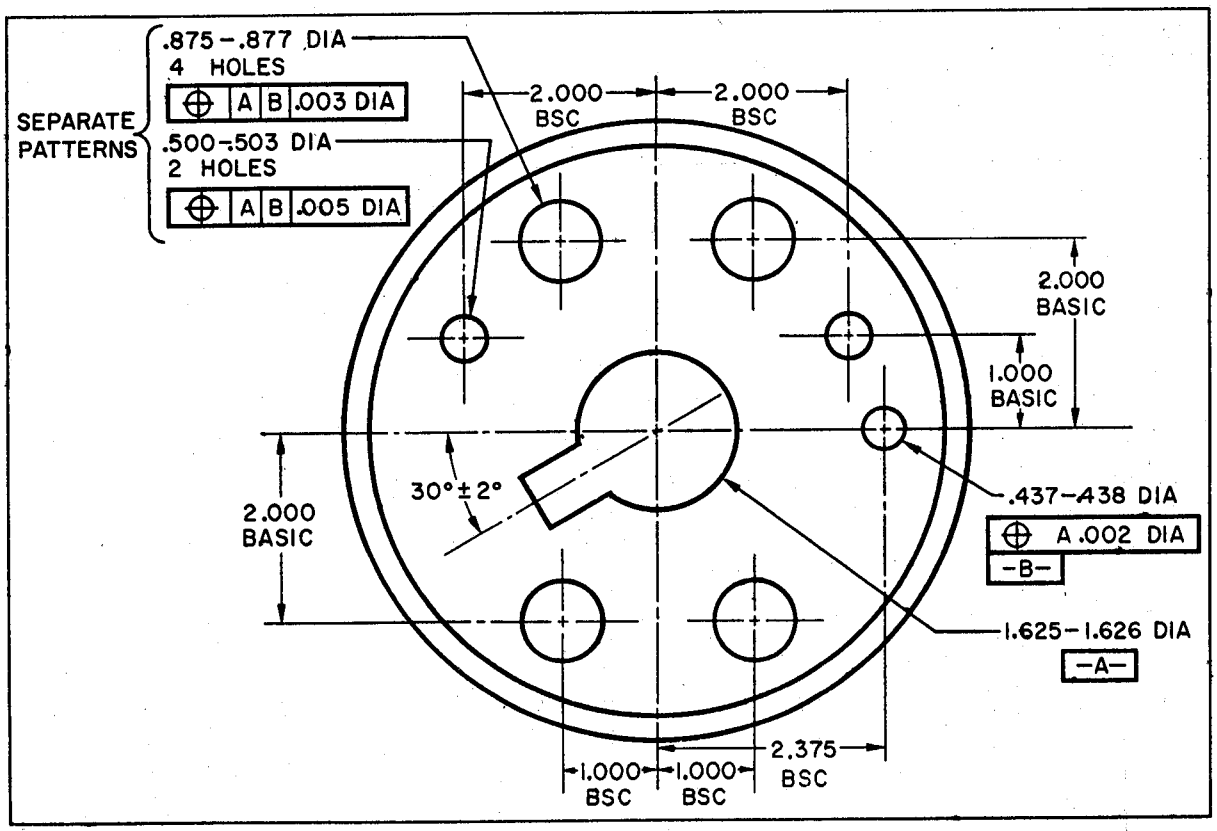


FIGURE 10–10. *True position.*

20. THE MEANING OF MMC AS RELATED TO POSITIONAL TOLERANCES. Maximum material condition by itself means that a feature of a finished product contains the maximum amount of material permitted by the toleranced size dimensions shown for that feature. Thus, for holes, slots, and other internal features, maximum material is the condition when these features are at their minimum allowable sizes, and for pads, shafts, bosses, thicknesses, and other external features, maximum material is the condition when these features are at their maximum sizes permitted by the toleranced dimensions. As a general rule, tolerances of position and the specified maximum material condition of mating features must be considered in relation

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to each other. In some cases the limits of location must be observed regardless of the actual finished sizes of the features concerned. However, in other applications, the limits of center distance may be exceeded if the features are not on their maximum material limits of size, because of the increased clearance between the mating features. This, then, is the interpretation (a) when the modifier "MMC" is used in conjunction with the location specification, and (b) under rule 1 of paragraph 7.3.4 when no modifier is specified in true position tolerancing.

20.1 To carry on interchangeable production efficiently, precautions must be taken to make sure that parts can be freely assembled. Fixed gages, made to simulate mating features at dimensional limits that make assembly most critical, can be used to assure free and proper assembly and permit simple and economical inspection methods. Figures 20–1 through 20–6 illustrate the use of fixed pin gages to check the locational accuracy of a two hole pattern, both when the holes are at maximum material condition (minimum size), and at maximum size.

20.1.1 Figurs 20–1 shows one method of specifying on a drawing a locational dimension for two holes, the allowable variation in location being expressed by limit dimensioning.

20.1.1.1 If the holes are exactly .500 in diameter (the maximum material condition, or the smallest size hole permitted by the drawing specification) and are centered exactly 2.000 apart, they will theoretically receive a gage consisting of two round pins fixed in a plate if the pins are centered 2.000 apart and are .500 in diameter.

20.1.1.2 However, the limits specified on the drawing permit the center distance between the holes to be from 1.995 to 2.005. The pins in the gage would therefore have to be .005 smaller, or .495 diameter, to enter the holes in their extreme positions, as shown in figure 20–2.

20.1.1.3 If the holes are exactly .500 in diameter but located at the maximum permissible center distance, the .495-diameter gage pins would contact the inner sides of the holes, since the distance between the inner sides of the gage pins is 1.505 (fig. 20–3).

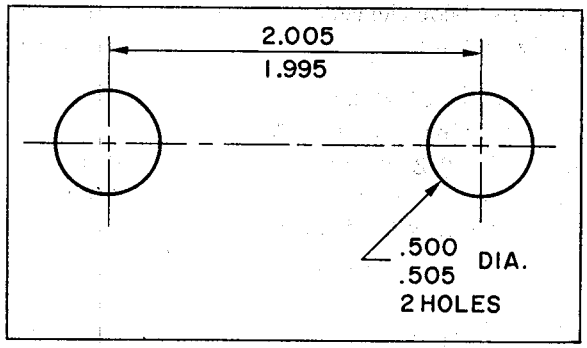


FIGURE 20–1. *The meaning of MMC as related to positional tolerances.*

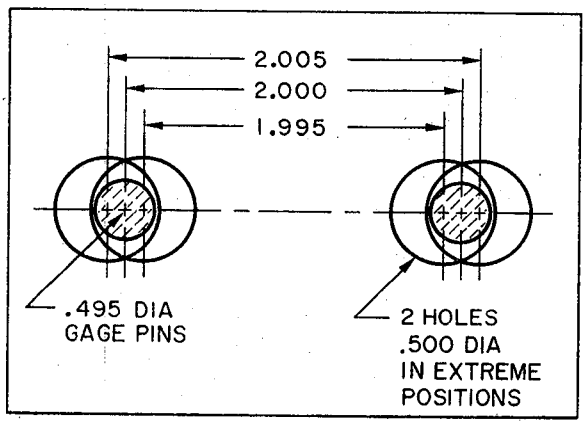


FIGURE 20–2. *The meaning of MMC as related to positional tolerances.*

20.1.1.4 Now if the holes are exactly .500 in diameter but located at the minimum permissible center distance, the gage pins would contact the outer sides of the holes, since the distance between the outer sides of the gage pins is 2.495 (fig. 20–4).

20.1.1.5 Neglecting gage-maker's tolerances, the gage pins would have to be .495 in diameter and the centers of the pins would have to be located exactly 2.000 apart. Thus the holes in the part, if they are .500 in size, will engage the gage pins if located within the limits specified on the drawing.

20.1.1.6 Now if the holes are at maximum size, .505 diameter, permitted by the drawing specification, they will be accepted by the gage when the inner sides of the holes contact the inner sides of the gage pins, which means that the holes will have a center distance of 2.010 (see figure 20–5).

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This is outside the center distance limits specified by figure 20–1.

20.1.1.7 These holes will also be accepted by the gage when their outer sides contact the outer sides of the gage pins, or a center distance of 1.990 (see fig. 20–6). This is again outside the center distance limits specified by figure 20–1.

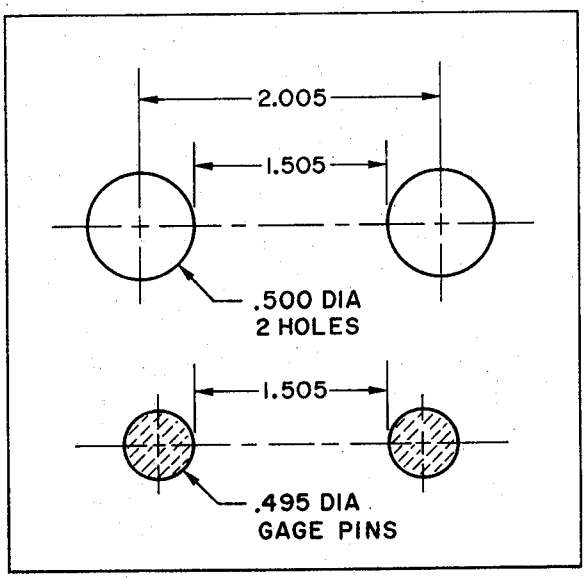


FIGURE 20–3. *The meaning of MMC as related to positional tolerances.*

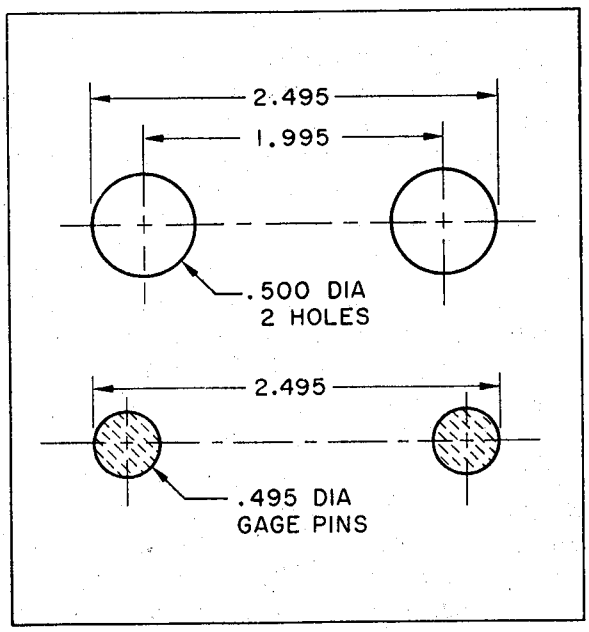


FIGURE 20–4. *The meaning of MMC as related to positional tolerances.*

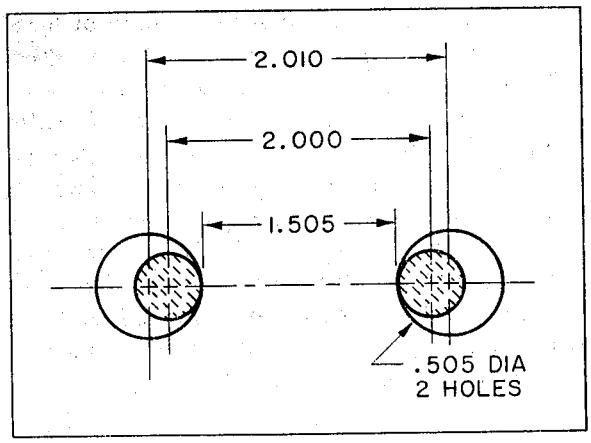


FIGURE 20–5. *The meaning of MMC as related to positional tolerances.*

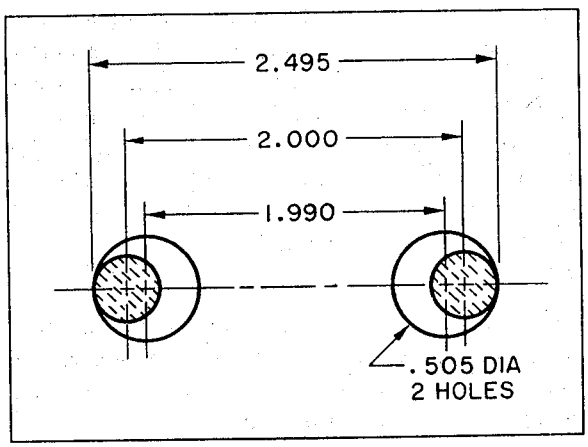


FIGURE 20–6. *The meaning of MMC as related to positional tolerances.*

20.1.1.8 When the condition described in paragraphs 20.1.1.6 and 20.1.1.7 is acceptable, the drawing may be dimensioned to state this precisely. Figure 20–7 (as contrasted with fig. 20–1) states specifically that the tolerance applies at MMC, and the cylindrical tolerance zones resulting from true position dimensioning are consistent with fixed pin gages.

20.2 It is desirable to have a generally accepted method of indicating on the toleranced center distances that the limits of center distance must be observed only if the features are at their maximum material limits of size, but may be exceeded when the features are finished away from their maximum material limits in the direction of minimum material. This standard provides such a method in paragraph 7.3.

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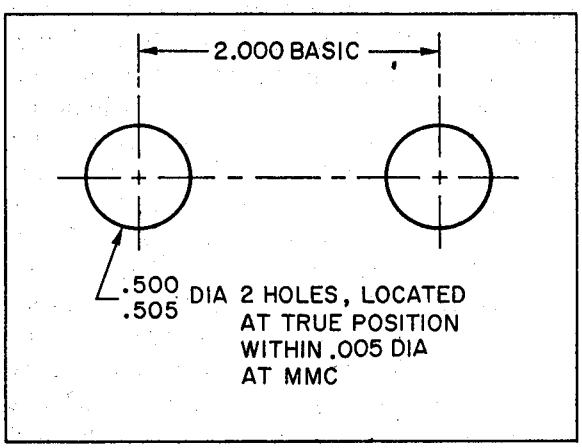


FIGURE 20-7. *The meaning of MMC as related to positional tolerances.*

20.2.1 Figure 20–8 illustrates the additional positional tolerance permissible at minimum material condition when positional tolerances apply only at maximum material condition. When the hole is at minimum material condition the total positional tolerance available is the specified true position tolerance plus the difference between permissible maximum size and minimum size.

20.2.2 Experience has indicated that in all but a few exceptional designs, this additional tolerance is acceptable and desirable. It permits the acceptance of parts which can be freely assembled whether or not the hole or other features are located within the specified positional tolerance. It will be noted that when a hole is larger than its minimum diameter, the total positional tolerance may be exceeded by an amount equal to the difference between the specified minimum size and the actual size of the hole (see fig. 20–8). If a particular hole in an actual part is at minimum diameter, the difference is zero, and so the actual location of that hole must be within the positional tolerance stated on the drawing.

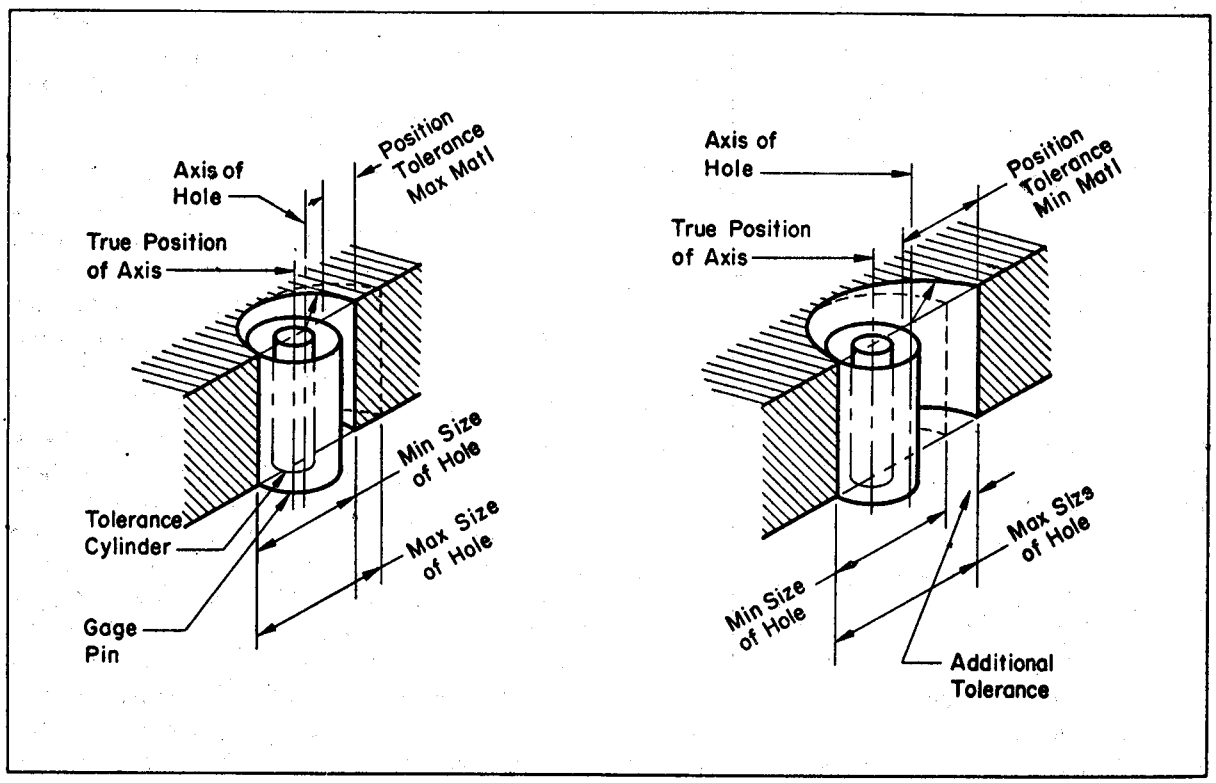


FIGURE 20–8. *The meaning of MMC as related to positional tolerances.*

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20.2.3 This principle of additional tolerance available when parts are at other than their maximum material condition has been recognized and used in manufacturing for many years. This is evident from the fixed-type pin gages which have been commonly employed to inspect parts in order to exercise control over the least favorable condition of assembly. However, the practice should not be applied indiscriminately. Its prime purpose should be to facilitate manufacture by permitting inspectors to accept parts that will assemble and function properly but that may otherwise be rejected because the positional tolerances specified on the drawing have been exceeded. When the additional tolerance provided by departures from MMC is not acceptable and the true position tolerance must apply regardless of feature size, the typical notes and symbols of paragraphs 7.5.4.2 and 7.5.4.4 should be changed to the following:

(a) 6 HOLES, .XXX DIA, LOCATED AT TRUE POSITION WITHIN .010 DIA REGARDLESS OF HOLE SIZE.

(b) 6 HOLES, .XXX DIA

⌖ .010 DIA Ⓢ

(c) 6 HOLES, .XXX DIA, LOCATED AT TRUE POSITION WITHIN .010 DIA RFS IN RELATION TO DATUM A RFS.

(d) 6 HOLES, .XXX DIA

⌖	A Ⓢ	.010 DIA Ⓢ

30. DIFFERENCES IN AMERICAN, BRITISH, AND CANADIAN STANDARDS. Figure 30–1 shows the differences that exist between American, British, and Canadian standards for form and positional tolerancing.

40. INTERRELATED GEOMETRIC CHARACTERISTICS AND THE TWO DATUM SYSTEM.

40.1 *Interrelated geometric characteristics.* The means of showing an interrelationship of geometric tolerances is prescribed by paragraph 7.2.1.7. A common example of interrelationship occurs in the case of a requirement for concentricity, when the functional cylindrical datum feature is very short and a joint face perpendicular to such cylinder has a much greater influence upon alignment of mating parts. Figure 40–1 illustrates this situation. Assuming the part shown to be one of a series of mating shells in an assembly, alignment is controlled primarily by the joint flanges, datums D and N. The cylindrical pilots, B and E, are more significant in controlling radial position (or axial offset at each joint). Datum E is required to be both concentric to B and perpendicular to D within .010 TIR as an interrelated requirement. Such a requirement could be checked with reference to an accurate gaging fixture, having a bore piloted on datum B and a flange held in contact with datum D. Concentricity and perpendicularity of bore E could then be checked simultaneously with an indicator while the gaging fixture is rotated about its axis.

40.1.1 Figure 40–2 illustrates the use of symbols for controlling the geometrical relationship of multiple surfaces constructed around and at right angles to common datum axes. Note that all surfaces used as datums for establishing the three axes are functional.

40.2 *True position with reference to two datums.* Figure 40–1 also illustrates the use of two datums in true position dimensioning of both individual features and patterns of holes. Bore F is located with respect to the cylindrical pilot B and the dowel hole C within a true position tolerance of .04 diameter. Datum B is the primary datum and governs the radial location of bore F, whereas datum C is the secondary datum (or orientation dowel hole) and governs the angular location of bore F. Referring to detail Y, dowel hole G is located radially with reference to primary datum F and angularly with reference to secondary datum C. Note that with this system of true position dimensioning, B, C, F, H, and G are not required to be exactly on the same vertical centerline, and that positions are specified with reference to bores, pilots, and dowel holes rather than in relation to nonfunctional and ambiguous centerlines.

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ITEM	MIL-STD-8C		BRITISH	CANADIAN
	NOTES	SYMBOLS		
ROUNDNESS	ROUND WITHIN .002 ON DIA	○ .002 ON DIA	ENTIRE SURFACE MUST LIE INSIDE ANNULAR TOLERANCE ZONE .001 WIDE	ENTIRE SURFACE MUST LIE INSIDE ANNULAR TOLERANCE ZONE .001 WIDE
TRUE POSITION DIMENSION	BASIC OR BSC	BASIC OR BSC	(TP)	(TP)
TRUE POSITION AT MMC	LOCATED AT TRUE POSITION WITHIN .010 DIA	⌖ .010 DIA	POSN TOL .010 DIA (MMC)	LOCATE WITHIN .010 DIA MMC
TRUE POSITION RFS WITH DATUM RFS	LOCATED AT TRUE POSITION WITHIN .010 DIA RFS IN RELATION TO DATUM A RFS	⌖ A Ⓢ .010 DIA Ⓢ	POSN TOL .010 DIA DATUM-A	LOCATE WITHIN .010 DIA DATUM-A
TRUE POSITION AT MMC WITH DATUM AT MMC	LOCATED AT TRUE POSITION WITHIN .010 DIA IN RELATION TO DATUM A	⌖ A.010 DIA	POSN TOL .010 DIA (MMC) DATUM-A (MMC)	LOCATE WITHIN .010 MMC DATUM A MMC
TRUE POSITION WITH TWO DATUMS	LOCATED AT TRUE POSITION WITHIN .005 DIA AT MMC IN RELATION TO DATUM A AT MMC AND DATUM B RFS	⌖ A Ⓜ B Ⓢ .005 DIA Ⓜ	POSN TOL .005 DIA (MMC) DATUM-A (MMC) AND B	LOCATE WITHIN .005 DIA MMC DATUMS A, MMC AND B
TRUE POSITION OF SLOTS OR TABS	LOCATED AT TRUE POSITION WITHIN .002 WIDE ZONE RFS IN RELATION TO DATUM A AT MMC	⌖ A Ⓜ .002 TOTAL Ⓢ	POSN TOL .002 WIDE, DATUM-A (MMC)	LOCATE WITHIN .002 WIDE ZONE WITH RESPECT TO DATUM A AT MMC

FIGURE 30-1. *Differences in American-British-Canadian standards (Sheet 1).*

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ITEM	MIL-STD-8C		BRITISH	CANADIAN
	NOTES	SYMBOLS		
AUXILIARY INFORMATION	REF		(XX)	REF
DEC DIM.	1.005		1·005	1·005
TRUE PROFILE DIMENSIONING	BASIC		(TP)	(TP)
STRAIGHTNESS	STRAIGHT WITHIN .003 TOTAL	⏤ | .003	STR TOL ·003	STRAIGHT WITHIN ·003
FLATNESS	FLAT WITHIN .003 TOTAL	⏥ | .003	FLAT TOL ·003 WIDE	FLAT WITHIN ·003
PARALLELISM	PARALLEL TO A WITHIN .005 TOTAL	∥ A .005	PAR TOL ·005 WIDE (OR DIA) DATUM-FACE A	PARALLEL TO FACE A DATUM
PERPENDICULARITY	PERPENDICULAR TO A WITHIN .002 TOTAL	⊥ A .002	SQ TOL ·002 WIDE (OR DIA) DATUM-FACE A	SQUARE WITH FACE A WITHIN ·002
ANGULARITY	ANGULAR TOL .006 TOTAL DATUM A	∠ A .006	ANG. TOL ·006 WIDE DATUM-FACE A	ANG. TOL ·006 DATUM-FACE A
CONCENTRICITY	CONCENTRIC TO A WITHIN .003 TIR	◎ A .003	CONC TOL ·003 DIA DATUM-A	CONC TO CYL A WITHIN ·003 FIM
SYMMETRY	SYMMETRICAL WITH A WITHIN .004 TOTAL	⌯ A .004	SYM TOL ·004 WIDE DATUM-A	SYMMETRICAL WITH WIDTH A WITHIN ·004

FIGURE 30–1. *Differences in American-British-Canadian standards (Sheet 2).*

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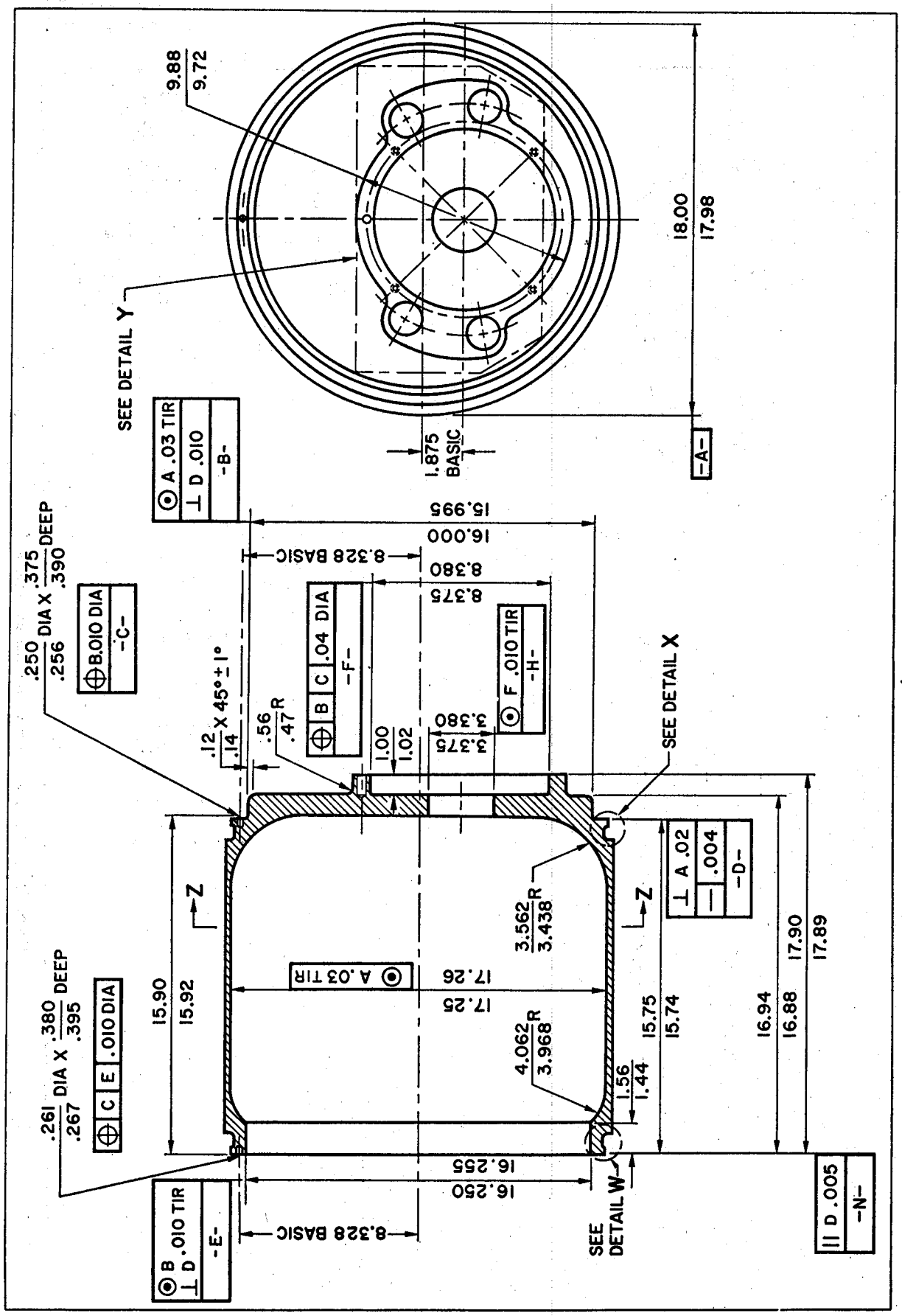


FIGURE 40-1. *True position and form tolerances using double datums (Sheet 1)*

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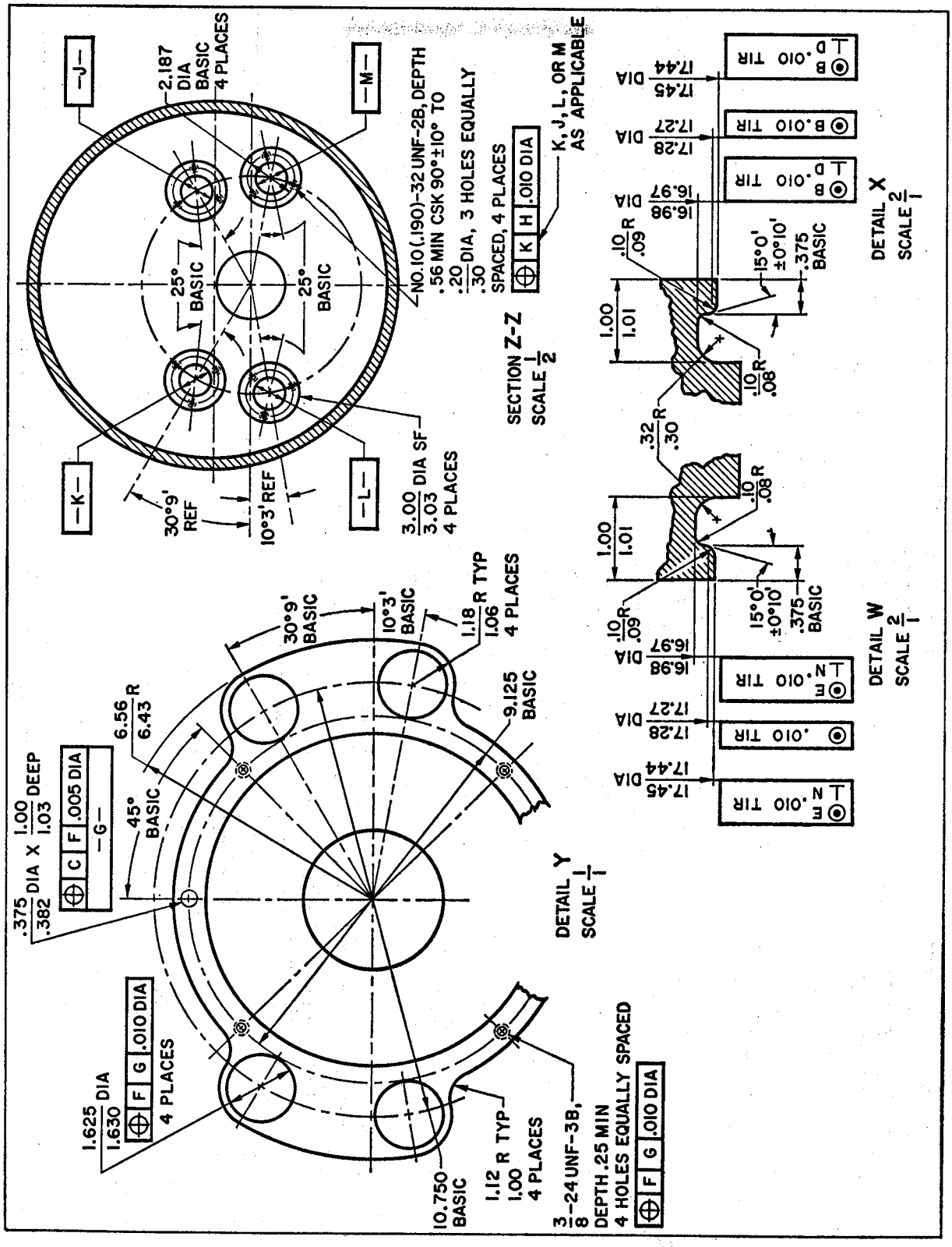


FIGURE 40-1. *True position and form tolerances using double datums (Sheet 2).*

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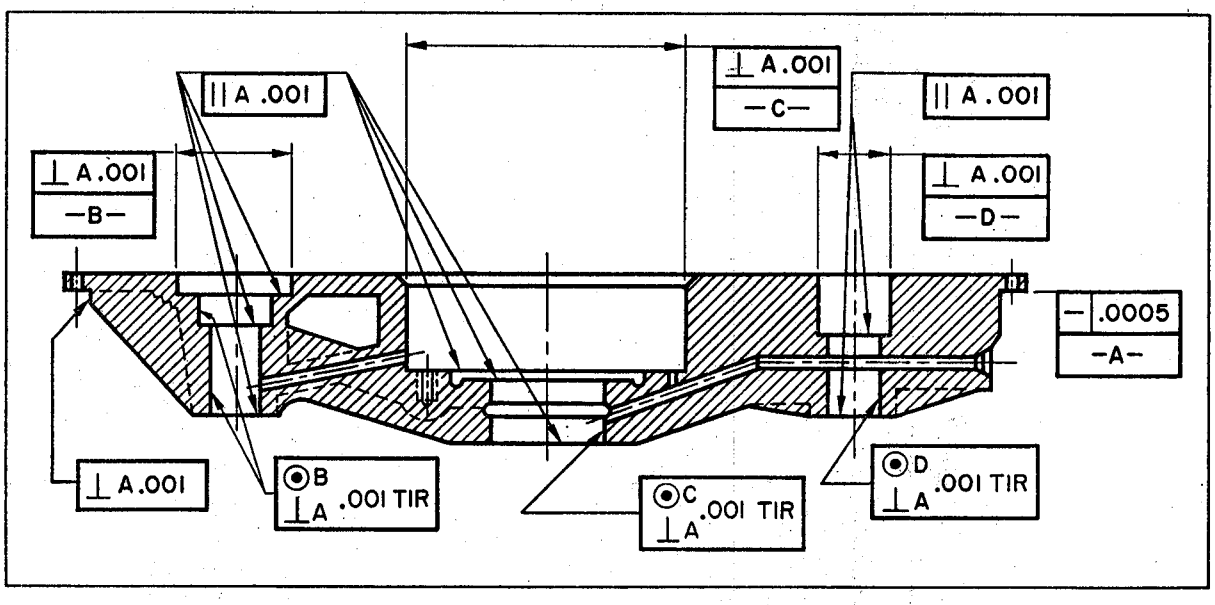


FIGURE 40–2. *Controlling geometrical relationships of multiple surfaces.*

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STANDARD CONVERSION TABLE

4ths	8ths	16ths	32nds	64ths	To 4 PLACES	To 3 PLACES	To 2 PLACES
				1/64	.0156	.016	.02
			1/32		.0312	.031	.03
				3/64	.0469	.047	.05
		1/16			.0625	.062	.06
				5/64	.0781	.078	.08
			3/32		.0938	.094	.09
				7/64	.1094	.109	.11
	1/8				.1250	.125	.12
				9/64	.1406	.141	.14
			5/32		.1562	.156	.16
				11/64	.1719	.172	.17
		3/16			.1875	.188	.19
				13/64	.2031	.203	.20
			7/32		.2188	.219	.22
				15/64	.2344	.234	.23
1/4					.2500	.250	.25
				17/64	.2656	.266	.27
			9/32		.2812	.281	.28
				19/64	.2969	.297	.30
		5/16			.3125	.312	.31
				21/64	.3281	.328	.33
			11/32		.3438	.344	.34
				23/64	.3594	.359	.36
	3/8				.3750	.375	.38
				25/64	.3906	.391	.39
			13/32		.4062	.406	.41
				27/64	.4219	.422	.42
		7/16			.4375	.438	.44
				29/64	.4531	.453	.45
			15/32		.4688	.469	.47
				31/64	.4844	.484	.48
					.5000	.500	.50

4ths	8ths	16ths	32nds	64ths	To 4 PLACES	To 3 PLACES	To 2 PLACES
				33/64	.5156	.516	.52
			17/32		.5312	.531	.53
				35/64	.5469	.547	.55
		9/16			.5625	.562	.56
				37/64	.5781	.578	.58
			19/32		.5938	.594	.59
				39/64	.6094	.609	.61
	5/8				.6250	.625	.62
				41/64	.6406	.641	.64
			21/32		.6562	.656	.66
				43/64	.6719	.672	.67
		11/16			.6875	.688	.69
				45/64	.7031	.703	.70
			23/32		.7188	.719	.72
				47/64	.7344	.734	.73
3/4					.7500	.750	.75
				49/64	.7656	.766	.77
			25/32		.7812	.781	.78
				51/64	.7969	.797	.80
		13/16			.8125	.812	.81
				53/64	.8281	.828	.83
			27/32		.8438	.844	.84
				55/64	.8594	.859	.86
	7/8				.8750	.875	.88
				57/64	.8906	.891	.89
			29/32		.9062	.906	.91
				59/64	.9219	.922	.92
		15/16			.9375	.938	.94
				61/64	.9531	.953	.95
			31/32		.9688	.969	.97
				63/64	.9844	.984	.98
					1.0000	1.000	1.00

TABLE I

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INDEX

	Page
Abbreviations	7
Aligned dimensions	6
Allowance, definition of	1
Angular dimensions	7
Angular surfaces, positioning of	23
Angularity	48
Average diameter	58
Basic dimensions, definition of	14
Bolt clearance holes	40
Chamfers	18
Center drills	17
Circular arcs, dimensioning of	16
Clearance, definition of	1
Clearance holes	40
Combined symbols	34
Common fractions, use of	8
Concentricity	55
Counterbored holes	17
Countersunk holes	17
Cylindrical or spherical surfaces	14
Dashes	7
Datums	10
accuracy of	12
avoiding ambiguity of	41
dimensions from	10
identifying symbols	33
implied	41
multiple	34
reference to	34
selection of	12
Decimal points	7
Decimals	
dimensioning	8
use of	8
Diameter, average	58
Diameters of concentric elements	14
Differences in American, British, and Canadian Standards	74
Dimension	
basic	14
definition of	1
lines	3
Dimensioning	1
angular, definition of	1
decimal	8
dovetails	25

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Check the source to verify that this is the current version before use.

INDEX (continued)

	Page
for end-product drawings	9
for fits	31
for process drawings	9
in limited space	5
of sheet metal parts	25
ordinate	12
round holes	17
special features	14
tabular	14
tapers	18
to hidden lines	7
within the outline of a view	7
Dimensions	9
angular	7
from datum	10
how indicated	2
establishing true position	38
position	10
reference	14
Distances along curved surfaces	15
Dovetails	25
Drafting methods	2
Eccentricity, definition of	1
Extension lines	3
crossing	4
through point locations	4
Feature, definition of	1
Fit	
actual, definition of	1
clearance, definition of	1
definition of	1
dimensioning for	31
interference, definition of	1
transition, definition of	1
Flatness	
specifying and interpreting	45
tolerance	45
Form tolerances, methods of expressing	32
Former symbols	34
Free-state variation	58
General tolerance notes	32
Geometric characteristic symbols	33
Implied datum surfaces	41
Interchangeability, definition of	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

INDEX (continued)

Page

Interference, definition of 2
Interrelated geometric characteristics 34, 73

Keys and keyways 18
Knurls 18

Leaders 5
  improper use of 5
  to circles and arcs 5
Limit dimensioning method 28
Limit numerals, placement of 31
Limits
  definition of 2
  general discussion 28
  interpretation of 30

MMC
  as related to positional tolerances 69
  general rules applicable to 35
  perfect form at 61
  significance of 35
Metric system 7
Multiple datums 34

Noncircular curves, dimensioning of 16

Ordinate dimensioning 12

Parallelism 53
Parts with rounded ends 15
Perpendicularity 50
Plain round holes 17
Position, dimensions establishing 10
Positional tolerances
  meaning of MMC as related to 69
  methods of expressing 32
  relation to other tolerances of form 61
Projected tolerance zones, use of 38

RFS
  general rules applicable to 35
  significance of 35
Radii of arcs 14
Reference dimensions 14
Reference to datum 34
Rigidity 58
Round holes
  dimensioning 17
  locating 10
Roundness 56

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

INDEX (continued)

Page

Shaft centers 17
Sheet metal parts, dimensioning of 25
Size 10
  actual 10
  basic 10
  design 10
  limits of 10
  nominal 10
Spherical surfaces 15
Spot-faced holes 17
Straightness
  of axis 45
  of elements 45
Surface coatings, effect of 9
Symbols 33
  combined 34
  datum identifying 33
  feature control 33
  former 34
Symmetry, definition of 56

Tabular dimensioning 14
Tapers 18
  conical 18
    basic dimensioning of 20
    form dimensioning of 19
  flat 23
Tapped holes 40
Tolerance numerals, placement of 31
Tolerances of form 32
  angularity 48
  concentricity 55
  flatness 45
  free-state variation 58
  meaning when not specified 62
  parallelism 53
  perpendicularity 50
  roundness 56
  symmetry 56
Tolerancing 28
  bilateral 28
  expressing allowable variations 28
  limit dimensioning 28
  methods of 28
  true position—see True position tolerancing
  unilateral 28

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

INDEX (continued)

	Page
True position tolerancing	37
dimensions establishing	38
interpretation of	40, 65
methods of specifying	38, 67, 73
tolerance formulas for holes	38
with reference to two datums	73
True radius, indication of	15
Two datum system	73
Typical, definition of	2
Unidirectional dimensions	6
Use of standards	2
Virtual size	36
Zone tolerances for contours	62

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